



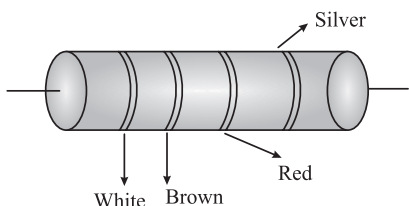
Physics

Current Electricity

Exercise -1

MICROSCOPIC ANALYSIS, DRIFT VELOCITY, OHM'S LAW AND CONDUCTIVITY

- There is a current of 40 ampere in a wire of 10^{-6} m^2 area of cross-section. If the number of free electron per m^3 is 10^{29} , then the drift velocity will be
 - $1.25 \times 10^3 \text{ m/s}$
 - $2.50 \times 10^{-3} \text{ m/s}$
 - $25.0 \times 10^{-3} \text{ m/s}$
 - $250 \times 10^{-3} \text{ m/s}$
- Every atom has one free electron in copper. If 1.1 ampere current is flowing in the wire of copper having 1 mm diameter, then the drift velocity (approx.) will be (Density of copper = $9 \times 10^3 \text{ kg m}^{-3}$ and atomic weight = 63g)
 - 0.3 mm/s
 - 0.1 mm/s
 - 0.2 mm/s
 - 0.2 cm/s
- Drift velocity, v_d , varies with the intensity of electric field E , as
 - $v_d \propto E$
 - $v_d \propto \frac{1}{E}$
 - $v_d \propto E^0$
 - $v_d \propto E^2$
- A certain wire has a resistance R . The resistance of another wire identical with the first except having twice the diameter is
 - $2R$
 - $0.25R$
 - $4R$
 - $0.5R$
- A metallic block has no potential difference applied across it, then the root mean velocity of free electrons is (T = absolute temperature of the block)
 - Proportional to T
 - Proportional to $\sqrt{T}$
 - Zero
 - Finite but independent of temperature
- In the figure a carbon resistor has bands of different colours on its body as mentioned in the figure. The value of the resistance is

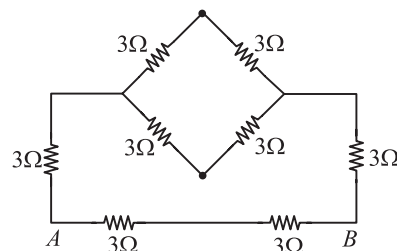


- $2.2 \text{ k}\Omega \pm 5\%$
- $3.3 \text{ k}\Omega \pm 10\%$
- $5.6 \text{ k}\Omega \pm 5\%$
- $9.1 \text{ k}\Omega \pm 10\%$

- A wire of cross-sectional area A , length L_1 , resistivity σ_1 and temperature coefficient of resistivity α_1 is connected to a second wire of length L_2 , resistivity σ_2 , temperature coefficient of resistivity α_2 and the same area A , so that both wires carry same current. Total resistance R is independent of temperature for small temperature change if (thermal expansion effect is negligible)
 - $\alpha_1 = -\alpha_2$
 - $\sigma_1 L_1 \alpha_1 + \sigma_2 L_2 \alpha_2 = 0$
 - $L_1 \alpha_1 + L_2 \alpha_2 = 0$
 - None of these

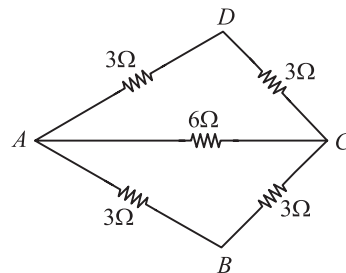
SERIES AND PARALLEL COMBINATION OF RESISTANCE

- A uniform wire of resistance R is uniformly compressed along its length, until its radius becomes n times the original radius. New resistance of the wire becomes
 - $\frac{R}{n^4}$
 - $\frac{R}{n^2}$
 - $\frac{R}{n}$
 - nR
- There are eight resistors of equal resistance R . Four groups of these are formed such that each group has two resistors connected in parallel. These groups are connected in series. The total resistance of the system will be
 - $R/2$
 - $2R$
 - $4R$
 - $8R$
- Equivalent resistance between A and B will be



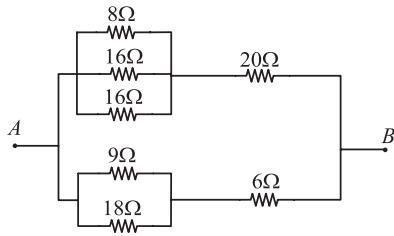
- 2Ω
- 18Ω
- 6Ω
- 3.6Ω

- The effective resistance between the points A and B in the figure is



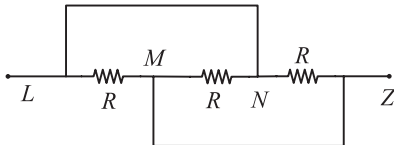
- 5Ω
- 2Ω
- 3Ω
- 4Ω

12. The equivalent resistance of the arrangement of resistances shown in adjoining figure between the points A and B is



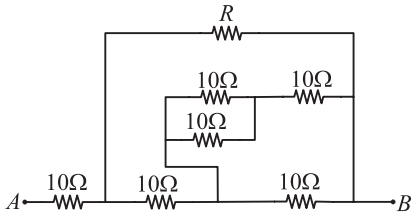
- (a) $6\ \Omega$ (b) $8\ \Omega$ (c) $16\ \Omega$ (d) $24\ \Omega$

13. Three equal resistances each of value R are joined as shown in the figure. The equivalent resistance between M and N is



- (a) R (b) $2R$ (c) $\frac{R}{2}$ (d) $\frac{R}{3}$

14. For what value of R the net resistance of the circuit will be $18\ \Omega$?



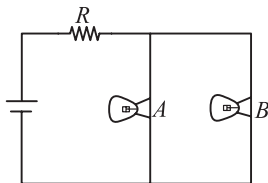
- (a) $8\ \Omega$ (b) $10\ \Omega$
(c) $16\ \Omega$ (d) $24\ \Omega$

15. Two wires of equal diameters, of resistivities ρ_1 and ρ_2 and lengths l_1 and l_2 , respectively, are joined in series. The equivalent resistivity of the combination is

- (a) $\frac{\rho_1 l_1 + \rho_2 l_2}{l_1 + l_2}$ (b) $\frac{\rho_1 l_2 + \rho_2 l_1}{l_1 - l_2}$
(c) $\frac{\rho_1 l_2 + \rho_2 l_1}{l_1 + l_2}$ (d) $\frac{\rho_1 l_1 - \rho_2 l_2}{l_1 - l_2}$

POWER AND HEAT DISSIPATION IN A CIRCUIT

16. A circuit consists of a battery, a resistor R and two light bulbs A and B as shown.



If the filament in light bulb A burns out, then which of the following is true for light bulb B ?

- (a) It is turned off (b) Its brightness does not change
(c) It gets dimmer (d) It gets brighter

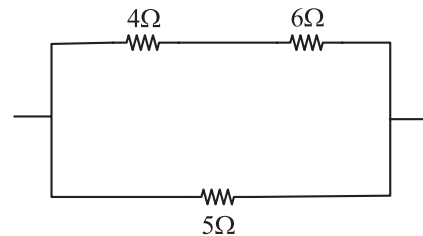
17. Power generated across a uniform wire connected across a supply is H . If the wire is cut into n equal parts and all the parts are connected in parallel across the same supply, the total power generated in the circuit is

- (a) $\frac{H}{n^2}$ (b) $n^2 H$
(c) nH (d) $\frac{H}{n}$

18. Three 60 W light bulbs are mistakenly wired in series and connected to a 120 V power supply. Assume the light bulbs are rated for single connection to 120 V . With the mistaken connection, the power dissipated by each bulb is

- (a) 6.7 W (b) 13.3 W (c) 20 W (d) 40 W

19. In the circuit shown in figure the rate of heat produced in the $5\ \Omega$ resistor due to a current flowing in it is $10\text{ calories per second}$. The rate of heat produced in the $4\ \Omega$ resistor is

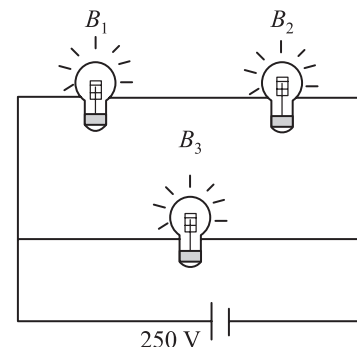


- (a) 1 cal s^{-1} (b) 2 cal s^{-1} (c) 3 cal s^{-1} (d) 4 cal s^{-1}

20. The charge flowing through a resistance R varies with time t as $Q = at - bt^2$. The total heat produced in R from $t = 0$ to the time when value of Q again becomes zero is

- (a) $\frac{a^3 R}{6b}$ (b) $\frac{a^3 R}{3b}$
(c) $\frac{a^3 R}{2b}$ (d) $\frac{a^3 R}{b}$

21. A 100 W bulb B_1 , and two 60 W bulbs B_2 and B_3 , are connected to a 250 V source, as shown in the figure. If W_1 , W_2 and W_3 are the output powers of the bulbs B_1 , B_2 and B_3 , respectively. Then,

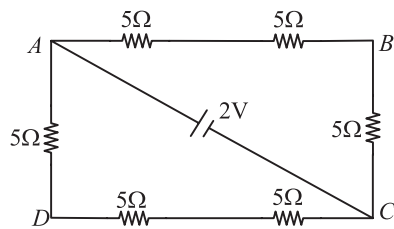


- (a) $W_1 > W_2 = W_3$
(b) $W_1 > W_2 > W_3$
(c) $W_1 < W_2 = W_3$
(d) $W_1 < W_2 < W_3$

22. If a cell produces the same amount of heat in two resistors R_1 and R_2 in the same time separately, the internal resistance of the cell is
- (a) $(R_1 + R_2)/2$ (b) $\sqrt{R_1 R_2}$
 (c) $\sqrt{(R_1 R_2)/2}$ (d) $(R_1 - R_2)/2$
23. Two heater wires of equal length are first connected in series and then in parallel. The ratio of heat produced in two cases will be
- (a) 1 : 2 (b) 1 : 4
 (c) 2 : 1 (d) 4 : 1
24. The effective wattage of 60 W, 220 V and 40 W, 220 V lamps connected in series with a 220 V power supply is equal to
- (a) 24 W (b) 20 W
 (c) 100 W (d) 80 W
25. A heater boils a certain quantity of water in time t_1 . Another heater boils the same quantity of water in time t_2 . If both heaters are connected in series, the combination will boil the same quantity of water in time
- (a) $\frac{1}{2}(t_1 + t_2)$
 (b) $(t_1 + t_2)$
 (c) $\frac{t_1 t_2}{(t_1 + t_2)}$
 (d) $\sqrt{t_1 t_2}$
26. The voltage across a bulb is decreased by 2%. Assuming that the resistance of the filament remains unchanged, the power of the bulb will
- (a) Decrease by 2% (b) Increase by 2%
 (c) Decrease by 4% (d) Increase by 4%
27. A 500 W heating unit is designed to operate on a 115 volt line. If the line voltage drops to 110 volt, the percentage drop in heat output will be
- (a) 10.20%
 (b) 4.1%
 (c) 8.5 %
 (d) 7.6%

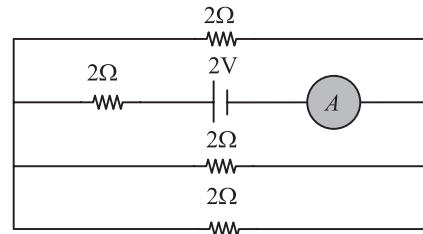
OHM'S LAW

28. The potential difference between points A and B of adjoining figure is



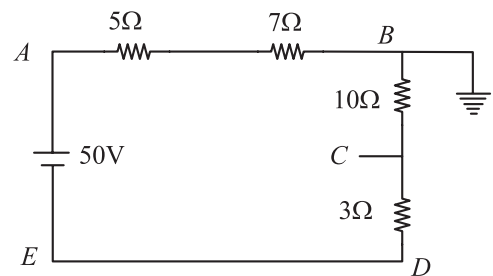
- (a) $\frac{2}{3}$ V (b) $\frac{8}{9}$ V (c) $\frac{4}{3}$ V (d) 2V

29. The reading of the ammeter in figure shown is



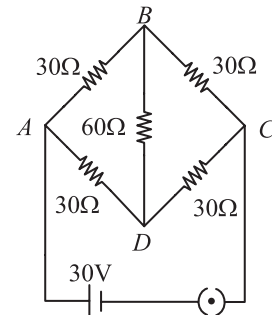
- (a) $\frac{1}{8}$ A
 (b) $\frac{3}{4}$ A
 (c) $\frac{1}{2}$ A
 (d) 2 A

30. In the circuit shown, the point 'B' is earthed. The potential at the point 'A' is



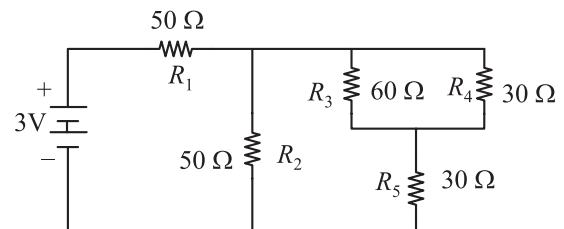
- (a) 14 V (b) 24 V (c) 26 V (d) 50 V

31. The current between B and D in the given figure is



- (a) 1 A (b) 2 A
 (c) Zero (d) 0.5 A

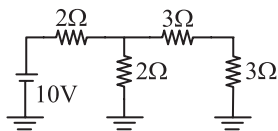
32. In circuit shown below, the resistances are given in ohms and the battery is assumed ideal with emf equal to 3 volt. The voltage across the resistance R_4 is



- (a) 0.4 V (b) 0.6 V
 (c) 1.2 V (d) 1.5 V

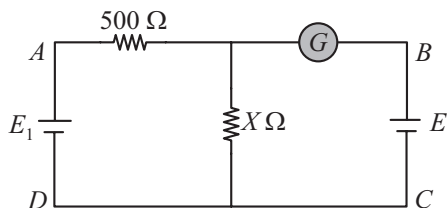
KIRCHHOFF'S LAWS AND EQUIVALENT BATTERY

33. Current in $3\ \Omega$ resistance is



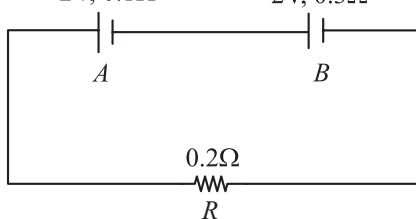
- (a) 1 A (b) $\frac{1}{7}$ A
(c) $\frac{5}{7}$ A (d) $\frac{15}{7}$ A

34. In the adjoining circuit, the battery E_1 has an emf of 12 volt and zero internal resistance while the battery E is ideal and has an emf of 2 volt. If the galvanometer G reads zero, then the value of the resistance X in ohm is



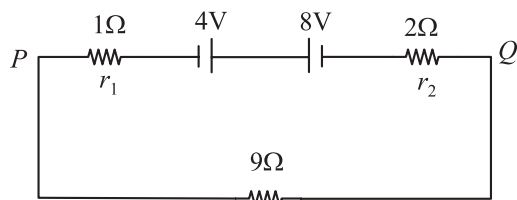
- (a) 10 (b) 100
(c) 500 (d) 200

35. The internal resistances of two cells shown are $0.1\ \Omega$ and $0.3\ \Omega$. If $R = 0.2\ \Omega$, the potential difference across the cell(s)



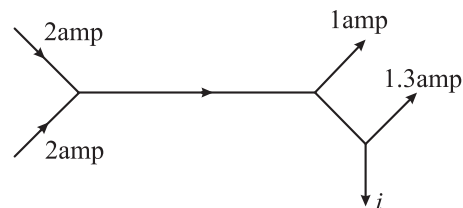
- (a) B will be zero
(b) A will be zero
(c) A and B will be 2 V
(d) A will be > 2 V and across the cell B will be < 2 V

36. Two batteries of emf 4 V and 8 V with internal resistances $1\ \Omega$ and $2\ \Omega$ are connected in a circuit with a resistance of $9\ \Omega$ as shown in figure. The current and potential difference between the points P and Q are



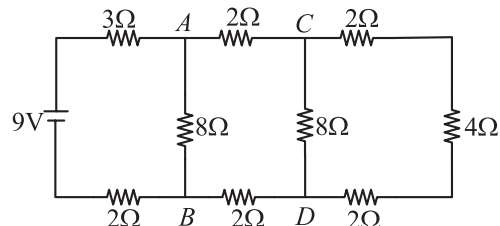
- (a) $\frac{1}{3}$ A and 3 V (b) $\frac{1}{6}$ A and 4 V
(c) $\frac{1}{9}$ A and 9 V (d) $\frac{1}{2}$ A and 12 V

37. The figure below shows currents in a part of an electric circuit. The current i is



- (a) 1.7 amp (b) 3.7 amp
(c) 1.3 amp (d) 1 amp

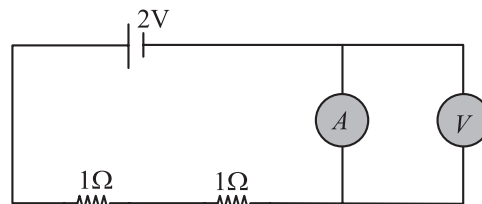
38. In the circuit shown in figure, the current through



- (a) the $3\ \Omega$ resistor is 0.50 A
(b) the $3\ \Omega$ resistor is 0.25 A
(c) the $4\ \Omega$ resistor is 0.50 A
(d) the $4\ \Omega$ resistor is 0.25 A

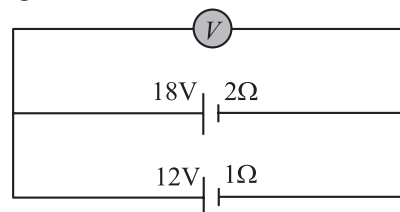
MEASURING INSTRUMENTS

39. In the circuit shown, A and V are ideal ammeter and voltmeter respectively. Reading of the voltmeter will be



- (a) 2 V (b) 1 V (c) 0.5 V (d) Zero

40. Two batteries, one of emf 18 volt and internal resistance $2\ \Omega$ and the other of emf 12 volt and internal resistance $1\ \Omega$, are connected as shown. The ideal voltmeter V will record a reading of



- (a) 15 volt
(b) 30 volt
(c) 14 volt
(d) 18 volt

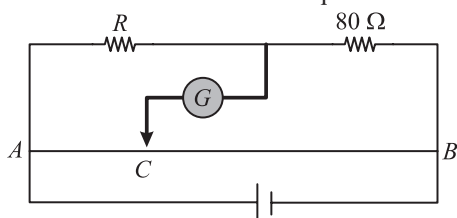
41. A 2 volt battery, a $15\ \Omega$ resistor and a potentiometer of wire length 100 cm, all are connected in series. If the resistance of potentiometer wire is $5\ \Omega$, then the potential gradient of the potentiometer wire is

(a) 0.005 V/cm (b) 0.05 V/cm
(c) 0.02 V/cm (d) 0.2 V/cm

42. An ammeter of 5 ohm resistance can read upto 5 mA. If it is to be used to read 100 volts, how much resistance is to be connected in series

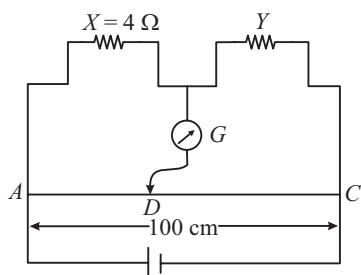
(a) 19.9995 Ω (b) 199.995 Ω
(c) 1999.95 Ω (d) 19995 Ω

43. AB is a wire of uniform resistance. The galvanometer G shows no current when the length $AC = 20$ cm and $CB = 80$ cm. The resistance R is equal to



(a) 2 Ω (b) 8 Ω (c) 20 Ω (d) 40 Ω

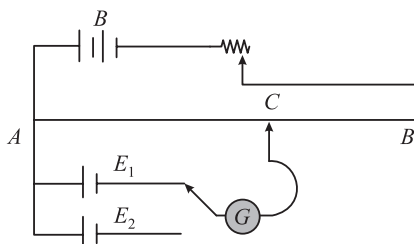
44. Figure shows a meter bridge. Wire AC has uniform cross-section. The length of wire AC is 100 cm. X is a standard resistor of $4\ \Omega$ and Y is a coil. When Y is immersed in melting ice the null point is at 40 cm from point A . When the coil Y is heated to 100°C , a $78\ \Omega$ resistor has to be connected in parallel with Y in order to keep the bridge balanced at the same point.



Temperature coefficient of resistance of the coil is

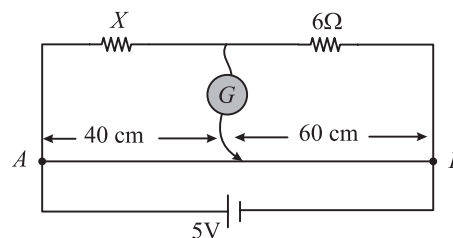
(a) $6.3 \times 10^{-4}\ \text{K}^{-1}$ (b) $4.3 \times 10^{-4}\ \text{K}^{-1}$
(c) $8.3 \times 10^{-4}\ \text{K}^{-1}$ (d) $2.3 \times 10^{-4}\ \text{K}^{-1}$

45. The circuit shown here is used to compare the emf of two cells E_1 and E_2 ($E_1 > E_2$). The null point is at C when the galvanometer is connected to E_1 . When the galvanometer is connected to E_2 , the null point will be



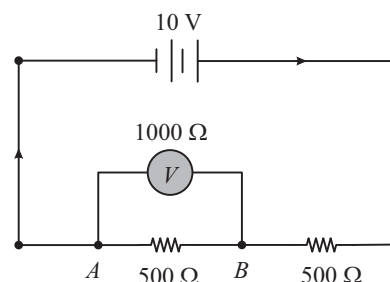
(a) To the left of C (b) To the right of C
(c) At C itself (d) Nowhere on AB

46. In the circuit shown, the meter bridge is in its balanced state. The meter bridge wire has a resistance $0.1\ \Omega/\text{cm}$. The value of unknown resistance X and the current drawn from the battery of negligible resistance is



(a) 6 Ω , 5 A (b) 10 Ω , 0.1 A
(c) 4 Ω , 1.0 A (d) 12 Ω , 0.5 A

47. What is the reading of voltmeter in the following figure?

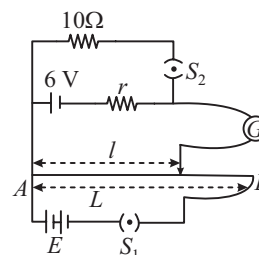


(a) 3 V (b) 2 V
(c) 5 V (d) 4 V

48. A galvanometer has a resistance of $20\ \Omega$ and reads full-scale when 0.2 V is applied across it. To convert it into a 10 A ammeter, the galvanometer coil should have a

(a) $0.01\ \Omega$ resistor connected across it in parallel
(b) $0.02\ \Omega$ resistor connected across it in parallel
(c) $200\ \Omega$ resistor connected in series with it
(d) $2000\ \Omega$ resistor connected in series with it

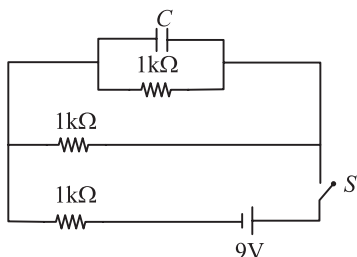
49. In the arrangement shown in figure when the switch S_2 is open, the galvanometer shows no deflection for $l = L/2$. When the switch S_2 is closed, the galvanometer shows no deflection for $l = 5L/12$. The internal resistance (r) of 6 V cell and the emf E of the other battery are respectively



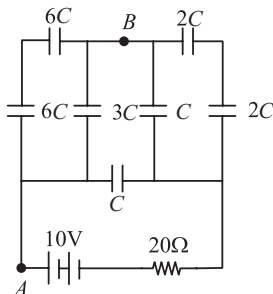
(a) 3 Ω , 8 V (b) 2 Ω , 12 V
(c) 2 Ω , 24 V (d) 3 Ω , 12 V

RC CIRCUIT

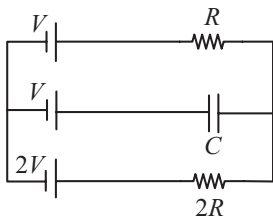
50. A capacitor $C = 100 \mu\text{F}$ is connected to three resistors each of resistance $1 \text{ k}\Omega$ and a battery of emf 9 V . The switch S has been closed for long time so as to charge the capacitor. When switch S is opened, the capacitor discharges with time constant



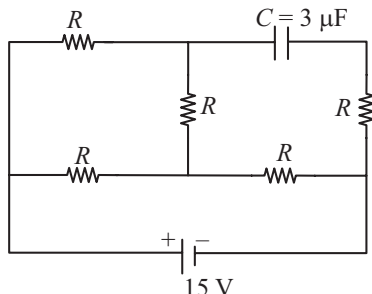
- (a) 33 ms (b) 5 ms
(c) 3.3 ms (d) 50 ms
51. For the circuit shown here, the potential difference between points A and B is (at steady state)



- (a) 2.5 V (b) 7.5 V
(c) 10 V (d) Zero
52. In the given circuit, with steady current the potential drop across the capacitor must be

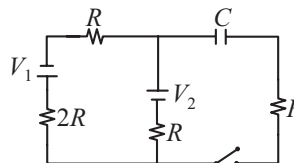


- (a) V (b) $\frac{V}{2}$ (c) $\frac{V}{3}$ (d) $\frac{2V}{3}$
53. In the circuit shown, the cell is ideal, with emf = 15 V . Each resistance is of 3Ω . The potential difference across the capacitor in steady state is



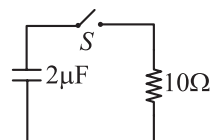
- (a) Zero (b) $9V$ (c) $12V$ (d) $15V$

54. In the circuit shown the time constant of the circuit is



- (a) $\frac{5}{3}RC$ (b) $\frac{5}{2}RC$ (c) $\frac{7}{4}RC$ (d) $\frac{7}{3}RC$

55. In the RC circuit shown in the figure, the total energy of $3.6 \times 10^{-3} \text{ J}$ is dissipated in the 10Ω resistor after the switch S is closed for a long time. The initial charge on the capacitor wire

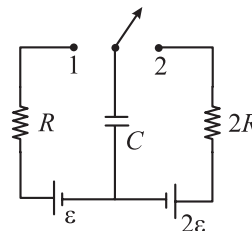


- (a) $60 \mu\text{C}$ (b) $120 \mu\text{C}$
(c) $60\sqrt{2} \mu\text{C}$ (d) $\frac{60}{\sqrt{2}} \mu\text{C}$

56. A capacitor of capacitance C begins to charge (at $t = 0$) by an ideal battery through a resistor of resistance R . The time after which potential difference across resistor becomes n times that across the capacitor is

- (a) $RC \ln\left(\frac{1+n}{n}\right)$ (b) $RC \ln\left(\frac{n}{1+n}\right)$
(c) $RC \ln\left(\frac{n}{n-1}\right)$ (d) $RC \ln(1+n)$

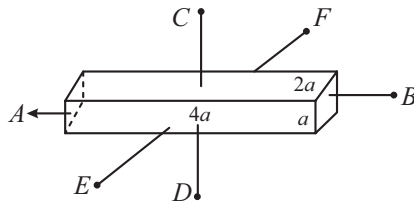
57. In the circuit shown, the switch was initially in position 1 for a long time. Then it is shifted from position 1 to 2 at $t = 0$. The graph between charge on capacitor C and time t is



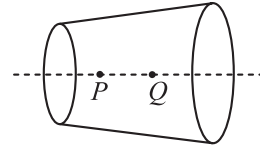
- (a) (b)
(c) (d)

Exercise -2

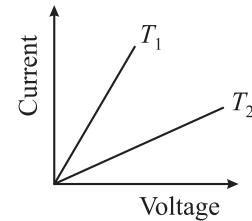
- In the presence of an applied electric field ($\vec{E}$) in a metallic conductor
 - The electrons move in the direction of $\vec{E}$
 - The electrons move in a direction opposite to $\vec{E}$
 - The electrons may move in any direction randomly, but slowly drift in the direction of $\vec{E}$
 - The electrons move randomly but slowly drift in a direction opposite to $\vec{E}$
- A current (I) flows through a uniform wire of diameter (d) when the mean drift velocity is v . The same current will flow through a wire of diameter $d/2$ made of the same material if the mean drift velocity of the electrons is
 - $v/4$
 - $v/2$
 - $4v$
 - $2v$
- A steady current is passing through a linear conductor of non-uniform cross-section. The current density in the conductor is
 - Independent of area of cross-section
 - Directly proportional to area of cross-section
 - Inversely proportional to area of cross-section
 - Inversely proportional to the square root of area of cross-section
- The resistance of a wire is $20\ \Omega$. The wire is stretched to three times its length. Then the resistance will now be
 - $6.67\ \Omega$
 - $60\ \Omega$
 - $120\ \Omega$
 - $180\ \Omega$
- A conductor with rectangular cross section has dimensions ($a \times 2a \times 4a$) as shown in figure. Resistance across AB is x , across CD is y and across EF is z . Then



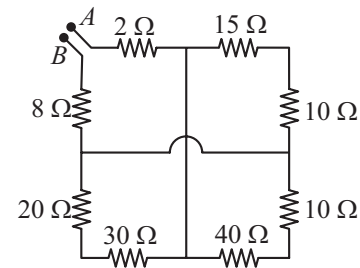
- $x = y = z$
 - $x > y > z$
 - $y > z > x$
 - $x > z > y$
- In a wire of cross-sectional radius r , free electrons travel with drift velocity v when a current I flows through the wire. What is the current in another wire of half the radius and of the same material when the drift velocity is $2v$?
 - $2I$
 - I
 - $I/2$
 - $I/4$
 - A wire has a non-uniform cross-section as shown in figure. A steady current flows through it. The drift speed of electrons at points P and Q is v_P and v_Q .



- $v_P = v_Q$
 - $v_P < v_Q$
 - $v_P > v_Q$
 - Data insufficient
- Two coils connected in series have resistances $600\ \Omega$ and $300\ \Omega$ at 20°C and temperature coefficient of resistivity $0.001\ \text{K}^{-1}$ and $0.004\ \text{K}^{-1}$ respectively. The resistance of the combination at temperature 50°C is
 - $426\ \Omega$
 - $954\ \Omega$
 - $1806\ \Omega$
 - $214\ \Omega$
 - Two coils connected in series have resistances $600\ \Omega$ and $300\ \Omega$ at 20°C and temperature coefficient of resistivity $0.001\ \text{K}^{-1}$ and $0.004\ \text{K}^{-1}$ respectively. The effective temperature coefficient of resistivity of the combination is
 - $\frac{1}{1000}\ \text{K}^{-1}$
 - $\frac{1}{250}\ \text{K}^{-1}$
 - $\frac{1}{500}\ \text{K}^{-1}$
 - $\frac{3}{1000}\ \text{K}^{-1}$
 - The current in a metallic conductor is plotted against voltage at two different temperatures T_1 and T_2 . Which of the following is correct?

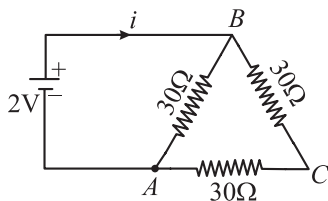


- $T_1 > T_2$
 - $T_1 < T_2$
 - $T_1 = T_2$
 - None of these
- The equivalent resistance between points A and B is



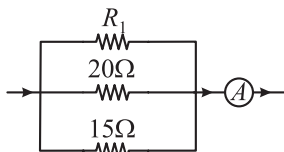
- $\frac{65}{2}\ \Omega$
- $\frac{45}{2}\ \Omega$
- $\frac{5}{2}\ \Omega$
- $\frac{91}{2}\ \Omega$

12. The current i in the circuit of figure is

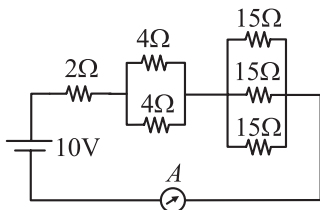


- (a) $\frac{1}{45}$ A (b) $\frac{1}{15}$ A (c) $\frac{1}{10}$ A (d) $\frac{1}{5}$ A

13. In the given circuit the current flowing through the resistance of 20 ohms is 0.3 ampere while the ammeter reads 0.8 ampere. What is the value of R_1 ?



- (a) 30 ohms (b) 40 ohms (c) 50 ohms (d) 60 ohms
14. The current through the ammeter shown in figure is 1 A. If each of the 4Ω resistor is replaced by 2Ω resistor, the current in circuit will become



- (a) $\frac{10}{9}$ A (b) $\frac{5}{4}$ A (c) $\frac{9}{8}$ A (d) $\frac{5}{8}$ A
15. Two non-ideal batteries are connected in parallel with like terminals joined together. Consider the following statements:
- (A) The equivalent emf is smaller than either of the two emfs.
- (B) The equivalent internal resistance is smaller than either of the two internal resistances.
- (a) Both (A) and (B) are correct
- (b) (A) is correct and (B) is wrong
- (c) (B) is correct but (A) is wrong
- (d) Both (A) and (B) are wrong

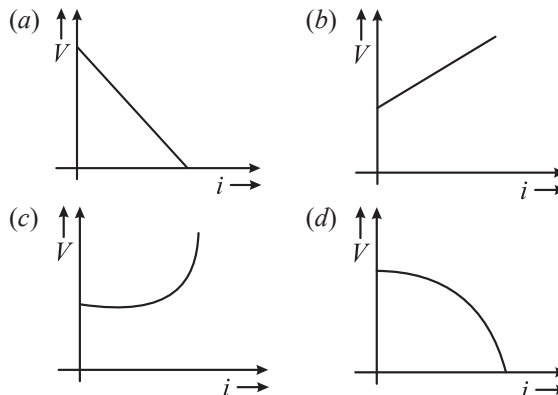
16. The same mass of copper is drawn into two wires A and B of radii r and $2r$ respectively. They are connected in series, and electric current is passed. The ratio of the heat produced in A and B is

- (a) 2 : 1 (b) 4 : 1 (c) 8 : 1 (d) 16 : 1

17. 24 identical cells, each of internal resistance 0.5Ω , are arranged in a parallel combination of n rows, each row containing m cells in series. The combination is connected across a resistor of 3Ω . In order to send maximum current through the resistor, we should have

- (a) $m = 12, n = 2$
- (b) $m = 8, n = 3$
- (c) $m = 2, n = 12$
- (d) $m = 3, n = 8$

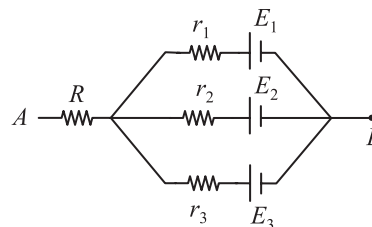
18. If internal resistance of a cell is proportional to current drawn from the cell. Then the best representation of terminal potential difference of a cell with current drawn from the cell will be



19. A storage battery is connected to a charger for charging with a voltage of 12.5 volts. The internal resistance of the storage battery is 1Ω . When the charging current is 0.5 A, the emf of the storage battery is

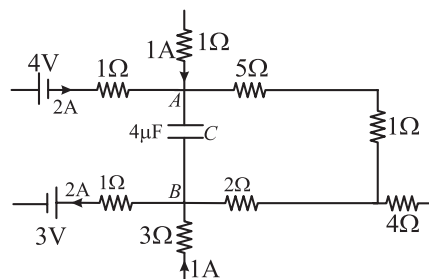
- (a) 13 Volts
- (b) 12.5 Volts
- (c) 12 Volts
- (d) 1.5 Volts

20. In the network shown the potential difference between A and B is ($R = r_1 = r_2 = r_3 = 1 \Omega$, $E_1 = 3 \text{ V}$, $E_2 = 2 \text{ V}$, $E_3 = 1 \text{ V}$)



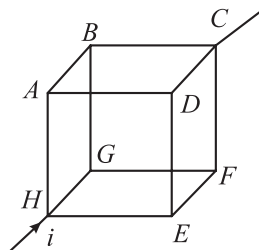
- (a) 1 V (b) 2 V (c) 3 V (d) 4 V

21. In following figure a part of circuit in steady state is shown. If energy stored in the capacitor is $x \times 10^{-4} \text{ J}$, then find value of x ?

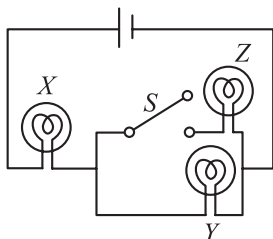


- (a) 5
- (b) 6
- (c) 7
- (d) 8

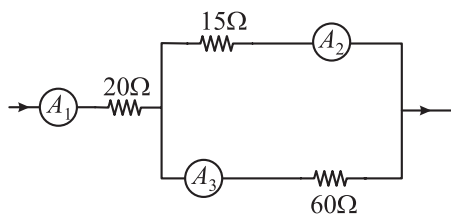
22. In the box shown current i enters at H and leaves at C . If $i_{AB} = \frac{i}{6}$, $i_{DC} = \frac{2i}{3}$, $i_{HA} = \frac{i}{2}$, $i_{GF} = \frac{i}{6}$, $i_{HE} = \frac{i}{6}$, choose the branch in which current is zero.



- (a) BG (b) FC
(c) ED (d) None of these
23. If X , Y and Z in figure are identical lamps, which of the following changes in the brightness of the lamps occur when switch S is closed?

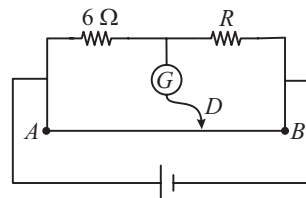


- (a) X stays the same, Y decreases
(b) X increases, Y decreases
(c) X increases, Y stays the same
(d) X decreases, Y increases
24. In a balanced wheatstone bridge, current in the galvanometer is zero. It remains zero when:
1. battery emf is increased
 2. all resistances are increased by 10 ohms
 3. all resistances are made five times
 4. the battery and the galvanometer are interchanged
- (a) Only 1 is correct (b) 1, 2 and 3 are correct
(c) 1, 3 and 4 are correct (d) 1 and 3 are correct
25. If the reading of ammeter A_3 in figure is 0.75 A. Neglecting the resistance of the ammeters, the reading of ammeter A_2 will be

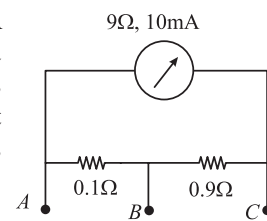


- (a) 1.5 A
(b) 3 A
(c) 4.5 A
(d) 6 A

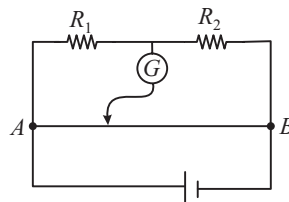
26. The meter-bridge wire AB shown in figure is 50 cm long. When $AD = 30$ cm, no deflection occurs in the galvanometer. Find R .



- (a) 1Ω (b) 2Ω
(c) 3Ω (d) 4Ω
27. If an ammeter is to be used in place of a voltmeter then we must connect with the ammeter
- (a) a low resistance in parallel
(b) a high resistance in parallel
(c) a high resistance in series
(d) a low resistance in series
28. A milliammeter of range 10 mA and resistance 9Ω is joined in a circuit as shown. The metre gives full-scale deflection for current I when A and B are used as its terminals, i.e., current enters at A and leaves at B (C is left isolated). The value of I is

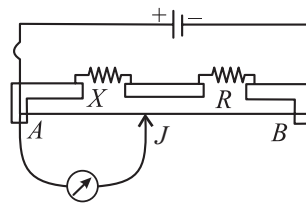


- (a) 100 mA (b) 900 mA
(c) 1 A (d) 1.1 A
29. In the figure shown for given values of R_1 and R_2 the balance point for jockey is at 40 cm from A . When R_2 is shunted by a resistance of 10Ω , balance shifts to 50 cm. R_1 and R_2 are ($AB = 1$ m)



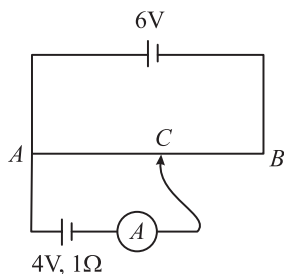
- (a) $\frac{10}{3} \Omega, 5 \Omega$ (b) $20 \Omega, 30 \Omega$
(c) $10 \Omega, 15 \Omega$ (d) $5 \Omega, \frac{15}{2} \Omega$

30. The figure shows a metre-bridge circuit, with $AB = 100$ cm, $X = 12 \Omega$ and $R = 18 \Omega$ and the jockey J in the position of balance. If R is now made 8Ω , through what distance will J have to be moved to obtain balance?

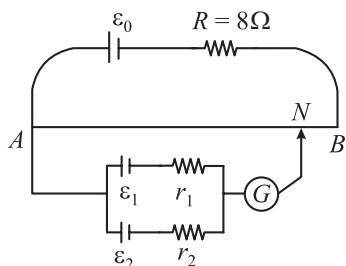


- (a) 10 cm (b) 20 cm (c) 30 cm (d) 40 cm

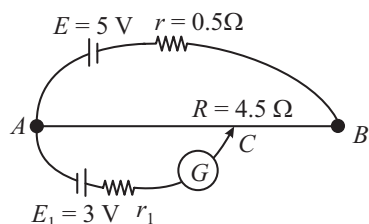
31. A 6V battery of negligible internal resistance is connected across a uniform wire of length 1m. The positive terminal of another battery of emf 4 V and internal resistance $1\ \Omega$ is joined to the point A as shown in figure. The ammeter shows zero deflection when the jockey touches the wire at the point C. Then AC is equal to



- (a) $\frac{2}{3}$ m (b) $\frac{1}{3}$ m (c) $\frac{3}{5}$ m (d) $\frac{1}{2}$ m
32. A potentiometer wire has length 10 m and resistance $10\ \Omega$. It is connected to a battery of emf 11 volt and internal resistance $1\ \Omega$, then the potential gradient of the wire is
- (a) 10 V/m (b) 1 V/m
(c) 0.1 V/m (d) None of these
33. The length of a potentiometer wire is l . A cell of emf E is balanced at a length $l/3$ from the positive end of the wire. If the length of the wire is increased by $l/2$. At what distance will the same cell give a balance point?
- (a) $\frac{2l}{3}$ (b) $\frac{l}{2}$ (c) $\frac{l}{6}$ (d) $\frac{l}{3}$
34. A battery of emf $\varepsilon_0 = 12\text{ V}$ is connected across a 4 m long uniform wire having resistance $4\ \Omega/\text{m}$. The cells of small emfs $\varepsilon_1 = 2\text{ V}$ and $\varepsilon_2 = 4\text{ V}$ having internal resistance $2\ \Omega$ and $6\ \Omega$ respectively, are connected as shown in the figure. If galvanometer shows no deflection at the point N, the distance of point N from the point A is equal to

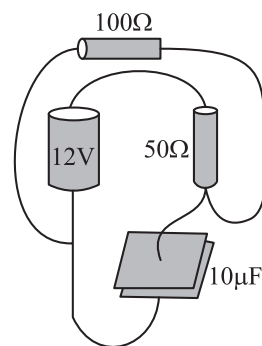


- (a) $\frac{1}{6}$ m (b) $\frac{1}{3}$ m (c) 125 cm (d) 150 cm
35. In the given potentiometer circuit length of the wire AB is 3 m and resistance is $R = 4.5\ \Omega$. The length AC for no deflection in galvanometer is



- (a) 2 m (b) 1.8 m
(c) dependent on r_1 (d) None of these

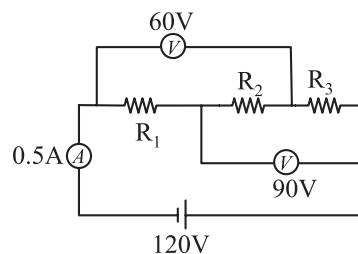
36. In the circuit shown in figure



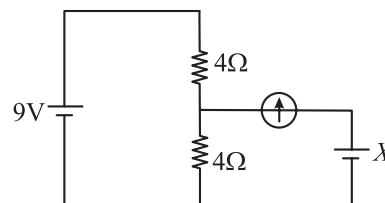
- (a) In steady state, there is no current in the $100\ \Omega$ resistor
(b) In steady state, the current in $100\ \Omega$ resistor is 0.08 A
(c) In steady state, there is no current in the $50\ \Omega$ resistor
(d) In steady state, the current in $50\ \Omega$ resistor is 0.04 A
37. An uncharged capacitor and a resistor of large resistance are connected in series to a battery of electromotive force 4.5 V. The voltage across the capacitor is 3 V one minute after closing the circuit. What will the voltage across the capacitor be 2 minutes after closing the circuit?
- (a) 6 V (b) 4.5 V
(c) 3.5 V (d) 4V

INTEGER TYPE QUESTIONS

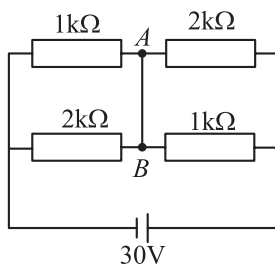
38. In the circuit shown, all voltmeters and the ammeter are ideal. The internal resistance of battery is $20\ \Omega$. The readings of voltmeters and ammeter are marked in the figure. What is the value of resistance R_2 (in ohm)?



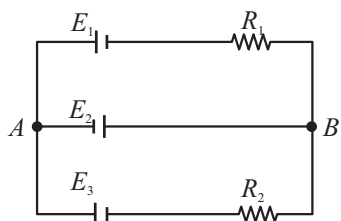
39. If each battery in figure has internal resistance of $1\ \Omega$, what should be the emf of X (in V) for there to be no deflection in the galvanometer?



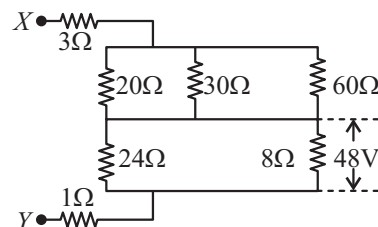
40. The current in the wire between points A and B is $I \times 10^{-4} \text{ A}$. Find I .



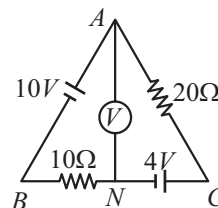
41. You are given several identical resistances each of value $R = 10 \, \Omega$ and each capable of carrying maximum current of 1 ampere. It is required to make a suitable combination of these resistances to produce a resistance of $5 \, \Omega$ which can carry a current of 4 amperes. The minimum number of resistances of the type R that will be required for this job are _____.
42. In the circuit shown here, $E_1 = E_2 = E_3 = 2 \text{ V}$ and $R_1 = R_2 = 4 \, \Omega$. What is the current (in A) flowing between points A and B through battery E_2 ?



43. The potential difference across $8 \, \Omega$ resistance is 48 volt as shown in the figure. Find the value of potential difference (in volts) across X and Y points.



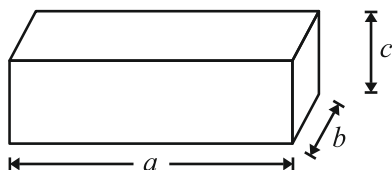
44. The reading of the ideal voltmeter in the adjoining diagram will be _____ V.



45. A wire of length L and 3 identical cells of negligible internal resistances are connected in series. Due to current, the temperature of the wire is raised by ΔT in a time t . A number N of similar cells is now connected in series with a wire of the same material and cross-section but of length $2L$. The temperature of the wire is raised by the same amount ΔT in the same time t . Find the value of N .

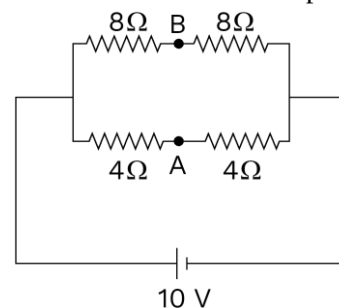
Exercise-3

1. All the edges of a block in cuboidal shape with parallel faces are unequal. Its longest edge is twice its shortest edge. The ratio of the maximum to minimum resistance between parallel faces is: ($a > b > c$)

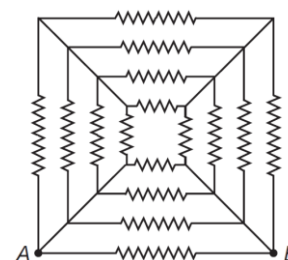


- (a) 2
(b) 4
(c) 8
(d) Indeterminate unless the length of the third edge is specified.
2. Two wires of same dimension but resistivities ρ_1 and ρ_2 are connected in series. The equivalent resistivity of the combination is-
- (a) $\rho_1 + \rho_2$
(b) $\left(\frac{\rho_1 + \rho_2}{2}\right)$
(c) $\sqrt{\rho_1 \rho_2}$
(d) $2(\rho_1 + \rho_2)$
3. 12 cells each having the same emf are connected in series and are kept in a closed box. Some of the cells are wrongly connected. This battery is connected in series with an ammeter and two cells identical with each other and identical with the previous cells. The current is 3 A when the external cells aid this battery and is 2 A when the cells oppose the battery. How many cells in the battery are wrongly connected?
- (a) One
(b) Two
(c) Three
(d) None
4. In a metre bridge experiment, null point is obtained at 20 cm from one end of the wire when resistance X is balanced against another resistance Y . If $X < Y$, then where will be the new position of the null point from the same end, if one decides to balance a resistance of $4X$ against Y ?
- (a) 50 cm (b) 80 cm
(c) 40 cm (d) 70 cm

5. The potential difference between points A and B is:

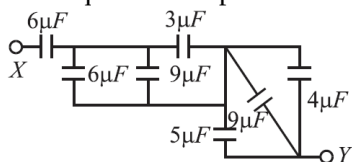


- (a) $\frac{20}{7}V$ (b) $\frac{40}{7}V$
(c) $\frac{10}{7}V$ (d) zero
6. In an ammeter 0.2% of main current passes through the galvanometer. If resistance of galvanometer is G , the resistance of ammeter will be:
- (a) $\frac{1}{499}G$ (b) $\frac{499}{500}G$
(c) $\frac{1}{500}G$ (d) $\frac{500}{499}G$
7. A battery consists of a variable number ' n ' of identical cells (having internal resistance ' r ' each) which are connected in series. The terminals of the battery are short-circuited and the current I is measured. Which of the graphs shows the correct relationship between I and n ?
- (a) (b) (c) (d)
8. Sixteen resistors each of resistance 16Ω are connected in the circuit as shown. The net resistance between AB is



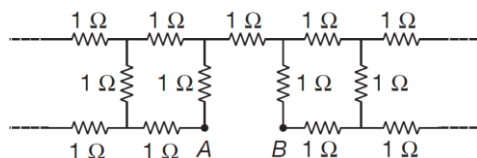
- (a) 1Ω (b) 2Ω
(c) 3Ω (d) 4Ω

9. Find equivalent capacitance between X and Y



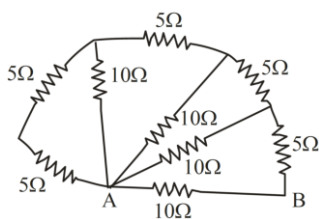
- (a) $3.6 \mu\text{F}$ (b) $6 \mu\text{F}$
(c) $1.5 \mu\text{F}$ (d) $9 \mu\text{F}$

10. Each resistor shown in figure is an infinite network of resistance 1Ω . The effective resistance between points A and B is



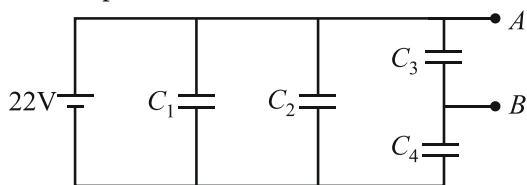
- (a) Less than 1Ω
(b) 1Ω
(c) More than 1Ω but less than 3Ω
(d) 3Ω

11. The equivalent resistance between points A and B in the given network is:



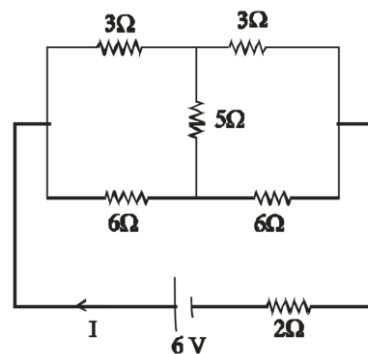
- (a) 65Ω
(b) 20Ω
(c) 5Ω
(d) 2Ω

12. In Fig., given $C_1 = 3 \mu\text{F}$, $C_2 = 5 \mu\text{F}$, $C_3 = 9 \mu\text{F}$ and $C_4 = 13 \mu\text{F}$. What is the potential difference between points A and B ?



- (a) 13 V
(b) 9 V
(c) 0 V
(d) 11 V

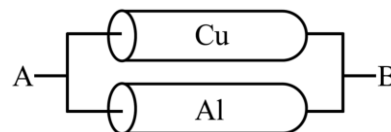
13. A battery of 6V is connected to the circuit as shown below. The current I drawn from the battery is:



- (a) 1A (b) 2A
(c) $\frac{6}{11}\text{A}$ (d) $\frac{4}{3}\text{A}$

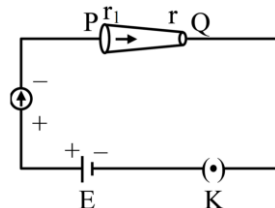
14. A Copper (Cu) rod of length 25 cm and cross-sectional area 3 mm^2 is joined with a similar Aluminium (Al) rod as shown in figure. Find the resistance of the combination between the ends A and B .

(Take Resistivity of Copper = $1.7 \times 10^{-8} \Omega\text{m}$
Resistivity of Aluminium = $2.6 \times 10^{-8} \Omega\text{m}$)



- (a) $2.170\text{ m}\Omega$
(b) $1.420\text{ m}\Omega$
(c) $0.0858\text{ m}\Omega$
(d) $0.858\text{ m}\Omega$

15. In the given figure, a battery of emf E is connected across a conductor PQ of length ' l ' and different area of cross-sections having radii r_1 and r_2 ($r_2 < r_1$).



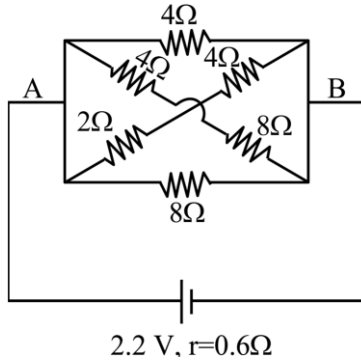
Choose the correct option as one moves from P to Q :

- (a) Drift velocity of electron increases.
(b) Electric field decreases.
(c) Electron current decreases.
(d) All of these

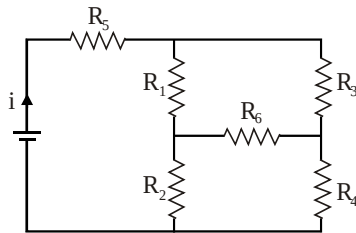
16. An electric bulb of 500 watt at 100 volt is used in a circuit having a 200 V supply. Calculate the resistance R to be connected in series with the bulb so that the power delivered by the bulb is 500 W .

- (a) 20Ω (b) 30Ω
(c) 5Ω (d) 10Ω

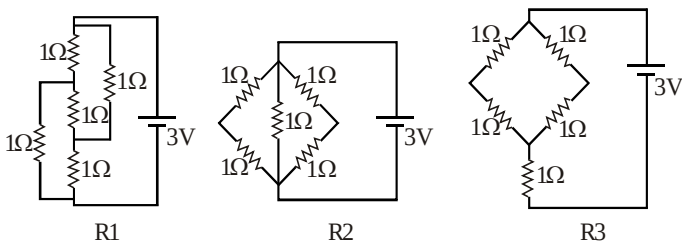
17. In the given figure, the emf of the cell is 2.2 V and if internal resistance is 0.6Ω . Calculate the power dissipated in the whole circuit :



- (a) 1.32 W
(b) 0.65 W
(c) 2.2 W
(d) 4.4 W
18. In the given circuit, it is observed that the current I is independent of the value of the resistance R_6 . Then, the resistance values must satisfy.

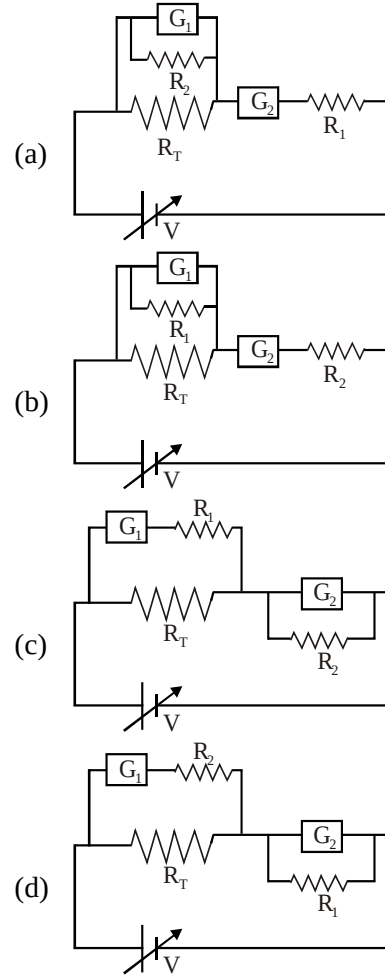


- (a) $R_1 R_2 R_5 = R_3 R_4 R_6$
(b) $\frac{1}{R_5} + \frac{1}{R_6} = \frac{1}{R_1 + R_2} + \frac{1}{R_3 + R_4}$
(c) $R_1 R_4 = R_2 R_3$
(d) $R_1 R_3 = R_2 R_4$
19. Figure shows three resistor configurations R_1 , R_2 and R_3 connected to 3V battery. If the power dissipated by the configuration R_1 , R_2 and R_3 is P_1 , P_2 and P_3 respectively, then

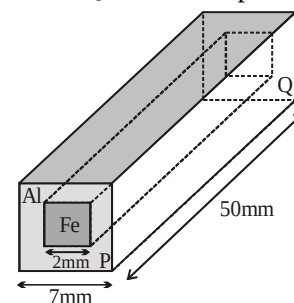


- (a) $P_1 > P_2 > P_3$
(b) $P_1 > P_3 > P_2$
(c) $P_2 > P_1 > P_3$
(d) $P_3 > P_2 > P_1$

20. To verify Ohm's law, a student is provided with a test resistor R_T , a high resistance R_1 , a small resistance R_2 , two identical galvanometers G_1 and G_2 , and a variable voltage source V . The correct circuit to carry out the experiment is:



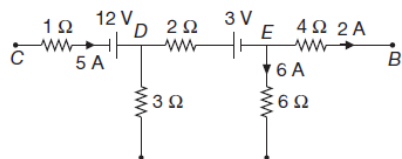
21. In an aluminum (Al) bar of square cross section, a square hole is drilled and is filled with iron (Fe) as shown in the figure. The electrical resistivities of Al and Fe are $2.7 \times 10^{-8} \Omega \text{ m}$ and $1.0 \times 10^{-7} \Omega \text{ m}$, respectively. The electrical resistance between the two faces P and Q of the composite bar is



- (a) $\frac{2475}{64} \mu\Omega$ (b) $\frac{1875}{64} \mu\Omega$
(c) $\frac{1875}{49} \mu\Omega$ (d) $\frac{2475}{132} \mu\Omega$

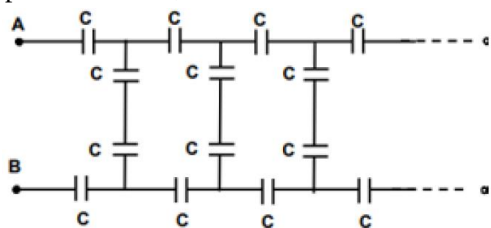
Integer Type Questions

22. Figure shows part of a circuit. Calculate the power, in watt, dissipated in 3Ω resistance. What is the potential difference $V_C - V_B$?

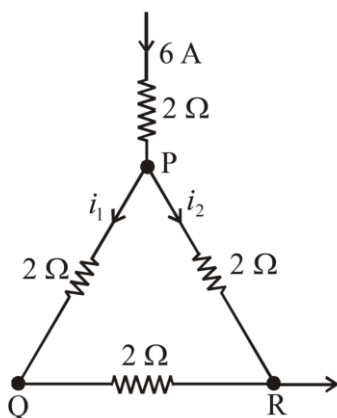


23. A parallel plate capacitor whose capacitance C is 14 pF is charged by a battery to a potential difference $V_0 = 12\text{V}$ between its plates. The charging battery is now disconnected and a porcelain plate with $k = 7$ is inserted between the plates, then the plate would oscillate back and forth between the plates with a constant mechanical energy of _____ pJ . (Assume no friction)

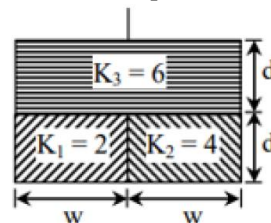
24. The equivalent capacitance of the shown infinite network across A and B , if each capacitor has a capacitance of C , is



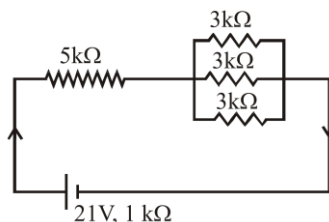
25. A current of 6 A enters one corner P of an equilateral triangle PQR having 3 wires of resistance 2Ω each and leaves by the corner R . The currents i_1 in ampere is _____.



26. A parallel plate capacitor of capacitance C (without dielectrics) is filled by dielectric slabs as shown in figure. Then the new capacitance of the capacitor is

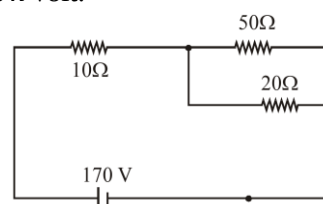


27. In the figure given, the electric current flowing through the $5\text{ k}\Omega$ resistor is ' x ' mA .



The value of x to the nearest integer is _____.

28. The voltage across the 10Ω resistor in the given circuit is x volt.



The value of ' x ' to the nearest integer is _____.

Exercise -4

Microscopic Analysis, Drift Velocity, Ohm's Law and Conductivity

1. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion (A): Alloys such as constantan and manganin are used in making standard resistance coils.

Reason (R): Constantan and manganin have very small value of temperature coefficient of resistance.

In the light of the above statements, choose the correct answer from the options given below.

[29 July, 2022 (Shift-II)]

- (a) Both (A) and (R) are true and (R) is the correct explanation of A
- (b) Both (A) and (R) are true but (R) is NOT the correct explanation of A
- (c) (A) is true but (R) is false
- (d) (A) is false but (R) is true

2. Resistance of the wire is measured as $2\ \Omega$ and $3\ \Omega$ at 10°C and 30°C respectively. Temperature co-efficient of resistance of the material of the wire is:

[28 June, 2022 (Shift-II)]

- (a) 0.033°C^{-1}
- (b) $-0.033^\circ\text{C}^{-1}$
- (c) 0.011°C^{-1}
- (d) 0.055°C^{-1}

3. A wire of resistance R_1 is drawn out so that its length is increased by twice of its original length. The ratio of new resistance to original resistance is

[28 July, 2022 (Shift-I)]

- (a) 9 : 1
- (b) 1 : 9
- (c) 4 : 1
- (d) 3 : 1

4. Two metallic wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of the these wires respectively, the effective conductivity of the combination is:

[29 July, 2022 (Shift-I)]

- (a) $\frac{\sigma_1\sigma_2}{\sigma_1 + \sigma_2}$
- (b) $\frac{2\sigma_1\sigma_2}{\sigma_1 + \sigma_2}$
- (c) $\frac{\sigma_1 + \sigma_2}{2\sigma_1\sigma_2}$
- (d) $\frac{\sigma_1 + \sigma_2}{\sigma_1\sigma_2}$

5. An aluminium wire is stretched to make its length, 0.4% larger. The percentage change in resistance is:

[26 June, 2022 (Shift-I)]

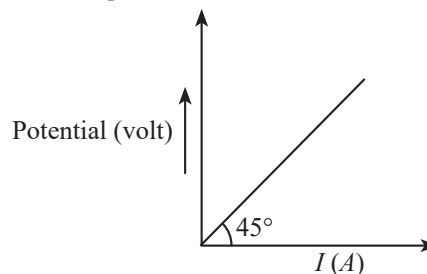
- (a) 0.4%
- (b) 0.2%
- (c) 0.8%
- (d) 0.6%

6. The variation of applied potential and current flowing through a given wire is shown in figure. The length of

wire is 31.4 cm. The diameter of wire is measured as 2.4 cm. The resistivity of the given wire is measured as $x \times 10^{-3}\ \Omega\text{cm}$. The value of x is _____.

[29 June, 2022 (Shift-I)]

[Take $\pi = 3.14$]



7. The current density in a cylindrical wire of radius 4 mm is $4 \times 10^6\ \text{Am}^{-2}$. The current through the outer portion of the wire between radial distance $\frac{R}{2}$ and R is _____ πA . [27

June, 2022 (Shift-I)]

8. The length of a given cylindrical wire is increased to double of its original length. The percentage increase in the resistance of the wire will be _____ %.

[25 June, 2022 (Shift-II)]

9. Eight copper wire of length l and diameter d are joined in parallel to form a single composite conductor of resistance R . If a single copper wire of length $2l$ have the same resistance (R) then its diameter will be _____ d .

[25 July, 2022 (Shift-I)]

10. The current density in a cylindrical wire of radius $r = 4.0\ \text{mm}$ is $1.0 \times 10^6\ \text{A/m}^2$. The current through the outer portion of the wire between radial distances $\frac{r}{2}$ and r is $x\pi\ \text{A}$; where x is _____

[27 June, 2022 (Shift-II)]

11. A 1 m long wire is broken into two unequal parts X and Y . The X part of the wire is stretched into another wire W . Length of W is twice the length of X and the resistance of W is twice that of Y . Find the ratio of length of X and Y .

[29 July, 2022 (Shift-II)]

- (a) 1 : 4
- (b) 1 : 2
- (c) 4 : 1
- (d) 2 : 1

12. A current of $10\ \text{A}$ exists in a wire of cross-sectional area of 5mm^2 with a drift velocity of $2 \times 10^{-3}\ \text{ms}^{-1}$. The number of free electrons in each cubic meter of the wire

[17 March, 2021 (Shift-I)]

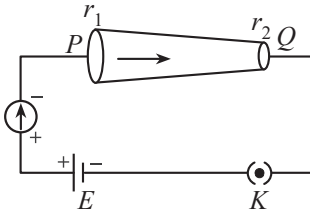
- (a) 2×10^{25}
- (b) 1×10^{23}
- (c) 2×10^6
- (d) 625×10^{25}

13. In the given figure, a battery of emf E is connected across a conductor PQ of length ' l ' and different area of cross-sections by having radii r_1 and r_2 ($r_2 < r_1$).

Choose the correct option as one moves from P to Q :

[27 July, 2021 (Shift-I)]

- (a) All of these.
 (b) Electron current decreases.
 (c) Drift velocity of electron increases.
 (d) Electric field decreases.



14. The resistance of a conductor at 15°C is 16Ω and at 100°C is 20Ω . What will be the temperature coefficient of resistance of the conductor ?

[27 July, 2021 (Shift-II)]

- (a) 0.003°C^{-1} (b) 0.042°C^{-1}
 (c) 0.033°C^{-1} (d) 0.010°C^{-1}

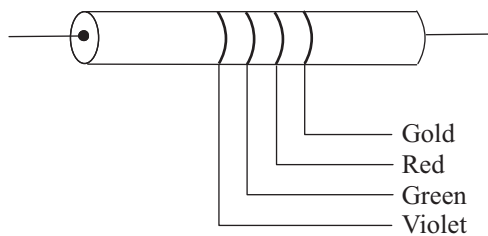
15. A current through a wire depends on time as $i = \alpha_0 t + \beta t^2$. Where $\alpha_0 = 20 \text{ A/s}$ and $\beta = 8 \text{ As}^{-2}$. Find the charge crossed through a section of the wire in 15 s.

[24 Feb, 2021 (Shift-I)]

- (a) 2100 C
 (b) 2250 C
 (c) 260 C
 (d) 11250 C

16. The colour coding on a carbon resistor is shown in the given figure. The resistance value of the given resistor is:

[27 Aug, 2021 (Shift-II)]



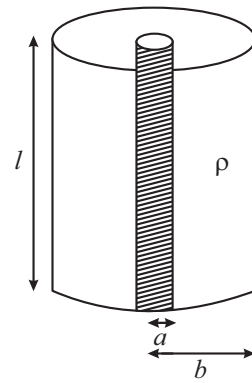
- (a) $(7500 \pm 375)\Omega$ (b) $(5700 \pm 285)\Omega$
 (c) $(7500 \pm 750)\Omega$ (d) $(5700 \pm 375)\Omega$

17. A cylindrical wire of radius 0.5 mm and conductivity $5 \times 10^7 \text{ S/m}$ is subjected to an electric field of 10 mV/m. The expected value of current in the wire will be $x^3 \pi \text{ mA}$. The value of x is _____.

[24 Feb, 2021 (Shift-II)]

18. Model a torch battery of length l to be made up of a thin cylindrical bar of radius ' a ' and a concentric thin cylindrical shell of radius ' b ' filled in between with an electrolyte of resistivity ρ (see figure). If the battery is connected to a resistance of value R , the maximum Joule heating in R will take place for

[3 Sep, 2020 (Shift-I)]



(a) $R = \frac{\rho}{\pi l} \ln\left(\frac{b}{a}\right)$ (b) $R = \frac{2\rho}{\pi l} \ln\left(\frac{b}{a}\right)$

(c) $R = \frac{\rho}{2\pi l} \ln\left(\frac{b}{a}\right)$ (d) $R = \frac{\rho}{2\pi l} \ln\left(\frac{a}{b}\right)$

19. Consider four conducting materials copper, tungsten, mercury and aluminium with resistivity ρ_C , ρ_T , ρ_M and ρ_A respectively. Then

[2 Sep, 2020 (Shift-I)]

- (a) $\rho_A > \rho_T > \rho_C$ (b) $\rho_M > \rho_A > \rho_C$
 (c) $\rho_C > \rho_A > \rho_T$ (d) $\rho_A > \rho_M > \rho_C$

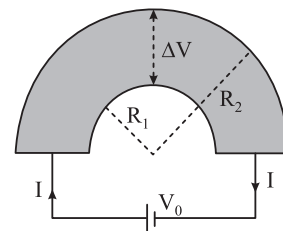
20. A circuit to verify Ohm's law uses ammeter and voltmeter in series or parallel connected correctly to the resistor. In the circuit:

[2 Sep, 2020 (Shift-I)]

- (a) ammeter is always used in parallel and voltmeter is series.
 (b) both ammeter and voltmeter must be connected in parallel.
 (c) ammeter is always connected in series and voltmeter in parallel.
 (d) both, ammeter and voltmeter must be connected in series.

21. Shown in the figure is a semicircular metallic strip that has thickness t and resistivity ρ . Its inner radius is R_1 and outer radius is R_2 . If a voltage V_0 is applied between its two ends, a current I flows in it. In addition, it is observed that a transverse voltage ΔV develops between its inner and outer surfaces due to purely kinetic effects of moving electrons (ignore any role of the magnetic field due to the current). Then (figure is schematic and not drawn to scale)

[JEE Adv, 2020]



(a) $I = \frac{V_0 t}{\pi \rho} \ln\left(\frac{R_2}{R_1}\right)$

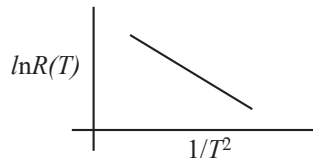
- (b) the outer surface is at a higher voltage than the inner surface
 (c) the outer surface is at a lower voltage than the inner surface
 (d) $\Delta V \propto I^2$

22. Drift speed of electrons, when 1.5 A of current flows in a copper wire of cross section 5 mm^2 , is v . If the electron density in copper is $9 \times 10^{28}/\text{m}^3$ the value of v in mm/s is close to (Take charge of electron to be $= 1.6 \times 10^{-19} \text{ C}$)

[9 Jan, 2019 (Shift-I)]

- (a) 0.02 (b) 3
(c) 2 (d) 0.2
23. In an experiment, the resistance of a material is plotted as a function of temperature (in some range). As shown in the figure, it is a straight line. One may conclude that:

[10 April, 2019 (Shift-I)]



- (a) $R(T) = R_0 e^{T^2/T_0^2}$ (b) $R(T) = \frac{R_0}{T^2}$
(c) $R(T) = R_0 e^{-T^2/T_0^2}$ (d) $R(T) = R_0 e^{-(T_0^2/T^2)}$
24. A carbon resistance has a following colour code. What is the value of the resistance? [9 Jan, 2019 (Shift-II)]



- (a) $530 \text{ k}\Omega \pm 5\%$ (b) $5.3 \text{ M}\Omega \pm 5\%$
(c) $6.4 \text{ M}\Omega \pm 5\%$ (d) $64 \text{ k}\Omega \pm 10\%$
25. Space between two concentric conducting spheres of radii a and b ($b > a$) is filled with a medium of resistivity ρ . The resistance between the two spheres will be

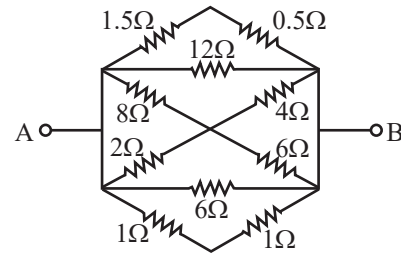
[10 April, 2019 (Shift-II)]

- (a) $\frac{\rho}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$ (b) $\frac{\rho}{2\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$
(c) $\frac{\rho}{4\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$ (d) $\frac{\rho}{4\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$
26. A current of 5 A passes through a copper conductor (resistivity $= 1.7 \times 10^{-8} \Omega\text{m}$) of radius of cross-section 5 mm. Find the mobility of the charges if their drift velocity is $1.1 \times 10^{-3} \text{ m/s}$. [10 April, 2019 (Shift-I)]

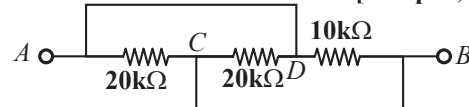
- (a) $1.8 \text{ m}^2/\text{Vs}$
(b) $1.0 \text{ m}^2/\text{Vs}$
(c) $1.3 \text{ m}^2/\text{Vs}$
(d) $1.5 \text{ m}^2/\text{Vs}$
27. A 200Ω resistor has a certain color code. If one replaces the red color by green in the code, the new resistance will be: [10 April, 2019 (Shift-I)]
- (a) 500Ω
(b) 400Ω
(c) 300Ω
(d) 100Ω

Series and Parallel Combination of Resistance

28. The equivalent resistance between A and B is _____. [30 Jan, 2023 (Shift-II)]



- (a) $\frac{2}{3} \Omega$ (b) $\frac{1}{2} \Omega$ (c) $\frac{3}{2} \Omega$ (d) $\frac{1}{3} \Omega$
29. The equivalent resistance between A and B as shown in figure is: [08 April, 2023 (Shift-II)]



- (a) $5 \text{ k}\Omega$
(b) $30 \text{ k}\Omega$
(c) $10 \text{ k}\Omega$
(d) $20 \text{ k}\Omega$
30. Equivalent resistance between the adjacent corners of a regular n -sided polygon of uniform wire of resistance R would be: [1 Feb, 2023 (Shift-II)]

- (a) $\frac{(n-1)R}{n^2}$
(b) $\frac{(n-1)R}{(2n-1)}$
(c) $\frac{n^2 R}{n-1}$
(d) $\frac{(n-1)R}{n}$

31. Given below are two statements:

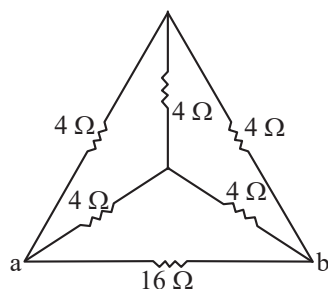
Statement-I: The equivalent resistance of resistors in a series combination is smaller than least resistance used in the combination.

Statement-II: The resistivity of the material is independent of temperature.

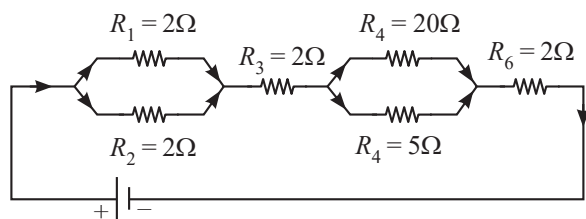
In the light of the above statements, choose the correct answer from the options given below: [15 April, 2023 (Shift-I)]

- (a) Statement-I is false but Statement-II is true
(b) Both Statement-I and Statement-II are false
(c) Statement-I is true but Statement-II is false
(d) Both Statement-I and Statement-II are true

32. The equivalent resistance of the circuit shown below between points a and b is: [10 April, 2023 (Shift-I)]

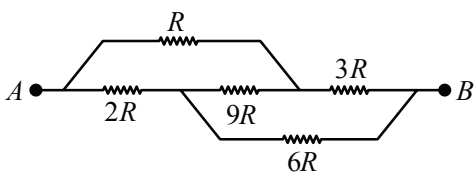


- (a) $24\ \Omega$ (b) $3.2\ \Omega$
(c) $20\ \Omega$ (d) $16\ \Omega$
33. As shown in the figure, a network of resistors is connected to a battery of 24 V with an internal resistance of $3\ \Omega$. The currents through the resistors R_4 and R_5 are I_4 and I_5 respectively. The values of I_4 and I_5 are: [24 Jan, 2023 (Shift-I)]

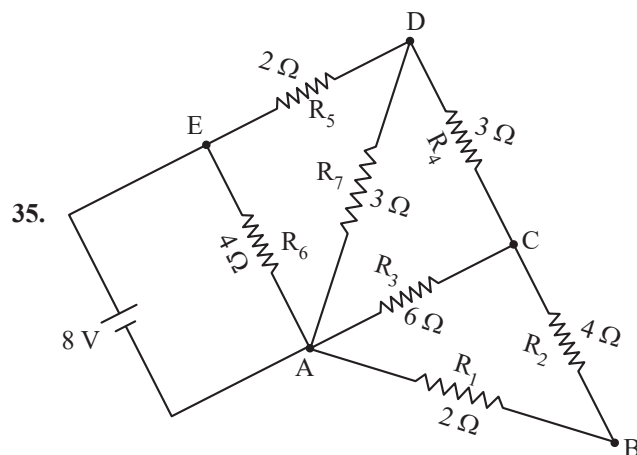


($24\text{V}, 3\ \Omega$)

- (a) $I_4 = \frac{8}{5}\text{ A}$ and $I_5 = \frac{2}{5}\text{ A}$
(b) $I_4 = \frac{24}{5}\text{ A}$ and $I = -\text{A}$
(c) $I_4 = \frac{6}{5}\text{ A}$ and $I_5 = \frac{24}{5}\text{ A}$
(d) $I_4 = \frac{2}{5}\text{ A}$ and $I_5 = \frac{8}{5}\text{ A}$
34. The equivalent resistance between A and B of the network shown in figure: [01 Feb, 2023 (Shift-I)]



- (a) $11\frac{2R}{3}$
(b) $14\ R$
(c) $21\ R$
(d) $\frac{8}{3}\ R$

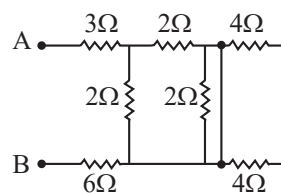


The current flowing through R_2 is:

[11 April, 2023 (Shift-II)]

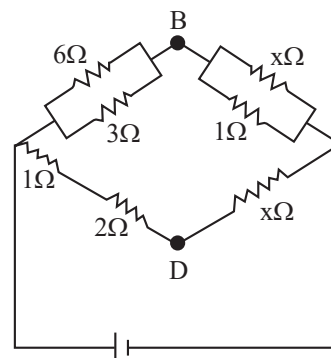
- (a) $\frac{2}{3}\text{ A}$
(b) $\frac{1}{4}\text{ A}$
(c) $\frac{1}{2}\text{ A}$
(d) $\frac{1}{3}\text{ A}$

36. In the given circuit, the equivalent resistance between the terminal A and B is _____ Ω . [25 Jan, 2023 (Shift-I)]



37. If the potential difference between B and D is zero, the value of x is $\frac{1}{n}\ \Omega$. The value of n is

[30 Jan, 2023 (Shift-II)]



38. 10 resistors each of resistance 10Ω can be connected in such as to get maximum and minimum equivalent resistance. The ratio of maximum and minimum equivalent resistance will be _____. [10 April, 2023 (Shift-I)]

39. What will be the most suitable combination of three resistors $A = 2\Omega$, $B = 4\Omega$, $C = 6\Omega$ so that $\left(\frac{22}{3}\right)$ is equivalent resistance of combination?

[24 June, 2022 (Shift-II)]

- (a) Parallel combination of A and C connected in series with B.
 (b) Parallel combination of A and B connected in series with C.
 (c) Series combination of A and C connected in parallel with B.
 (d) Series combination of B and C connected in parallel with A.

40. The current I in the given circuit will be:

[26 July, 2022 (Shift-I)]

- (a) 10 A
 (b) 20 A
 (c) 4 A
 (d) 40 A

41. Given below are two statements:

Statement-I: A uniform wire of resistance 80Ω is cut into four equal parts. These parts are now connected in parallel. The equivalent resistance of the combination will be 5Ω .

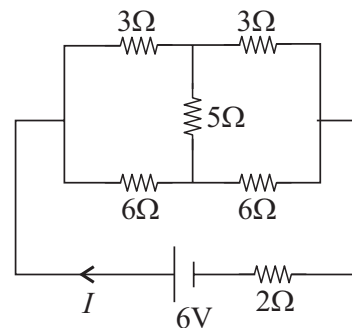
Statement-II: Two resistances $2R$ and $3R$ are connected in parallel in a electric circuit. The value of thermal energy developed in $3R$ and $2R$ will be in the ratio $3 : 2$.

In the light of the above statements, choose the most appropriate answer from the option given below

[28 July, 2022 (Shift-II)]

- (a) Both statement-I and statement-II are correct
 (b) Both statement-I and statement-II are incorrect
 (c) Statement-I is correct but statement-II is incorrect
 (d) Statement-I is incorrect but statement-II is correct
42. A battery of $6V$ is connected to the circuit as shown below. The current I drawn from the battery is:

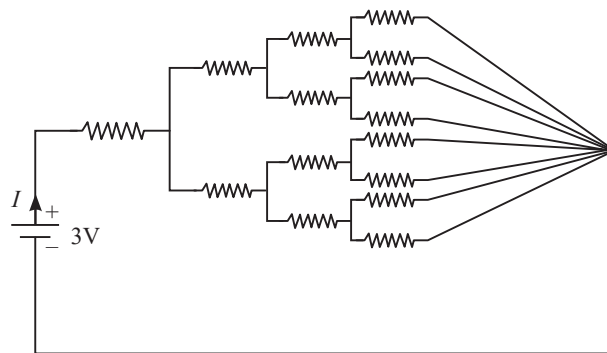
[26 July, 2022 (Shift-II)]



- (a) 1 A
 (b) 2 A
 (c) $\frac{6}{11}A$
 (d) $\frac{4}{3}A$

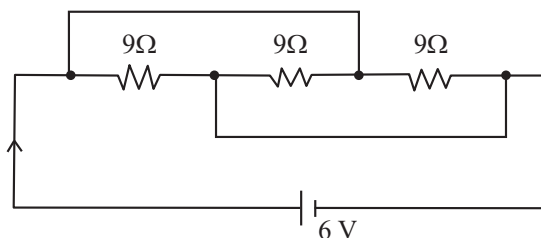
43. All resistances in figure are 1Ω each. The value of current 'I' is $\frac{a}{5}A$. The value of a is _____.

[28 June, 2022 (Shift-II)]



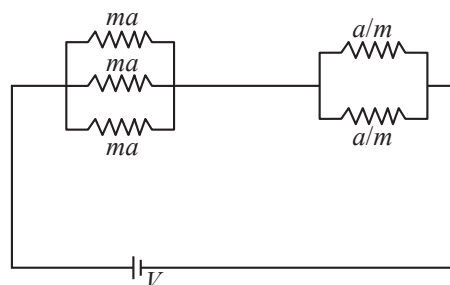
44. The current I flowing through the given circuit will be _____ A.

[29 July, 2022 (Shift-I)]



45. In the given circuit 'a' is an arbitrary constant. The value of m for which the equivalent circuit resistance is minimum, will be $\sqrt{\frac{x}{2}}$. The value of x is _____

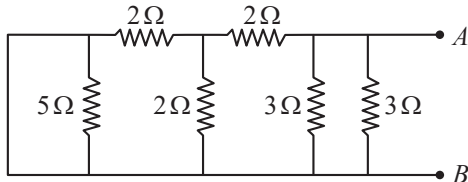
[27 June, 2022 (Shift-II)]



46. If you are provided a set of resistances 2Ω , 4Ω , 6Ω and 8Ω . Connect these resistances so as to obtain an equivalent resistance of $\frac{46}{3}\Omega$. [26 Aug, 2021 (Shift-II)]

- (a) 4Ω and 6Ω are in parallel with 2Ω and 8Ω in series.
 (b) 6Ω and 8Ω are in parallel with 2Ω and 4Ω in series.
 (c) 2Ω and 4Ω are in parallel with 6Ω and 8Ω in series.
 (d) 2Ω and 6Ω are in parallel with 4Ω and 8Ω in series.

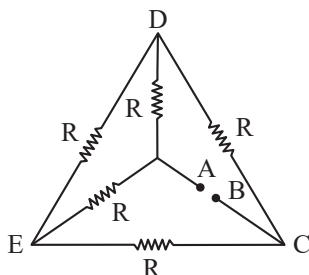
47. The equivalent resistance of the given circuit between the terminals A and B is: [31 Aug, 2021 (Shift-II)]



- (a) 0Ω (b) 1Ω
 (c) $9/2\Omega$ (d) 3Ω

48. Five equal resistances are connected in a network as shown in figure. The net resistance between the points A and B is: [26 Feb, 2021 (Shift-I)]

- (a) $\frac{3R}{2}$ (b) R (c) $\frac{R}{2}$ (d) $2R$

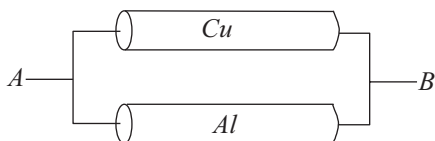


49. A Copper (Cu) rod of length 25cm and cross-sectional area 3mm^2 is joined with a similar aluminium (Al) rod as shown in figure. Find the resistance of the combination between the ends A and B .

[22 July, 2021 (Shift-II)]

(Take Resistivity of Copper = $1.7 \times 10^{-8}\Omega\text{m}$,

Resistivity of Aluminium = $2.6 \times 10^{-8}\Omega\text{m}$)



- (a) $0.858\text{m}\Omega$ (b) $0.0858\text{m}\Omega$
 (c) $1.420\text{m}\Omega$ (d) $2.170\text{m}\Omega$

50. What equal length of an iron wire and a copper-nickel alloy wire, each of 2mm diameter connected parallel to give an equivalent resistance of 3Ω ?

[26 Aug, 2021 (Shift-I)]

(Given resistivities of iron and copper-nickel alloy wire

are $12\mu\Omega\text{cm}$ and $51\mu\Omega\text{cm}$ respectively)

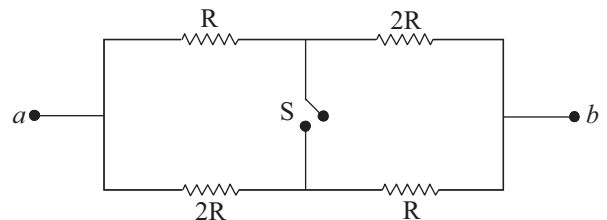
- (a) 110m (b) 90m
 (c) 82m (d) 97m

51. Two resistors $R_1 = (4 \pm 0.8)\Omega$ and $R_2 = (4 \pm 0.4)\Omega$ are connected in parallel. The equivalent resistance of their parallel combination will be: [1 Sep, 2021 (Shift-II)]

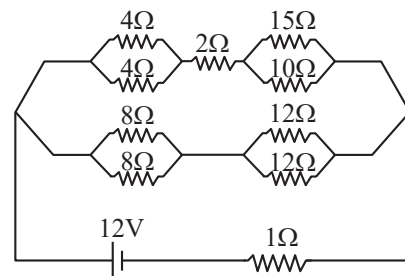
- (a) $(4 \pm 0.4)\Omega$
 (b) $(2 \pm 0.3)\Omega$
 (c) $(4 \pm 0.3)\Omega$
 (d) $(2 \pm 0.4)\Omega$

52. The ratio of the equivalent resistance of the network (shown in figure) between the points a and b when switch is open and switch is closed is $x : 8$. The value of x is _____.

[27 Aug, 2021 (Shift-II)]

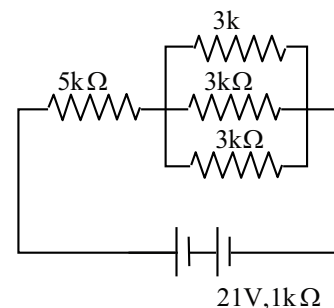


53. The voltage drop across 15Ω resistance in the given figure will be _____ V. [31 Aug, 2021 (Shift-I)]



54. A square shaped wire with resistance of each side 3Ω is bent to form a complete circle. The resistance between two diametrically opposite points of the circle in unit of Ω will be _____. [31 Aug, 2021 (Shift-I)]

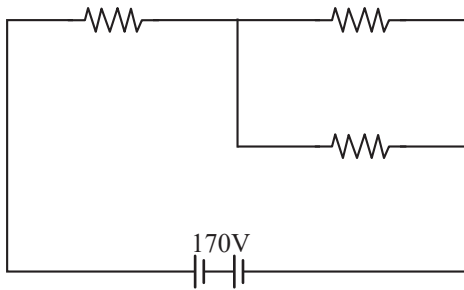
55. In the figure given, the electric current flowing through the $5\text{k}\Omega$ resistor is ' x ' mA. The value of x to the nearest integer is _____. [16 March, 2021 (Shift-I)]



56. The voltage across the 10Ω resistor in the given circuit is x volt.

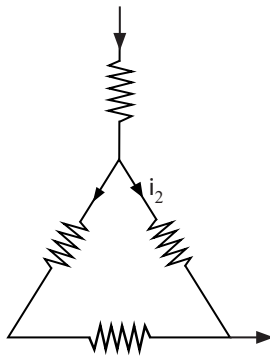
The value of 'x' to the nearest integer is _____.

[18 March, 2021 (Shift-I)]



57. A current of 6 A enters one corner P of an equilateral triangle PQR having 3 wires of resistance 2Ω each and leaves by the corner R . The currents i_1 in ampere is _____

[25 Feb, 2021 (Shift-II)]



58. First, a set of n equal resistors of 10Ω each are connected in series to a battery of emf 20V and internal resistance 10Ω . A current I is observed to flow. Then, the n resistors are connected in parallel to the same battery. It is observed that the current is increased 20 times, then the value of n is _____.

[27 Aug, 2021 (Shift-I)]

59. Two wires of same length and thickness having specific resistances $6\Omega\text{ cm}$ and $3\Omega\text{ cm}$ respectively are connected in parallel. The effective resistivity is $\rho\Omega\text{ cm}$. The value of ρ to the nearest integer, is _____.

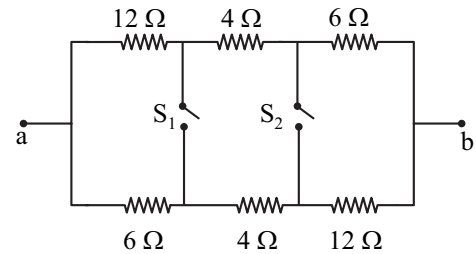
[18 March, 2021 (Shift-II)]

60. The equivalent resistance of series combination of two resistors is ' s '. When they are connected in parallel, the equivalent resistance is ' p '. If $s = np$, then the minimum value for n is _____ (Round off to the nearest integer)

[17 March, 2021 (Shift-I)]

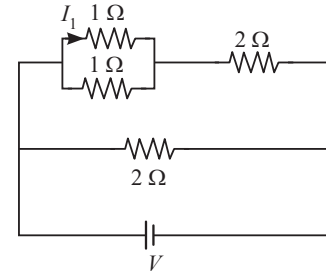
61. In the given figure switches S_1 and S_2 are in open condition. The resistance across ab when the switches S_1 and S_2 are closed is _____ Ω .

[20 July, 2021 (Shift-II)]



62. The current I_1 (in A) flowing through 1Ω resistor in the following circuit is _____

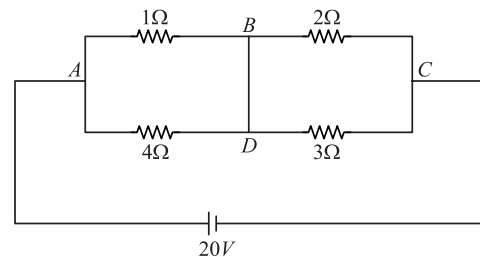
[7 Jan, 2020 (Shift-I)]



- (a) 0.25 (b) 0.4 (c) 0.2 (d) 0.5

63. In the given circuit diagram, a wire is joining points B and D . The current in this wire is _____

[9 Jan, 2020 (Shift-I)]



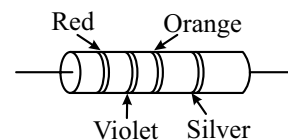
- (a) Zero (b) 2 A (c) 0.4 A (d) 4 A

64. Two resistors 400Ω and 800Ω are connected in series across a 6 V battery. The potential difference measured by a voltmeter of $10\text{ k}\Omega$ across 400Ω resistor is close to: [3 Sep, 2020 (Shift-II)]

- (a) 2 V (b) 2.05 V
(c) 1.95 V (d) 1.8 V

65. Four resistances of 15Ω , 12Ω , 14Ω and 10Ω respectively in cyclic order to form Whetstone's network. The resistance that is to be connected in parallel with the resistance of 10Ω to balance the network is _____.

66. A resistance is shown in the figure. Its value and tolerance are given respectively by: [9 Jan, 2019 (Shift-I)]



- (a) 270Ω , 10%
(b) $27\text{ k}\Omega$, 10%
(c) $27\text{ k}\Omega$, 20%
(d) 270Ω , 5%

67. A metal wire of resistance $3\ \Omega$ is elongated to make a uniform wire of double its previous length. The new wire is now bent and the ends joined to make a circle. If two points on this circle make an angle 60° at the centre, the equivalent resistance between these two points will be

[9 April, 2019 (Shift-II)]

- (a) $\frac{12}{5}\ \Omega$ (b) $\frac{5}{3}\ \Omega$
(c) $\frac{5}{2}\ \Omega$ (d) $\frac{7}{2}\ \Omega$

68. A uniform metallic wire has a resistance of $18\ \Omega$ and is bent into an equilateral triangle. Then, the resistance between any two vertices of the triangle is:

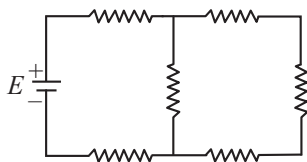
[10 Jan, 2019 (Shift-I)]

- (a) $4\ \Omega$ (b) $8\ \Omega$
(c) $12\ \Omega$ (d) $2\ \Omega$

69. In the figure shown, what is the current (in Ampere) drawn from the battery? You are given

$R_1 = 15\ \Omega$, $R_2 = 10\ \Omega$, $R_3 = 20\ \Omega$, $R_4 = 5\ \Omega$, $R_5 = 25\ \Omega$, $R_6 = 30\ \Omega$, $E = 15\text{V}$

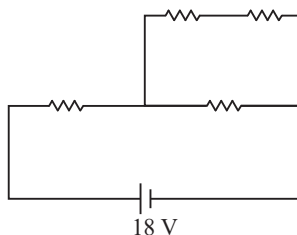
[8 April, 2019 (Shift-II)]



- (a) $13/24$ (b) $7/18$ (c) $9/32$ (d) $20/3$

70. In the given circuit the internal resistance of the 18V cell is negligible. If $R_1 = 400\ \Omega$, $R_3 = 100\ \Omega$ and $R_4 = 500\ \Omega$ and the reading of an ideal voltmeter across R_4 is 5V , then the value of R_2 will be

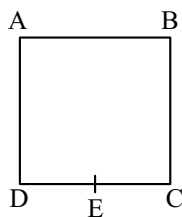
[9 Jan, 2019 (Shift-II)]



- (a) $300\ \Omega$ (b) $450\ \Omega$ (c) $550\ \Omega$ (d) $230\ \Omega$

71. A wire of resistance R is bent to form a square ABCD as shown in the figure. The effective resistance between E and C is: (E is mid-point of arm CD)

[9 April, 2019 (Shift-I)]



- (a) $\frac{1}{16}R$ (b) $\frac{7}{64}R$ (c) $\frac{3}{4}R$ (d) R

Power and Heat Dissipated in Electric Circuits

72. The H amount of thermal energy is developed by a resistor in 10s when a current of 4A is passed through it. If the current is increased to 16A , the thermal energy developed by the resistor in 10s will be:

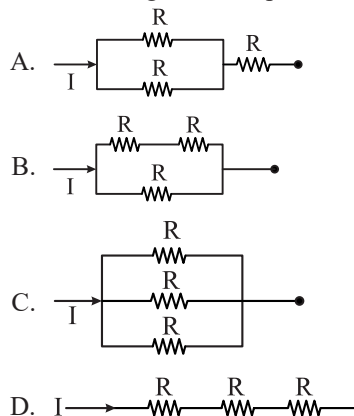
[31 Jan, 2023 (Shift-II)]

- (a) H (b) $16H$ (c) $\frac{H}{4}$ (d) $4H$

73. Different combination of 3 resistors of equal resistance R are shown in the figures.

[13 April, 2023 (Shift-I)]

The increasing order for power dissipation is:



- (a) $P_A < P_B < P_C < P_D$
(b) $P_C < P_D < P_A < P_B$
(c) $P_B < P_C < P_D < P_A$
(d) $P_C < P_B < P_A < P_D$

74. Ratio of thermal energy released in two resistors R and $3R$ connected in parallel in an electric circuit is :

[29 Jan, 2023 (Shift-I)]

- (a) $3 : 1$
(b) $1 : 1$
(c) $1 : 3$
(d) $1 : 27$

75. Two identical heater filaments are connected first in parallel and then in series. At the same applied voltage, the ratio of heat produced in same time for parallel to series will be:

[11 April, 2023 (Shift-I)]

- (a) $4 : 1$ (b) $2 : 1$ (c) $1 : 2$ (d) $1 : 4$

76. A potential V_0 is applied across a uniform wire of resistance R . The power dissipation is P_1 . The wire is then cut into two equal halves and a potential of V_0 is applied across the length of each half. The total power dissipation across two wires is P_2 . The ratio $P_2 : P_1$ is $\sqrt{x} : 1$. The value of x is _____.

[13 April, 2023 (Shift-I)]

77. An electrical bulb rated 220 V, 100 W, is connected in series with another bulb rated 220 V, 60 W. If the voltage across combination is 220 V, the power consumed by the 100 W bulb will be about _____ W.
[28 July, 2022 (Shift-II)]

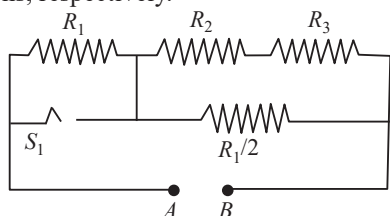
78. A resistor develops 300 J of thermal energy in 15 s, when a current of 2 A is passed through it. If the current increases to 3 A, the energy developed in 10 s is _____ J

[25 June, 2022 (Shift-I)]

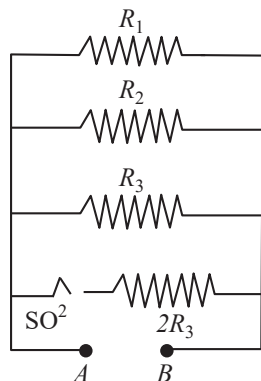
79. In Circuit-1 and Circuit-2 shown in the figures, $R_1 = 1\ \Omega$, $R_2 = 2\ \Omega$ and $R_3 = 3\ \Omega$.

P_1 and P_2 are the power dissipations in Circuit-1 and Circuit-2 when the switches S_1 and S_2 are in open conditions, respectively.

Q_1 and Q_2 are the power dissipations in Circuit-1 and Circuit-2 when the switches S_1 and S_2 are in closed conditions, respectively.



Circuit-1



Circuit-2

Which of the following statements is(are) correct?

[JEE Adv, 2022]

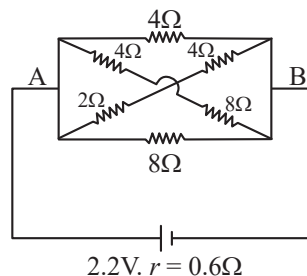
- (a) When a voltage source of 6 V is connected across A and B in both circuits, $P_1 < P_2$
 (b) When a constant current source of 2 A is connected across A and B in both circuits, $P_1 > P_2$
 (c) When a voltage source of 6 V is connected across A and B in Circuit-1, $Q_1 > P_1$
 (d) When a constant current source of 2 A is connected across A and B in both circuits, $Q_2 < Q_1$
80. A refrigerator consumes an average 35 W power to operate between temperature -10°C to 25°C . If there is no loss of energy then how much average heat per second does it transfer?
[26 Aug, 2021 (Shift-II)]
- (a) 350 J/s (b) 298 J/s (c) 35 J/s (d) 263 J/s

81. An electric bulb of 500 watt at 100 volt is used in a circuit having a 200 V supply. Calculate the resistance R to be connected in series with the bulb so that the power delivered by the bulb is 500 W.
[26 Aug, 2021 (Shift-II)]

(a) $20\ \Omega$ (b) $5\ \Omega$ (c) $10\ \Omega$ (d) $30\ \Omega$

82. In the given figure, the emf of the cell is 2.2 V and if internal resistance is $0.6\ \Omega$. Calculate the power dissipated in the whole circuit :
[26 Aug, 2021 (Shift-I)]

(a) 2.2 W (b) 0.65 W (c) 4.4 W (d) 132 W



83. An electric appliance supplies 6000 J/min heat to the system. If the system delivers a power of 90 W. How long it would take to increase the internal energy by $2.5 \times 10^3\ \text{J}$?
[26 Aug, 2021 (Shift-I)]

(a) $4.1 \times 10^1\ \text{s}$ (b) $2.4 \times 10^3\ \text{s}$
 (c) $2.5 \times 10^1\ \text{s}$ (d) $2.5 \times 10^2\ \text{s}$

84. A resistor develops 500J of thermal energy in 20 s when a current of 1.5A is passed through it. If the current is increased from 1.5 A to 3 A, what will be the energy developed in 20 s.
[16 March, 2021 (Shift-II)]

(a) 2000 J (b) 500 J
 (c) 1000 J (d) 1500 J

85. A resistor dissipates 192 J of energy in 1 s when a current of 4A is passed through it. Now, when the current is doubled, the amount of thermal energy dissipated in 5 s in _____ J.
[31 Aug, 2021 (Shift-II)]

86. A uniform heating wire of resistance $36\ \Omega$ is connected across a potential difference of 240 V. The wire is then cut into half and potential difference of 240 V is applied across each half separately. The ratio of power dissipation in first case to the total power dissipation in the second case would be 1 : x, where x is _____.

[1 Sep, 2021 (Shift-II)]

87. The energy dissipated by a resistor is 10 mJ in 1 s when an electric current of 2 mA flows through it. The resistance is _____ Ω . (Round off to the Nearest Integer)

[16 March, 2021 (Shift-II)]

88. An electric bulb rated as 200 W at 100 V is used in a circuit having 200 V supply. The resistance 'R' that must be put in series with the bulb so that the bulb delivers the same power is _____ Ω .
[25 July, 2021 (Shift-I)]

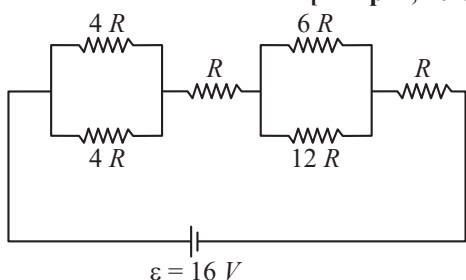
89. In a building there are 15 bulbs of 45 W, 15 bulbs of 100 W, 15 small fans of 10 W and 2 heaters of 1 kW. The voltage of electric main is 220 V. The minimum fuse capacity (rated value) of the building will be

[7 Jan, 2020 (Shift-II)]

- (a) 10 A (b) 20 A
(c) 25 A (d) 15 A
90. A battery of 3.0 V is connected to a resistor dissipating 0.5 W of power. If the terminal voltage of the battery is 2.5 V, the power dissipated within the internal resistance is [4 Sep, 2020 (Shift-I)]

- (a) 0.10 W (b) 0.072 W
(c) 0.50 W (d) 0.125 W

91. The resistive network shown below is connected to a D.C. source of 16 V. The power consumed by the network is 4 Watt. The value of R is: [12 April, 2019 (Shift-I)]



- (a) 8 Ω (b) 6 Ω (c) 16 Ω (d) 1 Ω
92. Two electric bulbs, rated at (25 W, 220 V) and (100 W, 220 V), are connected in series across a 220 V voltage source. If the 25 W and 100 W bulbs draw powers P_1 and P_2 respectively, then: [12 Jan, 2019 (Shift-I)]

- (a) $P_1 = 16 \text{ W}, P_2 = 4 \text{ W}$ (b) $P_1 = 16 \text{ W}, P_2 = 9 \text{ W}$
(c) $P_1 = 9 \text{ W}, P_2 = 16 \text{ W}$ (d) $P_1 = 4 \text{ W}, P_2 = 16 \text{ W}$

93. A cell of internal resistance r drives current through an external resistance R . The power delivered by the cell to the external resistance will be maximum when:

[10 Jan, 2019 (Shift-I)]

- (a) $R = 0.001 r$ (b) $R = 1000 r$
(c) $R = 2r$ (d) $R = r$

94. A current of 2 mA was passed through an unknown resistor which dissipated a power of 4.4 W. Dissipated power when an ideal power supply of 11 V is connected across it is :

[10 Jan, 2019 (Shift-II)]

- (a) $11 \times 10^{-5} \text{ W}$ (b) $11 \times 10^{-3} \text{ W}$
(c) $11 \times 10^{-4} \text{ W}$ (d) $11 \times 10^5 \text{ W}$

95. Two equal resistances when connected in series to a battery, consume electric power of 60 W. If these resistances are now connected in parallel combination to the same battery, the electric power consumed will be.

[11 Jan, 2019 (Shift-I)]

- (a) 60 W (b) 240 W (c) 120 W (d) 30 W

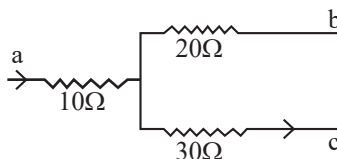
96. A 2 W carbon resistor is color coded with green, black, red and brown respectively. The maximum current which can be passed through this resistor is: [10 Jan, 2019 (Shift-I)]

- (a) 20 mA (b) 100 mA
(c) 0.4 mA (d) 63 mA

Kirchhoff's Law and Equivalent E.M.F.

97. Figure shows a part of an electric circuit. The potentials at points a , b and c are 30 V, 12 V and 2 V respectively. The current through the 20Ω resistor will be.

[6 April, 2023 (Shift-II)]



- (a) 0.4 A (b) 0.2 A (c) 0.6 A (d) 1.0 A

98. Two identical cells each of emf 1.5 V are connected in parallel across a parallel combination of two resistors each of resistance 20Ω . A voltmeter connected in the circuit measures 1.2 V. The internal resistance of each cell is:

[24 June, 2022 (Shift-I)]

- (a) 2.5 Ω (b) 4 Ω (c) 5 Ω (d) 10 Ω

99. Two cells of same emf but different internal resistance r_1 and r_2 are connected in series with a resistance R . The value of resistance R , for which the potential difference across second cell is zero, is:

[25 June, 2022 (Shift-II)]

- (a) $r_2 - r_1$ (b) $r_1 - r_2$ (c) r_1 (d) r_2

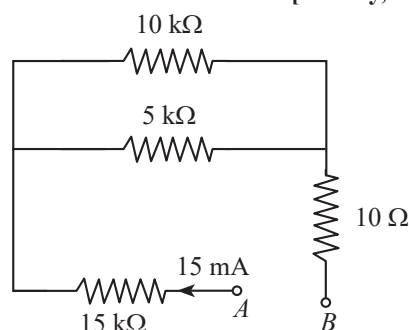
100. The combination of two identical cells, whether connected in series or parallel combination provides the same current through an external resistance of 2Ω . The value of internal resistance of each cell is:

[29 June, 2022 (Shift-II)]

- (a) 2 Ω
(b) 4 Ω
(c) 6 Ω
(d) 8 Ω

101. A current of 15 mA flows in the circuit as shown in figure. The value of potential difference between the points A and B will be:

[25 July, 2022 (Shift-II)]



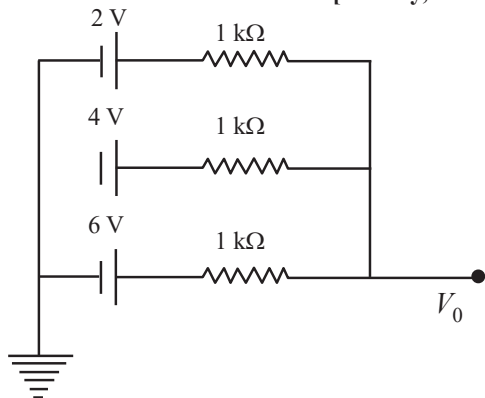
- (a) 50 V (b) 75 V (c) 150 V (d) 275 V

102. Two sources of equal emfs are connected in series. This combination is connected to an external resistance R . The internal resistances of the two sources are r_1 and r_2 ($r_1 > r_2$). If the potential difference across the source of internal resistance r_1 is zero, then the value of R will be [27 July, 2022 (Shift-I)]

(a) $r_1 - r_2$ (b) $\frac{r_1 r_2}{r_1 + r_2}$ (c) $\frac{r_1 + r_2}{2}$ (d) $r_2 - r_1$

103. In the given figure, the value of V_0 will be _____ V.

[25 July, 2022 (Shift-I)]

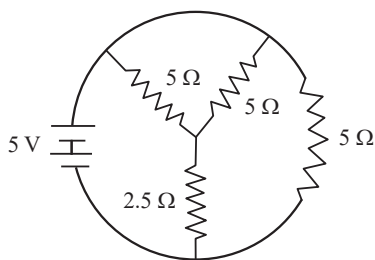


104. A 72Ω galvanometer is shunted by a resistance of 8Ω . The percentage of the total current which passes through the galvanometer is: [27 June, 2022 (Shift-I)]

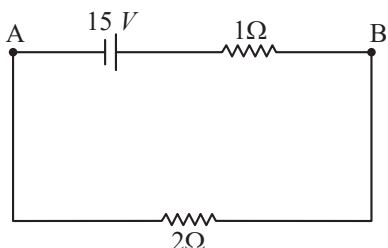
- (a) 0.1%
(b) 10%
(c) 25%
(d) 0.25%

105. The total current supplied to the circuit as shown in figure by the 5V battery is _____ A.

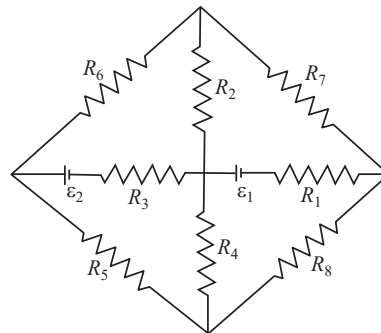
[25 June, 2022 (Shift-I)]



106. For the network shown below, the value of $V_B - V_A$ is _____ V. [29 June, 2022 (Shift-I)]



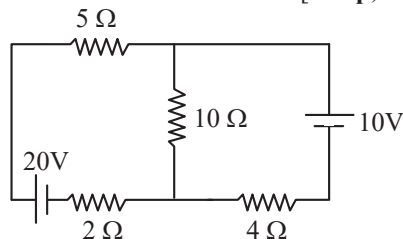
107. The figure shows a circuit having eight resistances of 1Ω each, labelled R_1 to R_8 , and two ideal batteries with voltages $\epsilon_1 = 12\text{ V}$ and $\epsilon_2 = 6\text{ V}$. [JEE Adv, 2022]



Which of the following statements is(are) correct?

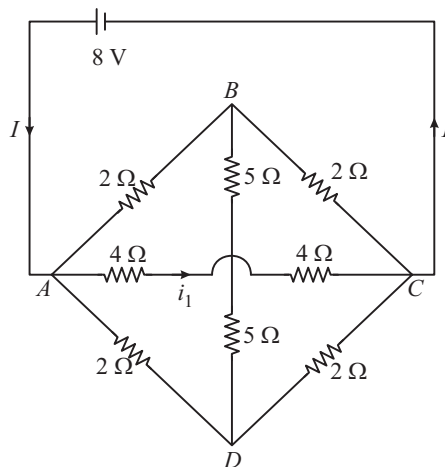
- (a) The magnitude of current flowing through R_1 is 7.2 A
(b) The magnitude of current flowing through R_2 is 1.2 A
(c) The magnitude of current flowing through R_3 is 4.8 A
(d) The magnitude of current flowing through R_5 is 2.4 A

108. In the figure shown, the current in the 10 V battery is close to: [6 Sep, 2020 (Shift-II)]



- (a) 0.71 A from positive to negative terminal
(b) 0.42 A from positive to negative terminal
(c) 0.21 A from positive to negative terminal.
(d) 0.36 A from negative to positive terminal.

109. The value of current i_1 flowing from A to C in the circuit diagram is [4 Sep, 2020 (Shift-II)]

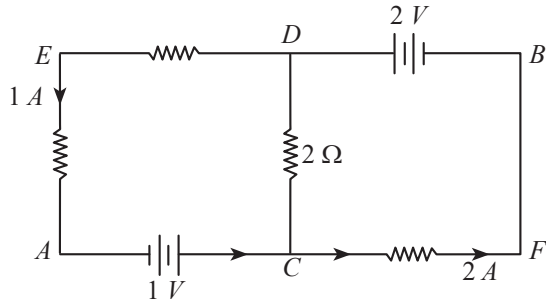


- (a) 1 A
(b) 4 A
(c) 5 A
(d) 2 A

110. In the circuit, given in the figure currents in different branches and value of one resistor are shown. Then potential at point B with respect to the point A is:

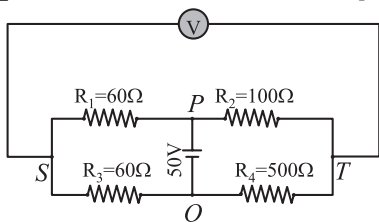
[5 Sep, 2020 (Shift-II)]

- (a) $+1V$ (b) $-1V$ (c) $+2V$ (d) $-2V$



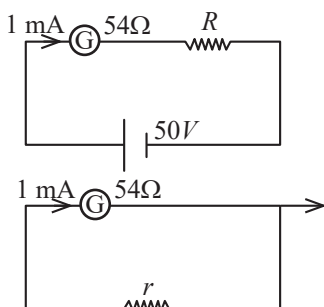
111. The series combination of two batteries, both of the same emf $10V$, but different internal resistance of 20Ω and 5Ω , is connected to the parallel combination of two resistors 30Ω and $R\Omega$. The voltage difference across the battery of internal resistance 20Ω is zero, the value of R (in Ω) is _____. [8 Jan, 2020 (Shift-II)]

112. In the balanced condition, the values of the resistances of the four arms of a Wheatstone bridge are shown in the figure below. The resistance R_3 has temperature coefficient $0.0004^\circ\text{C}^{-1}$. If the temperature of R_3 is increased by 100°C , the voltage developed between S and T will be _____ volt. [JEE Adv, 2020]



Current Measuring Instrument

113. For designing a voltmeter of range $50V$ and an ammeter of range 10mA using a galvanometer which has a coil of resistance 54Ω showing a full scale deflection for 1mA as in figure. [15 April, 2023 (Shift-I)]



- (A) for voltmeter $R \approx 50\text{k}\Omega$
 (B) for ammeter $r \approx 0.2\Omega$
 (C) for ammeter $r \approx 6\Omega$

- (D) for voltmeter $R \approx 5\text{k}\Omega$
 (E) for voltmeter $R \approx 500\Omega$

Choose the correct answer from the options given below:

- (a) (C) and (E) (b) (C) and (D)
 (c) (A) and (C) (d) (A) and (B)

114. Given below are two statement:

Statement-I : If the number of turns in the coil of a moving coil galvanometer is doubled then the current sensitivity becomes double.

Statement-II : Increasing current sensitivity of a moving coil galvanometer by only increasing the number of turns in the coil will also increase its voltage sensitivity in the same ratio:

In the light of the above statement, choose the correct answer from the options given below:

[10 April, 2023 (Shift-I)]

- (a) Statement-I is false but Statement-II is true
 (b) Both Statement-I and Statement-II are true
 (c) Both Statement-I and Statement-II are false
 (d) Statement-I is true but Statement-II is false

115. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): For measuring the potential difference across a resistance of 600Ω , the voltmeter with resistance 1000Ω will be preferred over voltmeter with resistance 4000Ω .

Reason (R): Voltmeter with higher resistance will draw smaller current than voltmeter with lower resistance.

In the light of the above statements, choose the most appropriate answer from the options given below. [1 Feb, 2023 (Shift-II)]

- (a) (A) is not correct but (R) is correct
 (b) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 (c) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 (d) (A) is correct but R is not correct

116. With the help of a potentiometer, we can determine the value of emf of a given cell. The sensitivity of the potentiometer [29 Jan, 2023 (Shift-II)]

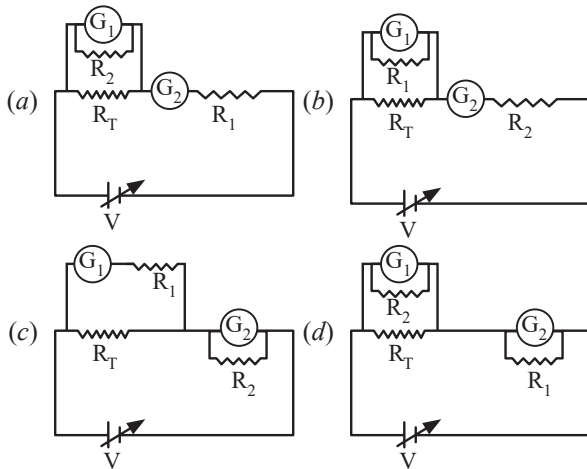
- A. is directly proportional to the length of the potentiometer wire
 B. is directly proportional to the potential gradient of the wire
 C. is inversely proportional to the potential gradient of the wire
 D. is inversely proportional to the length of the potentiometer wire

Choose the correct option for the above statements:

- (a) B and D only (b) A and C only
 (c) A only (d) C only

117. A student is provided with a variable voltage source V , a test resistor $R_T = 10\Omega$, two identical galvanometers G_1 and G_2 and two additional resistors, $R_1 = 10M\Omega$ and $R_2 = 0.001\Omega$. For conducting an experiment to verify ohms law, the most suitable circuit is:

[6 April, 2023 (Shift-II)]

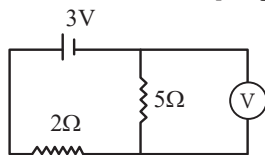


118. When two resistances R_1 and R_2 are connected in series and introduced into the left gap of a meter bridge and a resistance of 10Ω is introduced into the right gap, a null point is found at 60 cm from the left side. When R_1 and R_2 are connected in parallel and introduced into the left gap, a resistance of 3Ω is introduced into the right-gap to get null point at 40 cm from left the end. The product of $R_1 R_2$ is _____ Ω^2

[29 Jan, 2023 (Shift-II)]

119. As shown in the figure, the voltmeter reads 2V across 5Ω resistor. The resistance of the voltmeter is _____ Ω .

[6 April, 2023 (Shift-II)]



120. In an experiment to find emf of a cell using potentiometer, the length of null point for a cell of emf 1.5 V is found to be 60 cm. If this cell is replaced by another cell of emf E , the length of null point increases by 40 cm. The value of E is $\frac{x}{10}$ V. The value of x is _____.

[1 Feb, 2023 (Shift-I)]

121. When a resistance of 5Ω is shunted with a moving coil galvanometer, it shows a full scale deflection for a current of 250 mA, however when 1050Ω resistance is connected with it in series, it gives full scale deflection for 25 volt. The resistance of galvanometer is _____ Ω .

[13 April, 2023 (Shift-I)]

122. In a metre bridge experiment the balance point is obtained if the gaps are closed by 2Ω and 3Ω . A shunt of $X\Omega$ is added to 3Ω resistor to shift the balancing point by 22.5 cm. The value of X is _____.

[29 Jan, 2023 (Shift-I)]

123. A null point is found at 200 cm in potentiometer when cell in secondary circuit is shunted by 5Ω . When a resistance of

15Ω is used for shunting null point moves to 300 cm. The internal resistance of the cell is _____ Ω .

[29 Jan, 2023 (Shift-II)]

124. A 72Ω galvanometer is shunted by a resistance of 8Ω . The percentage of the total current which passes through the galvanometer is:

[27 June, 2022 (Shift-I)]

- (a) 0.1%
(b) 10%
(c) 25%
(d) 0.25%

125. A teacher in his physics laboratory allotted an experiment to determine the resistance (G) of a galvanometer. Students took the observations for $\frac{1}{3}$ deflection in the galvanometer.

Which of the below is true for measuring value of G ?

[25 June, 2022 (Shift-I)]

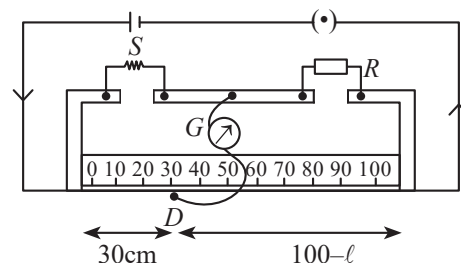
- (a) $\frac{1}{3}$ deflection method cannot be used for determining the resistance of the galvanometer.
(b) $\frac{1}{3}$ deflection method can be used and in this case the G equals to twice the value of shunt resistance(s).
(c) $\frac{1}{3}$ deflection method can be used and in this case, the G equals to three times the value of shunt resistance(s).
(d) $\frac{1}{3}$ deflection method can be used and in this case, the G value equals to the shunt resistance(s).

126. If n represents the actual number of deflections in a converted galvanometer of resistance G and shunt resistance S . Then the total current I when its figure of merit is K will be

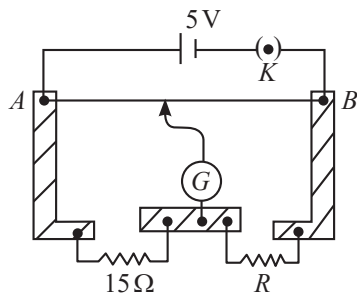
- (a) $\frac{KS}{(S+G)}$ (b) $\frac{(G+S)}{nKS}$
(c) $\frac{nKS}{(G+S)}$ (d) $\frac{nK(G+S)}{S}$

127. In a meter bridge experiment, for measuring unknown resistance ' S ', the null point is obtained at a distance 30 cm from the left side as shown at point D . If R is $5.6\text{ k}\Omega$, then the value of unknown resistance ' S ' will be _____ Ω .

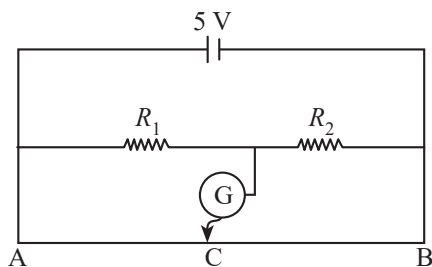
[27 July, 2022 (Shift-I)]



128. A meter bridge setup is shown in the figure. It is used to determine an unknown resistance R using a given resistor of 15Ω . The galvanometer (G) shows null deflection when tapping key is at 43 cm mark from end A . If the end correction for end A is 2 cm. Then the determined value of R will be _____ Ω . [28 June, 2022 (Shift-I)]

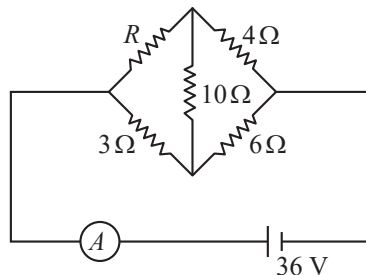


129. In the given figure of meter bridge experiment, the balancing length AC corresponding to null deflection of the galvanometer is 40 cm. The balancing length, if the radius of the wire AB is doubled, will be _____ cm. [27 July, 2022 (Shift-II)]

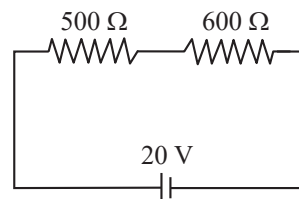


130. A potentiometer wire of length 10 m and resistance 20Ω is connected in series with a 25V battery and an external resistance 30Ω . A cell of emf E in secondary circuit is balanced by 250cm long potentiometer wire. The value of E (in volt) is $\frac{x}{10}$. The value of x is _____. [24 June, 2022 (Shift-II)]

131. Current measured by the ammeter (A) in the reported circuit when no current flows through 10 resistance, will be _____ A. [28 June, 2022 (Shift-I)]



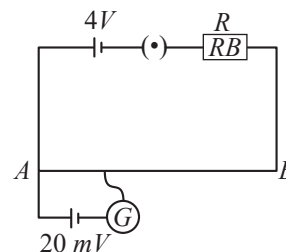
132. Two resistors are connected in series across a battery as shown in figure. If a voltmeter of resistance 2000Ω is used to measure the potential difference across 500Ω resistor, the reading of the voltmeter will be _____ V. [29 June, 2022 (Shift-II)]



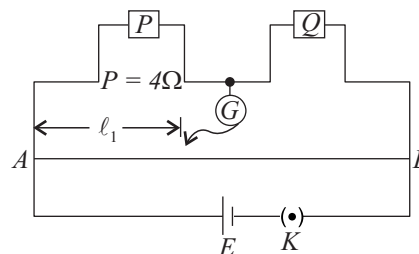
133. A cell, shunted by a 8Ω resistance, is balanced across a potentiometer wire of length 3 m. The balancing length is 2 m when cell is shunted by 4Ω resistance. The value of internal resistance of the cell will be _____ Ω . [27 June, 2022 (Shift-I)]

134. In a potentiometer arrangement, a cell of emf 1.20 V gives a balance point at 36cm length of wire. This cell is now replaced by another cell of emf 1.80 V. The difference in balancing length of potentiometer wire in above conditions will be _____ cm. [25 July, 2022 (Shift-II)]

135. As shown in the figure, a potentiometer wire of resistance 20Ω and length 300 cm is connected with resistance box (R.B) and a standard cell of emf 4 V. For a resistance 'R' of resistance box introduced into the circuit, the null point for a cell of 20mV is found to be 60 cm. The value of 'R' is _____ Ω . [28 July, 2022 (Shift-I)]



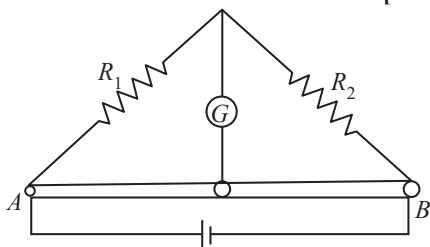
136. Resistances are connected in a meter bridge circuit as shown in the figure. The balancing length ℓ_2 is 40 cm. Now an unknown resistance x is connected in series with P and new balancing length is found to be 80cm measured from the same end. Then the value of x will be _____ Ω . [26 July, 2022 (Shift-I)]



137. In a potentiometer arrangement, a cell gives a balancing point at 75 cm length of wire. This cell is now replaced by another cell of unknown emf. If the ratio of the emf's of two cells respectively is 3:2, the difference in the balancing length of the potentiometer wire in above two cases will be _____ cm. [24 June, 2022 (Shift-I)]

138. A potentiometer wire of length 300 cm is connected in series with a resistance of 780Ω and a standard cell of emf 4V. A constant current flows through potentiometer wire. The length of the null point for cell of emf 20 mV is found to be 60 cm. The resistance of the potentiometer wire is _____ Ω .
[26 July, 2022 (Shift-II)]

139. Two resistances $R_1 = X\Omega$ and $R_2 = 1\Omega$ are connected to a wire AB of uniform resistivity, as shown in the figure. The radius of the wire varies linearly along its axis from 0.2 mm at A to 1 mm at B. A galvanometer (G) connected to the center of the wire, 50 cm from each end along its axis, shows zero deflection when A and B are connected to a battery. The value of x is
[JEE Adv, 2022]



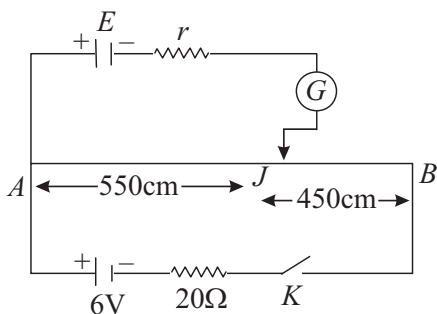
140. In the given potentiometer circuit arrangement, the balancing length AC is measured to be 250 cm. When the galvanometer connection is shifted from point (1) to point (2) in the given diagram, the balancing length becomes 400 cm. The ratio of the emf of two cells, $\frac{\epsilon_1}{\epsilon_2}$ is:
[25 July, 2021 (Shift-II)]

- (a) $\frac{4}{3}$ (b) $\frac{8}{5}$ (c) $\frac{3}{2}$ (d) $\frac{5}{3}$

141. For full scale deflection of total 50 divisions, 50 mV voltage is required in galvanometer. The resistance of galvanometer if its current sensitivity is 2 div/mA will be:
[27 Aug, 2021 (Shift-II)]

- (a) 2Ω (b) 4Ω
(c) 1Ω (d) 5Ω

142. In the given figure, there is a circuit of potentiometer of length AB = 10m. The resistance per unit length is 0.1Ω per cm. Across AB, a battery of emf E and internal resistance 'r' is connected. The maximum value of emf measured by this potentiometer is:
[25 July, 2021 (Shift-I)]



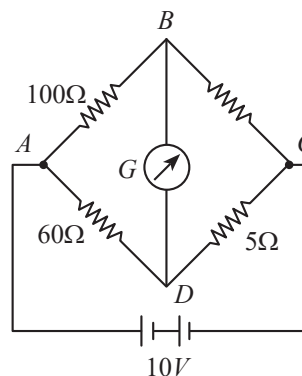
- (a) 2.75 V (b) 6 V
(c) 5 V (d) 2.25 V

143. Consider a galvanometer shunted with 5Ω resistance and 2% of current passes through it. What is the resistance of the given galvanometer?
[31 Aug, 2021 (Shift-I)]

- (a) 300Ω
(b) 344Ω
(c) 226Ω
(d) 245Ω

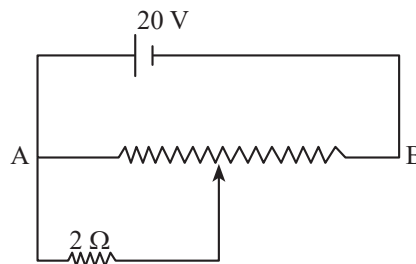
144. The four arms of a Wheatstone bridge have resistances as shown in the figure. A galvanometer of 15Ω resistance is connected across BD. Calculate the current through the galvanometer when a potential difference of 10 V is maintained across AC.
[17 March, 2021 (Shift-II)]

- (a) $4.87\mu A$
(b) 2.44 mA
(c) 4.87 mA
(d) $2.44\mu A$



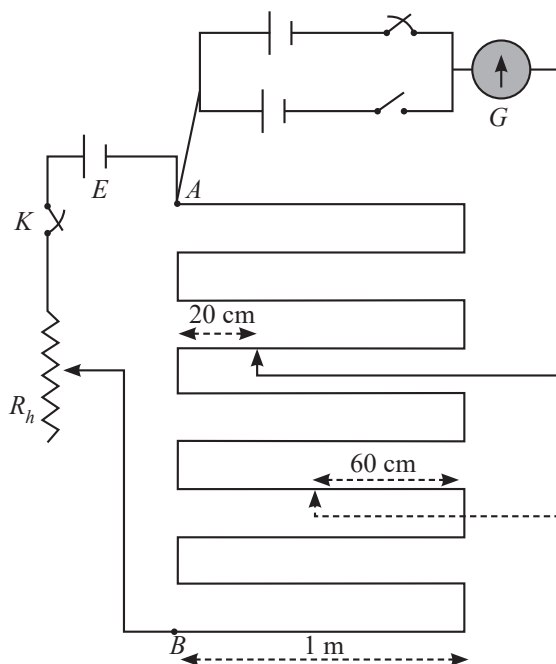
145. The given potentiometer has its wire of resistance 10Ω . When the sliding contact is in the middle of the potentiometer wire, the potential drop across 2Ω resistor is:
[25 July, 2021 (Shift-II)]

- (a) $\frac{40}{11}V$
(b) 10 V
(c) $\frac{40}{9}V$
(d) 5 V

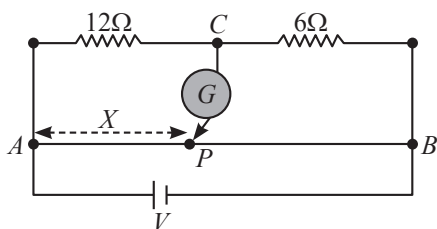


146. In the given circuit of potentiometer, the potential difference E across AB (10 m length) is larger than E_1 and E_2 as well. For key K_1 (closed), the jockey is adjusted to touch the wire at point J_1 so that there is no deflection in the galvanometer. Now the first battery (E_1) is replaced by second battery (E_2) for working by making K_1 open and K_2 closed. The galvanometer gives then null deflection at J_2 . The value of $\frac{E_1}{E_2}$ is $\frac{a}{b}$, where $a =$ _____

[25 Feb, 2021 (Shift-I)]



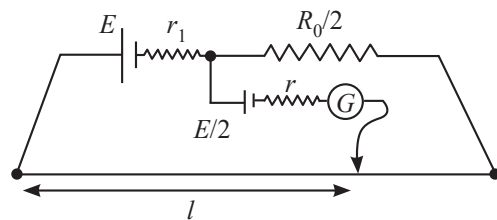
147. Consider a 72 cm long wire AB as shown in the figure. The galvanometer jockey is placed at P on AB at a distance x cm from A . The galvanometer shows zero deflection.



The value of x , to nearest integer, is _____

[18 March, 2021 (Shift-II)]

148. In order to measure the internal resistance r_1 of a cell of emf E , a meter bridge of wire resistance $R_0 = 50 \Omega$, a resistance $R_0/2$, another cell of emf $E/2$ (internal resistance r) and a galvanometer G are used in a circuit, as shown in the figure. If the null point is found at $l = 72$ cm, then the value of $r_1 = \dots \Omega$. [JEE Adv, 2021]



149. The length of a potentiometer wire is 1200 cm and it carries a current of 60 mA. For a cell of emf 5 V and internal resistance of 20Ω , the null point on it is found to be at 1000 cm. The resistance of whole wire is:

[8 Jan, 2020 (Shift-I)]

- (a) 60Ω (b) 100Ω (c) 80Ω (d) 120Ω

150. A galvanometer of resistance G is converted into a voltmeter of range $0 - 1$ V by connecting a resistance R_1 in series with it. The additional resistance that should be connected in series with R_1 to increase the range of the voltmeter to $0 - 2$ V will be:

[5 Sep, 2020 (Shift-I)]

- (a) $R_1 + G$ (c) G (b) R_1 (d) $R_1 - G$

151. A galvanometer is used in laboratory for detecting the null point in electrical experiments. If, on passing a current of 6 mA it produces a deflection of 2° , its figure of merit is close to:

[5 Sep, 2020 (Shift-II)]

- (a) 666° A/div. (b) $6 \times 10^{-3} \text{ A/div}$
(c) 333° A/div. (d) $3 \times 10^{-3} \text{ A/div}$

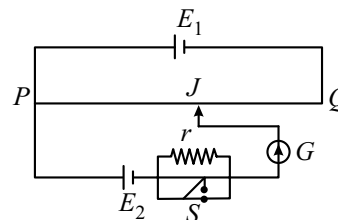
152. A galvanometer having a coil resistance 100Ω gives a full scale deflection when a current of 1 mA is passed through it. What is the value of the resistance which can convert this galvanometer into a voltmeter giving full scale deflection for a potential difference of 10 V.

[8 Jan, 2020 (Shift-II)]

- (a) $9.9 \text{ k}\Omega$
(b) $7.9 \text{ k}\Omega$
(c) $10 \text{ k}\Omega$
(d) $8.9 \text{ k}\Omega$

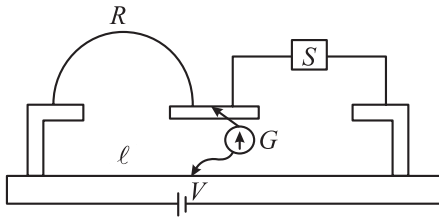
153. 20. A potentiometer wire PQ of 1 m length is connected to a standard cell E_1 . Another cell E_2 of emf 1.02 V is connected with a resistance ' r ' and switch S (as shown in figure). With switch S open, the null position is obtained at a distance of 49 cm from Q . The potential gradient in the potentiometer wire is:

[2 Sep, 2020 (Shift-II)]

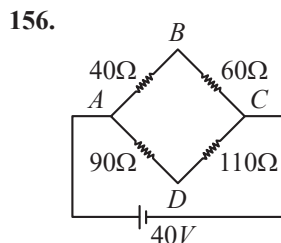
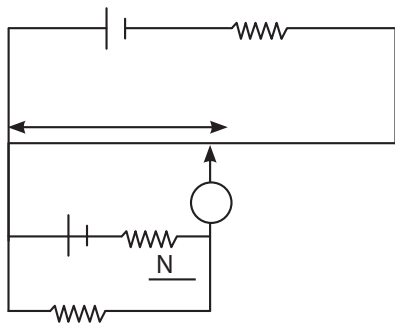
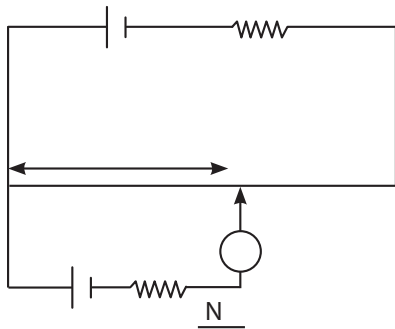


- (a) 0.04 V/cm
(b) 0.01 V/cm
(c) 0.02 V/cm
(d) 0.03 V/cm

154. In a meter bridge experiment S is a standard resistance. R is a resistance wire. It is found that balancing length is $\ell = 25$ cm. If R is replaced by a wire of half-length and half diameter that of R of same material, then the balancing distance ℓ (in cm) will now be [9 Jan, 2020 (Shift-II)]



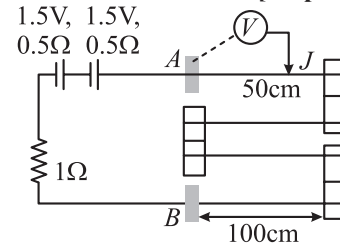
155. The balancing length for a cell is 560 cm in a potentiometer experiment. When an external resistance of $10\ \Omega$ is connected in parallel to the cell, the balancing length changes by 60 cm. If the internal resistance of the cell is $\frac{N}{10}\ \Omega$, where N is an integer then value of N is _____. [7 Jan, 2020 (Shift-II)]



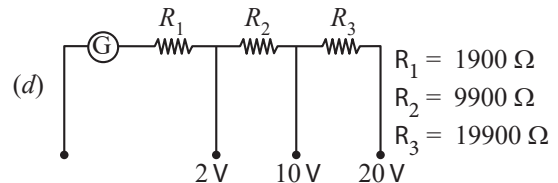
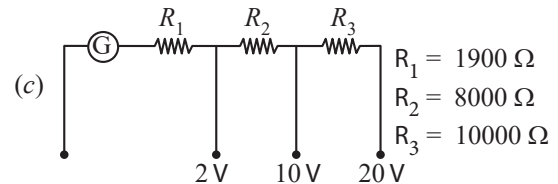
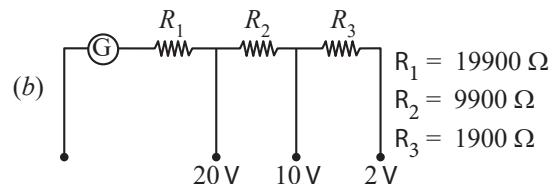
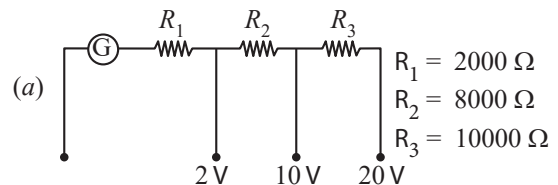
Four resistance $40\ \Omega$, $60\ \Omega$, $90\ \Omega$, and $110\ \Omega$, make the arms of a quadrilateral $ABCD$. Across AC is a battery of emf $40\ V$ and internal resistance negligible. The potential difference across BD in V is ____ [4 Sep, 2020 (Shift-II)]

157. In the circuit shown, a four wire potentiometer is made of a 400 cm long wire, which extends between A and B .

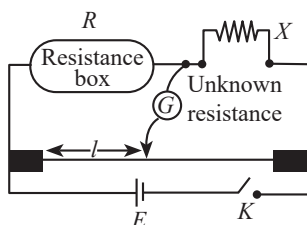
The resistance per unit length of the potentiometer wire is $r = 0.01\ \Omega/\text{cm}$. If an ideal voltmeter is connected as shown with jockey J at 50 cm from end A , the expected reading of the voltmeter will be [8 April, 2019 (Shift-II)]



- (a) 0.20 V (b) 0.25 V
(c) 0.75 V (d) 0.50 V
158. The resistance of a galvanometer is $50\ \text{ohm}$ and the maximum current which can be passed through it is $0.002\ \text{A}$. What resistance must be connected to it in order to convert it into an ammeter of range $0 - 0.5\ \text{A}$? [9 April, 2019 (Shift-II)]
- (a) 0.2 ohm (b) 0.002 ohm
(c) 0.02 ohm (d) 0.5 ohm
159. A moving coil galvanometer has resistance $50\ \Omega$ and it indicates full deflection at 4mA current. A voltmeter is made using this galvanometer and a $5\ \text{k}\Omega$ resistance. The maximum voltage, that can be measured using this voltmeter, will be close to: [9 April, 2019 (Shift-I)]
- (a) 15 V (b) 20 V (c) 10 V (d) 40 V
160. A galvanometer of resistance $100\ \Omega$ has 50 divisions on its scale and has sensitivity of $20\ \mu\text{A}/\text{division}$. It is to be converted to a voltmeter with three ranges, of $0 - 2\ V$, $0 - 10\ V$ and $0 - 20\ V$. The appropriate circuit to do so is: [12 April, 2019 (Shift-I)]



161. In a meter bridge experiment, the circuit diagram and the corresponding observation table are shown in figure.



Sl. No.	R (Ω)	l (cm)
1.	1000	60
2.	100	13
3.	10	1.5
4.	1	1.0

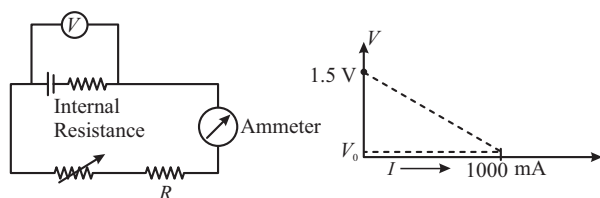
Which of the readings is inconsistent?

[10 April, 2019 (Shift-I)]

- (a) 4 (b) 3
(c) 2 (d) 1

162. To verify Ohm's law, a student connects the voltmeter across the battery as, shown in the figure. The measured voltage is plotted as a function of the current, and the following graph is obtained. If V_0 is almost zero, identify the correct statement

[12 April, 2019 (Shift-I)]



- (a) The value of the resistance R is 1.5Ω
(b) The emf of the battery is 1.5 V and the value of R is 1.5Ω
(c) The emf of the battery is 1.5 V and its internal resistance is 1.5Ω
(d) The potential difference across the battery is 1.5 V when it sends a current of 1000 mA

163. 3. An ideal battery of 4 V and resistance R are connected in series in the primary circuit of a potentiometer of length 1 m and resistance 5Ω . The value of R , to give a difference of 5 mV across 10 cm of potentiometer wire, is:

[12 Jan, 2019 (Shift-I)]

- (a) 490Ω (b) 480Ω (c) 395Ω (d) 495Ω

164. A moving coil galvanometer allows a full scale current of 10^{-4} A . A series resistance of $49 \text{ k}\Omega$ is required to convert the above galvanometer into a voltmeter of range $0 - 5 \text{ V}$. Therefore the value of shunt resistance (approximately) required to convert the above galvanometer into a ammeter of range $0 - 10 \text{ mA}$ is:

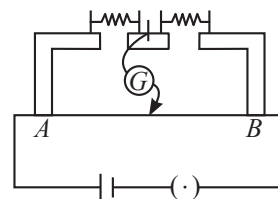
[10 April, 2019 (Shift-I)]

- (a) 200Ω (b) 100Ω (c) 10Ω (d) 500Ω

165. A moving coil galvanometer, having a resistance G , produces full scale deflection when a current I_g flows through it. This galvanometer can be converted into (i) an ammeter of range 0 to I_0 ($I_0 > I_g$) by connecting a shunt resistance R_A to it and (ii) into a voltmeter of range 0 to V ($V = GI_0$) by connecting a series resistance R_V to it. Then, [12 April, 2019 (Shift-II)]

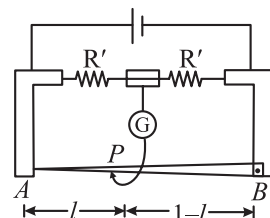
- (a) $R_A R_V = G^2 \left(\frac{I_g}{I_0 - I_g} \right)$ and $\frac{R_A}{R_V} = \left(\frac{I_0 - I_g}{I_g} \right)^2$
(b) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$
(c) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \frac{I_g}{I_0 - I_g}$
(d) $R_A R_V = G^2 \left(\frac{I_0 - I_g}{I_g} \right)$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$

166. In the experimental setup of metre bridge shown in the figure, the null point is obtained at a distance of 40 cm from A . If a 10Ω resistor is connected in series with R_1 , the null point shifts by 10 cm . The resistance that should be connected in parallel with $(R_1 + 10) \Omega$ such that the null point shifts back to its initial position is _____ :



- (a) 20Ω (b) 40Ω
(c) 60Ω (d) 30Ω

167. In a meter bridge, the wire of length 1 m has a non-uniform cross-section such that the variation $\frac{dR}{dl}$ of its resistance R with length l is $\frac{dR}{dl} \propto \frac{1}{\sqrt{l}}$. Two equal resistances are connected as shown in the figure. The galvanometer has zero deflection when the jockey is at point P . What is the length AP ? [12 Jan, 2019 (Shift-I)]



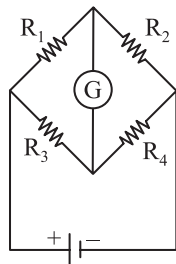
- (a) 0.2 m (b) 0.3 m
(c) 0.25 m (d) 0.35 m

168. A galvanometer, whose resistance is 50 ohm, has 25 divisions in it. When a current of 4×10^{-4} A passes through it, its needle (pointer) deflects by one division. To use this galvanometer as a voltmeter of range 2.5 V, it should be connected to a resistance of : [12 Jan, 2019 (Shift-II)]

(a) 250 ohm (b) 200 ohm
(c) 6200 ohm (d) 6250 ohm

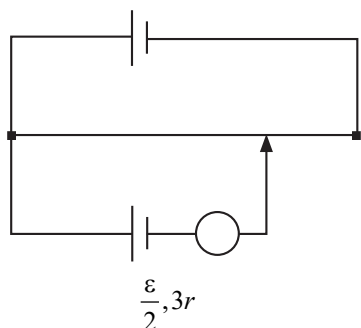
169. The Wheatstone bridge shown in Figure, gets balanced when the carbon resistor used as R_1 has the colour code (Orange, Red, Brown). The resistors R_2 and R_4 are 80Ω and 40Ω , respectively. Assuming that the colour code for the carbon resistors given their accurate values, the colour code for the carbon resistor, used as R_3 would be

[10 Jan, 2019 (Shift-II)]



- (a) Brown, Blue, Brown
(b) Brown, Blue, Black
(c) Red, Green, Brown
(d) Grey, Black, Brown
170. A potentiometer wire AB having length L and resistance $12r$ is joined to a cell D of emf ε and internal resistance r . A cell C having emf $\varepsilon/2$ and internal resistance $3r$ is connected. The length AJ at which the galvanometer as shown in figure shows no deflection is:

[10 Jan, 2019 (Shift-I)]



- (a) $\frac{11}{12}L$ (b) $\frac{11}{24}L$
(c) $\frac{13}{24}L$ (d) $\frac{5}{12}L$
171. A galvanometer having a resistance of 20Ω and 30 divisions on both sides has figure of merit 0.005 ampere /division. The resistance that should be connected in series such that it can be used as a voltmeter upto 15 volt is:

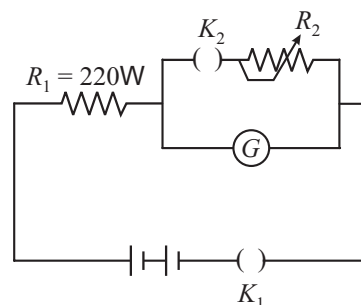
[11 Jan, 2019 (Shift-II)]

(a) 100Ω (b) 120Ω
(c) 80Ω (d) 125Ω

172. The galvanometer deflection, when key K_1 is closed but K_2 is open, equals θ_0 (see figure). On closing K_2 also and adjusting R_2 to 5Ω , the deflection in galvanometer becomes $\frac{\theta_0}{5}$. The resistance of the galvanometer is, then, given by

[Neglect the internal resistance of battery]:

[12 Jan, 2019 (Shift-I)]



(a) 5Ω (b) 22Ω
(c) 25Ω (d) 12Ω

173. In a Wheatstone bridge (see figure) Resistance P and Q are approximately equal. When $R = 400 \Omega$, the bridge is balanced. On interchanging P and Q , the value of R , for balance is 405Ω . The value of X is close to:

[11 Jan, 2019 (Shift-I)]

(a) 401.5 ohm (b) 404.5 ohm
(c) 403.5 ohm (d) 402.5 ohm

174. The resistance of the meter bridge AB in given figure is 4Ω . With a cell of emf $\varepsilon = 005$ V and rheostat resistance $R_h = 2 \Omega$ the null point is obtained at some point J . When the cell is replaced by another one of emf $\varepsilon = \varepsilon_2$ the same null point J is found for $R_h = 6 \Omega$. The emf ε_2 is:

[11 Jan, 2019 (Shift-I)]

(a) 0.4 V (b) 0.3 V
(c) 0.6 V (d) 0.5 V

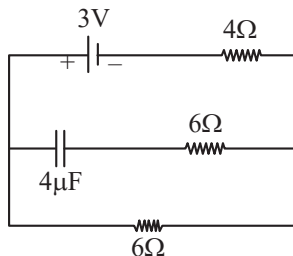
175. Two identical moving coil galvanometer have 10Ω resistance and full scale deflection at $2 \mu\text{A}$ current. One of them is converted into a voltmeter of 100 mV full scale reading and the other into an Ammeter of 1 mA full scale current using appropriate resistors. These are then used to measure the voltage and current in the Ohm's law experiment with $R = 1000 \Omega$ resistor by using an ideal cell. Which of the following statements is/are correct?

[JEE Adv, 2019]

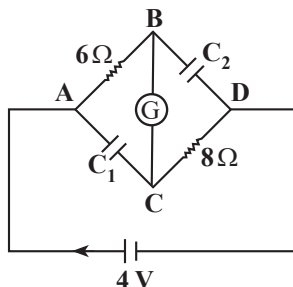
- (a) The measured value of R will be $978 \Omega < R < 982 \Omega$
(b) The resistance of the Voltmeter will be $100 \text{ k}\Omega$
(c) The resistance of the Ammeter will be 0.02Ω (round off to 2nd decimal place)
(d) If the ideal cell is replaced by a cell having internal resistance of 5Ω then the measured value of R will be more than 1000Ω

RC Circuit

176. In the network shown below, the charge accumulated in the capacitor in steady state will be : [13 April, 2023 (Shift-II)]

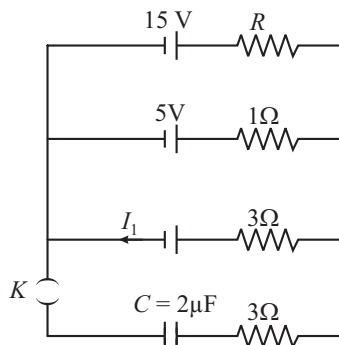


- (a) $7.2 \mu C$ (b) $4.8 \mu C$
(c) $10.3 \mu C$ (d) $12 \mu C$
177. In this figure the resistance of the coil of galvanometer G is 2Ω . The emf of the cell is $4 V$. The ratio of potential difference across C_1 and C_2 is: [8 April, 2023 (Shift-I)]



- (a) 1 (b) $\frac{4}{5}$
(c) $\frac{3}{4}$ (d) $\frac{5}{4}$
178. In a circuit shown in the figure, the capacitor C is initially uncharged and the key K is open. In this condition, a current of $1 A$ flows through the 1Ω resistor. The key is closed at time $t = t_0$. Which of the following statements is are correct? [JEE Adv, 2023]

[Given: $e^{-1} = 0.36$]



- (a) The value of the resistance R is 3Ω .
(b) For $t < t_0$, the value of current I_1 is $2 A$.
(c) At $t = t_0 + 7.2 \mu s$, the current in the capacitor is $0.6 A$.

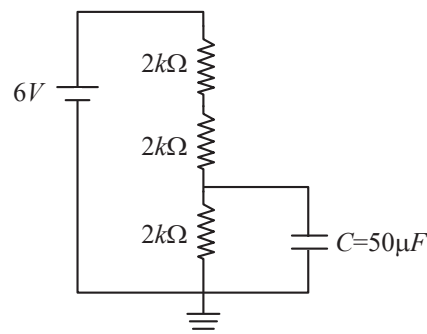
(d) For $t \rightarrow \infty$, the charge on the capacitor is $12 \mu C$.

179. A capacitor is connected to a $20 V$ battery through a resistance of 10Ω . It is found that the potential difference across the capacitor rises to $2 V$ in $1 \mu s$. The capacitance of the capacitor is _____ μF .

Given $\ln\left(\frac{10}{9}\right) = 0.105$ [1 Sep, 2021 (Shift-II)]

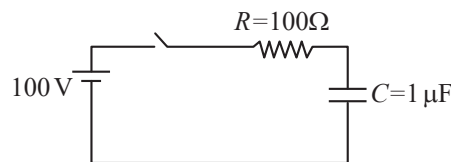
- (a) 1.85 (b) 0.95
(c) 0.105 (d) 9.52

180. A capacitor of $50 \mu F$ is connected in a circuit as shown in figure. The charge on the upper plate of the capacitor is _____ μC . [31 Aug, 2021 (Shift-I)]



181. A capacitor of capacitance $C = 1 \mu F$ is suddenly connected to a battery of 100 volt through a resistance $R = 100 \Omega$. The time taken for the capacitor to be charged to get $50 V$ is:

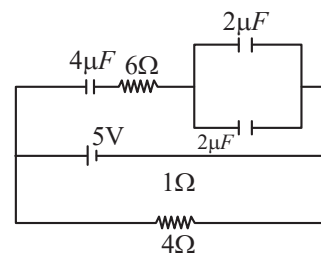
[Take $\ln 2 = 0.69$] [27 July, 2021 (Shift-I)]



- (a) $3.33 \times 10^{-4} s$ (b) $1.44 \times 10^{-4} s$
(c) $0.30 \times 10^{-4} s$ (d) $0.69 \times 10^{-4} s$

182. Calculate the amount of charge on capacitor of $4 \mu F$. The internal resistance of battery is 1Ω .

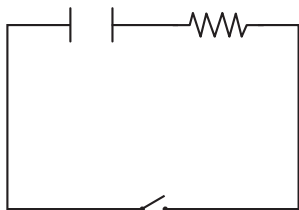
[27 Aug, 2021 (Shift-I)]



- (a) $16 \mu C$ (b) zero
(c) $4 \mu C$ (d) $8 \mu C$

183. The circuit shown in the figure consists of a charged capacitor of capacity $3\mu\text{F}$ and a charge of $30\mu\text{C}$. At time $t = 0$, when the key is closed, the value of current flowing through the $5\text{M}\Omega$ resistor is ' x ' μA .

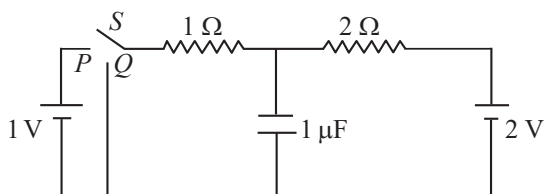
[18 March, 2021 (Shift-I)]



The value of ' x ' to the nearest integer is _____.

Comprehension (Q.184 to 185) In the circuit shown below, the switch S is connected to position P for a long time so that the charge on the capacitor becomes $q_1\mu\text{C}$. Then S is switched to position Q . After a long time, the charge on the capacitor is $q_2\mu\text{C}$.

[JEE Adv, 2021]

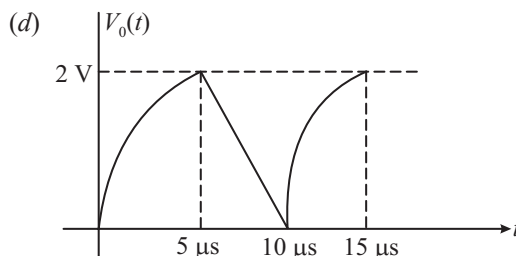
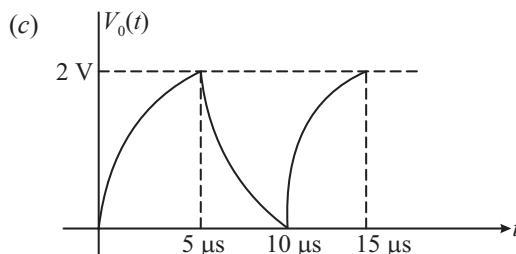
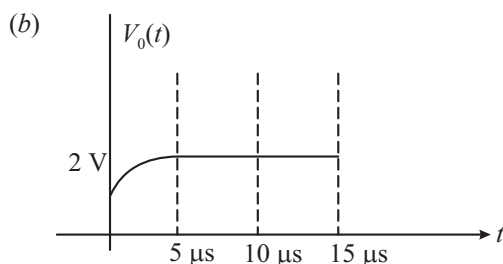
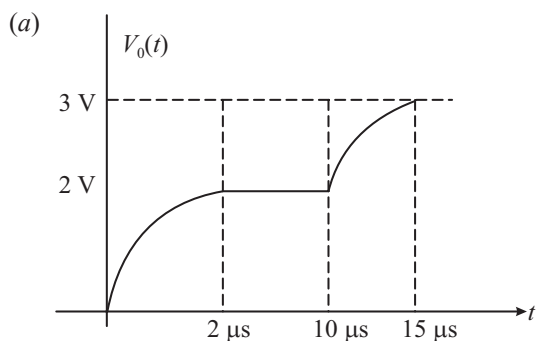
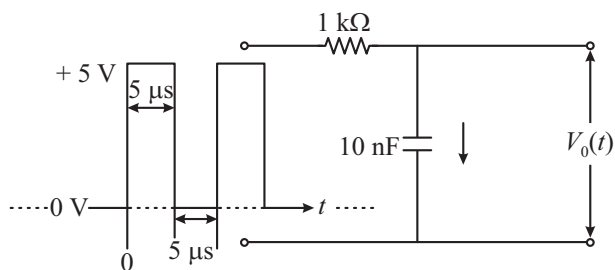


184. The magnitude of q_1 is _____.

185. The magnitude of q_2 is _____.

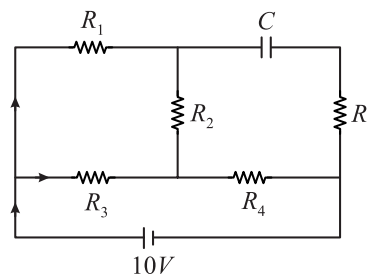
186. For the given input voltage waveform $V_{\text{in}}(t)$, the output voltage waveform $V_0(t)$, across the capacitor is correctly depicted by

[6 Sep, 2020 (Shift-I)]



187. An ideal cell of emf 10V is connected in circuit shown in figure. Each resistance is 2Ω . The potential difference (in V) across the capacitor when it is fully charged is _____.

[2 Sep, 2020 (Shift-II)]

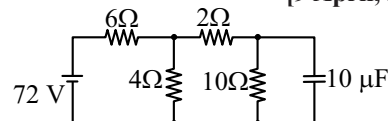


188. A 60pF capacitor is fully charged by a 20V supply. It is then disconnected from the supply and is connected to another uncharged 60pF capacitor in parallel. The electrostatic energy that is lost in this process by the time the charge is redistributed between them is (in nJ) _____.

[7 Jan, 2020 (Shift-II)]

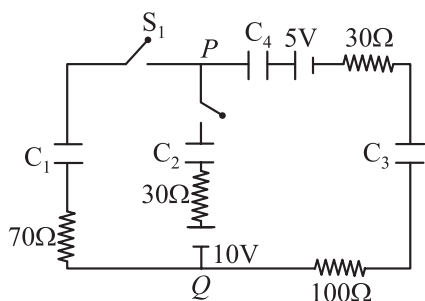
189. Determine the charge on the capacitor in the following circuit:

[9 April, 2019 (Shift-I)]



- (a) $200\mu\text{C}$
 (b) $60\mu\text{C}$
 (c) $10\mu\text{C}$
 (d) $2\mu\text{C}$

190. In the circuit shown, initially there is no charge on capacitors and keys S_1 and S_2 are open. The values of the capacitors are $C_1 = 10 \mu\text{F}$, $C_2 = 30 \mu\text{F}$ and $C_3 = C_4 = 80 \mu\text{F}$. [JEE Adv, 2019]



Which of the statements is/are correct?

- (a) The key S_1 is kept closed for long time such that capacitors are fully charged. Now key S_2 is closed, at this time, the instantaneous current across 30Ω resistor (between points P and Q) will be 0.2 A (round off to 1st decimal place)
- (b) If key S_1 is kept closed for long time such that capacitors are fully charged, the voltage difference between points P and Q will be 10 V
- (c) At time $t = 0$, the key S_1 is closed, the instantaneous current in the closed circuit will be 25 mA
- (d) If key S_1 is kept closed for long time such that capacitors are fully charged, the voltage across the capacitors C_1 will be 4 V

Answer Key (Manthan)

Exercise -1

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (b) | 3. (a) | 4. (b) | 5. (b) | 6. (d) | 7. (b) | 8. (a) | 9. (b) | 10. (d) |
| 11. (b) | 12. (b) | 13. (d) | 14. (c) | 15. (a) | 16. (d) | 17. (b) | 18. (a) | 19. (b) | 20. (b) |
| 21. (d) | 22. (b) | 23. (b) | 24. (a) | 25. (b) | 26. (c) | 27. (c) | 28. (c) | 29. (b) | 30. (b) |
| 31. (c) | 32. (a) | 33. (c) | 34. (b) | 35. (a) | 36. (a) | 37. (a) | 38. (d) | 39. (d) | 40. (c) |
| 41. (a) | 42. (d) | 43. (c) | 44. (c) | 45. (a) | 46. (c) | 47. (d) | 48. (b) | 49. (b) | 50. (d) |
| 51. (a) | 52. (c) | 53. (c) | 54. (c) | 55. (b) | 56. (a) | 57. (b) | | | |

Exercise -2

- | | | | | | | | | | |
|---------|---------|-----------|---------|---------|---------|---------|----------|---------|----------|
| 1. (d) | 2. (c) | 3. (c) | 4. (d) | 5. (d) | 6. (c) | 7. (c) | 8. (b) | 9. (c) | 10. (b) |
| 11. (b) | 12. (c) | 13. (d) | 14. (a) | 15. (c) | 16. (d) | 17. (a) | 18. (d) | 19. (c) | 20. (b) |
| 21. (d) | 22. (b) | 23. (b) | 24. (c) | 25. (b) | 26. (d) | 27. (c) | 28. (c) | 29. (a) | 30. (b) |
| 31. (a) | 32. (b) | 33. (b) | 34. (c) | 35. (d) | 36. (b) | 37. (d) | 38. [80] | 39. [4] | 40. [75] |
| 41. [8] | 42. [2] | 43. [160] | 44. [8] | 45. [6] | | | | | |

Exercise -3

- | | | | | | | | | | |
|---------|----------|-----------|---------|---------|---------|---------|----------|---------|---------|
| 1. (b) | 2. (b) | 3. (a) | 4. (a) | 5. (d) | 6. (c) | 7. (a) | 8. (c) | 9. (a) | 10. (c) |
| 11. (c) | 12. (a) | 13. (a) | 14. (d) | 15. (a) | 16. (a) | 17. (c) | 18. (c) | 19. (c) | 20. (c) |
| 21. (2) | 22. (20) | 23. (864) | 24. (3) | 25. (2) | 26. (2) | 27. (3) | 28. (70) | | |

Exercise -4

- | | | | | | | | | | |
|-------------------|-----------|---------------------|-------------|------------|-----------|-------------------|-------------|---------------|-----------|
| 1. (a) | 2. (a) | 3. (a) | 4. (b) | 5. (c) | 6. [144] | 7. [48] | 8. [300] | 9. [4] | 10. [12] |
| 11. (b) | 12. (d) | 13. (c) | 14. (a) | 15. (d) | 16. (a) | 17. [5] | 18. (c) | 19. (b) | 20. (c) |
| 21. (a, c, d) | 22. (a) | 23. (d) | 24. (a) | 25. (d) | 26. (b) | 27. (a) | 28. (a) | 29. (a) | 30. (a) |
| 31. (b) | 32. (b) | 33. (d) | 34. (d) | 35. (d) | 36. [10] | 37. [2] | 38. [100] | 39. (b) | 40. (a) |
| 41. (a) | 42. (a) | 43. [8] | 44. [2] | 45. [3] | 46. (c) | 47. (b) | 48. (b) | 49. (a) | 50. (d) |
| 51. (b) | 52. [9] | 53. [6] | 54. [3] | 55. [3] | 56. [70] | 57. [2] | 58. [20] | 59. [4] | 60. [4] |
| 61. [10] | 62. (c) | 63. (b) | 64. (c) | 65. [10] | 66. (b) | 67. (b) | 68. (a) | 69. (c) | 70. (a) |
| 71. (b) | 72. (b) | 73. (d) | 74. (a) | 75. (a) | 76. [16] | 77. [14] | 78. [450] | 79. (a, b, c) | 80. (d) |
| 81. (a) | 82. (a) | 83. (d) | 84. (a) | 85. [3840] | 86. [4] | 87. [2500] | 88. [50] | 89. (b) | 90. (a) |
| 91. (a) | 92. (a) | 93. (d) | 94. (a) | 95. (b) | 96. (a) | 97. (a) | 98. (c) | 99. (a) | 100. (a) |
| 101. (d) | 102. (d) | 103. [4] | 104. (b) | 105. [2] | 106. [10] | 107. (a, b, c, d) | | 108. (c) | 109. (a) |
| 110. (a) | 111. [30] | 112. [0.26 to 0.28] | | 113. (c) | 114. (d) | 115. (a) | 116. (b) | 117. (c) | 118. [30] |
| 119. [20] | 120. [25] | 121. [50] | 122. [2] | 123. [5] | 124. (b) | 125. (b) | 126. (d) | 127. [2400] | 128. [19] |
| 129. [40] | 130. [25] | 131. [10] | 132. [8] | 133. [8] | 134. [18] | 135. [780] | 136. [20] | 137. [25] | 138. [20] |
| 139. [5] | 140. (d) | 141. (a) | 142. (c) | 143. (d) | 144. (c) | 145. (c) | 146. [1] | 147. [48] | 148. [3] |
| 149. (b) | 150. (a) | 151. (d) | 152. (a) | 153. (c) | 154. [40] | 155. [12] | 156. [2] | 157. (b) | 158. (a) |
| 159. (b) | 160. (c) | 161. (a) | 162. (c) | 163. (c) | 164. (c) | 165. (b) | 166. (c) | 167. (c) | 168. (b) |
| 169. (a) | 170. (d) | 171. (c) | 172. (b) | 173. (d) | 174. (b) | 175. (a, c) | 176. (a) | 177. (b) | |
| 178. (a, b, c, d) | | 179. (b) | 180. [100] | 181. (d) | 182. (d) | 183. [2] | 184. [1.33] | 185. [0.67] | 186. (a) |
| 187. [8] | 188. [6] | 189. (a) | 190. (c, d) | | | | | | |



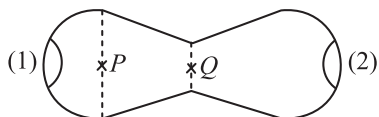
Physics

Current Electricity

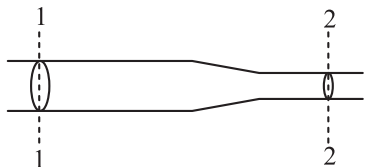
Exercise -1

MULTIPLE CORRECT TYPE QUESTIONS

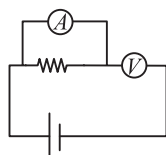
1. A metallic conductor of irregular cross-section is as shown in the figure. A constant potential difference is applied across the ends (1) and (2). Then



- (a) The current at the cross-section P equals the current at the cross-section Q
 (b) The electric field intensity at P is less than that at Q
 (c) The rate of heat generated per unit time at Q is greater than that at P
 (d) The number of electrons crossing per unit area of cross-section at P is less than that at Q
2. The area of cross section of a current carrying conductor is A_0 and $\frac{A_0}{4}$ at sections (1) and (2) respectively. If $v_{\alpha 1}, v_{\alpha 2}$ and E_1, E_2 be the drift velocities and electric fields at sections 1 and 2 respectively, then



- (a) $v_{\alpha 1} : v_{\alpha 2} = 1 : 4$ (b) $v_{\alpha 1} : v_{\alpha 2} = 4 : 1$
 (c) $E_1 : E_2 = 4 : 1$ (d) $E_1 : E_2 = 1 : 4$
3. By mistake, a voltmeter is placed in series and an ammeter in parallel with a resistance in an electric circuit, with a cell in series.

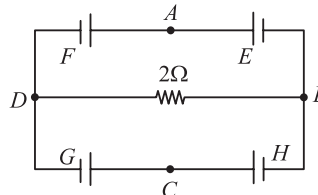


- (a) The main current in the circuit will be very low and almost all current will flow through the ammeter, if resistance of ammeter is much smaller than the resistance in parallel.
 (b) If the devices are ideal, a large current will flow through the ammeter and it will be damaged.
 (c) If the devices are ideal, ammeter will read zero current and voltmeter will read the emf of cell.
 (d) The devices may get damaged if emf of the cell is very high and the meters are non-ideal.

4. A micro-ammeter has a resistance of 100Ω and full scale range of $50 \mu\text{A}$. It can be used as a voltmeter and an ammeter of a higher range provided a resistance is added to it. Pick the correct range and resistance combination(s).

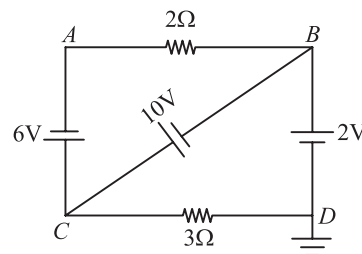
- (a) 50 V range with $10 \text{ k}\Omega$ resistance in series
 (b) 10 V range with $200 \text{ k}\Omega$ resistance in series
 (c) 5 mA range with 1Ω resistance in parallel
 (d) 10 mA range with 1Ω resistance in parallel

5. In the circuit shown E, F, G and H are cells of e.m.f. 2 V, 1 V, 3 V and 1 V respectively and their internal resistances are $2 \Omega, 1 \Omega, 3 \Omega$ and 1Ω respectively.



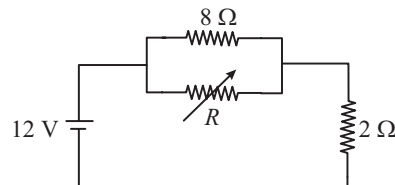
- (a) $V_D - V_B = -2/13 \text{ V}$
 (b) $V_D - V_B = 2/13 \text{ V}$
 (c) $V_G = 21/13 \text{ V}$ = potential difference across G
 (d) $V_H = 19/13 \text{ V}$ = potential difference across H

6. In the shown circuit



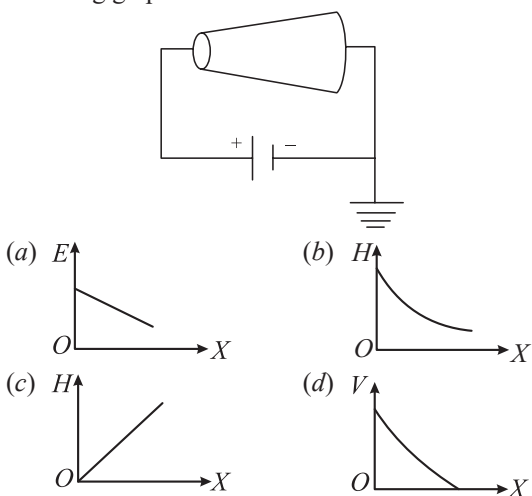
- (a) current passing through 2Ω resistance is 2 A
 (b) current passing through 3Ω resistance is 4 A
 (c) current in the wire between D to earth is zero
 (d) potential of point A is 10 V

7. The value of the resistance R in figure is adjusted such that power dissipated in the 2Ω resistor is maximum. Under this condition

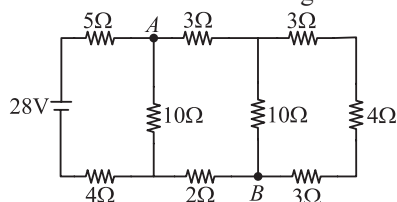


- (a) $R = 0$
 (b) $R = 8 \Omega$
 (c) Power dissipated in the 2Ω resistor is 72 W
 (d) Power dissipated in the 2Ω resistor is 8 W

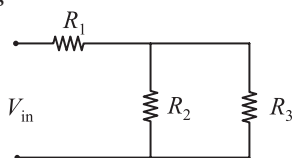
8. A conductor is made of an isotropic material and has shape of a truncated cone. A battery of constant emf is connected across it in figure. If at a section at a distance X from left the electric field intensity, potential and the rate of generation of heat per unit length are E , V and H respectively, which of the following graphs is/are correct?



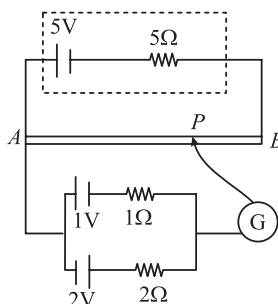
9. Consider the circuit shown in the figure.



- (a) The current in the 5Ω resistor is 2 A.
 (b) The current in the 5Ω resistor is 1 A.
 (c) The potential difference $V_A - V_B$ is 8 V.
 (d) The potential difference $V_A - V_B$ is -5 V.
10. For ensuring dissipation of same energy in all three resistors (R_1, R_2, R_3) connected as shown in figure, their values must be related as



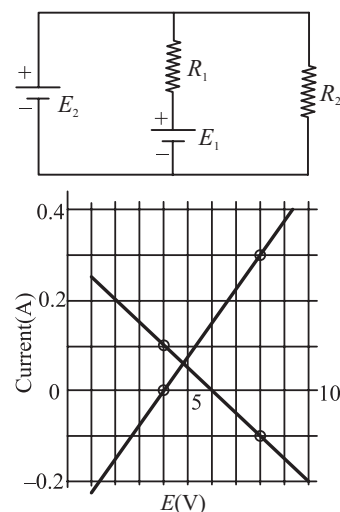
- (a) $R_1 = R_2 = R_3$ (b) $R_2 = R_3$ and $R_1 = 4R_2$
 (c) $R_2 = R_3$ and $R_1 = \frac{1}{4}R_2$ (d) $R_1 = R_2 + R_3$
11. A battery of emf $\epsilon_0 = 5\text{ V}$ and internal resistance 5Ω is connected across a long uniform wire AB of length 1 m and resistance per unit length $5\Omega\text{ m}^{-1}$. Two cells of $\epsilon_1 = 1\text{ V}$ and $\epsilon_2 = 2\text{ V}$ are connected as shown in the figure.



- (a) The null point is at A
 (b) If the jockey is touched to point B the current in the galvanometer will be going towards B
 (c) When jockey is connected to point A no current is flowing through 1V battery
 (d) The null point is at distance of $8/15\text{ m}$ from A

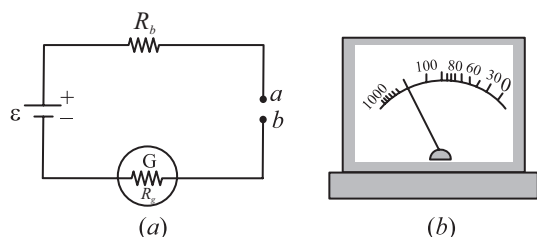
COMPREHENSION BASED QUESTIONS

Comprehension (Q. 12 to 14): In the circuit given below, both batteries are ideal. Emf E_1 of battery 1 has a fixed value, but emf E_2 of battery 2 can be varied between 1.0 V and 10.0 V. The graph gives the currents through the two batteries as a function of E_2 , but are not marked as which plot corresponds to which battery. But for both plots, current is assumed to be negative when the direction of the current through the battery is opposite the direction of battery's emf. (direction of emf is from negative to positive)



12. The value of emf E_1 is
 (a) 8 V (b) 6 V (c) 4 V (d) 2 V
13. The resistance R_1 has value
 (a) 10Ω (b) 20Ω (c) 30Ω (d) 40Ω
14. The resistance R_2 is equal to
 (a) 10Ω (b) 20Ω (c) 30Ω (d) 40Ω

Comprehension (Q. 15 to 17): A simple ohmmeter consists of a battery connected in series with a galvanometer and a resistor, as shown in figure. The resistance R_b is chosen such that when the terminals a and b are shorted (put in electrical contact, with negligible resistance between them), the current through the galvanometer gives a full-scale deflection. Thus, a full scale deflection indicates no resistance between terminals a and b . A zero deflection indicates an infinite resistance between the terminals. When the terminals are connected across an unknown resistance R , the current through the galvanometer depends on R , so the scale can be calibrated to give a direct reading of R , as shown in figure. Because an ohmmeter sends a current through the resistance to be measured, some caution must be exercised when using this instrument. For example, you would not want to try to measure the resistance of sensitive ammeter with an ohmmeter, because the current provided by the battery in the ohmmeter would probably damage the ammeter. Let us use a galvanometer with resistance of 20Ω and maximum current of 10 mA, $\epsilon = 1\text{ V}$. For a current more than 10 mA, galvanometer would be damaged.



15. The scale is non linear because
 (a) The resistance changes with change in temperature.
 (b) Of internal resistance of the battery.
 (c) The current in circuit decreases with increase in resistance.
 (d) Of internal resistance of galvanometer.
16. The minimum resistance, R_b , required for rest of the circuit (other than galvanometer) is
 (a) 20Ω
 (b) 30Ω
 (c) 80Ω
 (d) 100Ω
17. The reading in galvanometer when resistance connected between a and b is 100Ω is
 (a) 5 mA (b) 3.3 mA (c) 2 mA (d) 8 mA

MATCH THE COLUMN TYPE QUESTIONS

18. The following table gives the lengths of four copper rods at the same temperature, their diameters, and the potential differences between their ends.

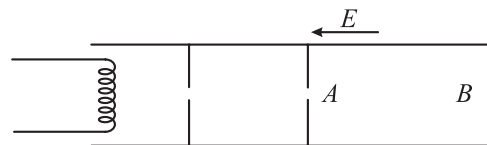
Rod	Length	Diameter	Potential Difference
1	L	$3d$	V
2	$2L$	d	$3V$
3	$3L$	$2d$	$2V$
4	$3L$	d	V

Correctly match the physical quantities mentioned in the Column-I with the rods as marked in Column-II.

Column-I		Column-II	
A.	Greatest drift speed of the electrons	p.	Rod 1
B.	Greatest current	q.	Rod 2
C.	Greatest rate of thermal energy produced	r.	Rod 3
D.	Greatest electric field	s.	Rod 4

- (a) A-(p); B-(q); C-(p); D-(r)
 (b) A-(q); B-(p); C-(p); D-(q)
 (c) A-(r); B-(s); C-(p); D-(q)
 (d) A-(q); B-(r); C-(p); D-(s)

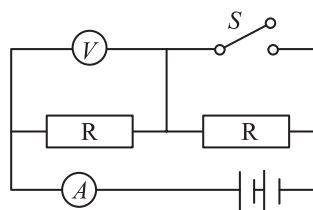
19. Electrons are emitted by a hot filament and are accelerated by an electric field as shown in figure. The two stops at the left ensure that the electron beam has a uniform cross-section. Match the entries of Column-I with Column-II as electrons move from A to B .



Column-I		Column-II	
A.	Speed of an electron	p.	Increases
B.	Number of free electrons per unit volume	q.	Decreases
C.	Current density	r.	Remains same
D.	Electric potential	s.	Any of the above is possible

- (a) A-(p); B-(q); C-(r); D-(p)
 (b) A-(q); B-(p); C-(p); D-(q)
 (c) A-(p, q); B-(s); C-(r); D-(q)
 (d) A-(s); B-(q); C-(p); D-(r)

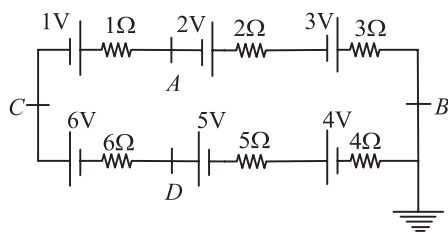
20. The diagram shows a circuit with two identical resistors. The battery has a negligible internal resistance. What will the effect on the ammeter and the voltmeter be if the switch S is closed?



Column-I		Column-II	
A.	Ammeter reading	p.	Increases
B.	Voltmeter reading	q.	Decreases
C.	Equivalent resistance of circuit	r.	Does not change
D.	Power dissipated across R in right branch	s.	Becomes zero

- (a) A-(s, p); B-(q); C-(p); D-(q)
 (b) A-(p); B-(p); C-(p); D-(q)
 (c) A-(p); B-(p); C-(q); D-(s, q)
 (d) A-(s); B-(q, r); C-(p); D-(r)

21. Six batteries of increasing emfs and increasing internal resistances are as shown in figure. Match the following:



Column-I		Column-II	
A.	Potential of point A	p.	Zero
B.	Potential of point B	q.	2 V
C.	Potential of point C	r.	4 V
D.	Potential of point D	s.	6 V

- (a) A-(s); B-(q); C-(r); D-(q)
 (b) A-(p); B-(q); C-(p); D-(q)
 (c) A-(p); B-(p); C-(p); D-(p)
 (d) A-(p); B-(q); C-(p); D-(p)
22. Three bulbs A, B and C are having rated powers P_A , P_B and P_C respectively. Each bulb is designed to operate at rated voltage V . It is given that $P_A > P_B > P_C$. In Column-I the three bulbs are arranged in different configurations, while in Column-II the information about intensities of bulbs is mentioned. Match the Column-I with Column-II. Neglect the variation in resistance due to change in temperature.

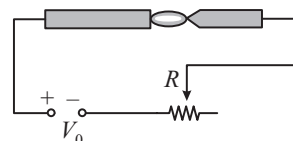
Column-I		Column-II	
A.		p.	A is having the minimum intensity
B.		q.	C is glowing with maximum brightness
C.		r.	B is glowing with minimum brightness
D.		s.	A is glowing with maximum brightness

- (a) A-(s); B-(q); C-(r); D-(p)
 (b) A-(p); B-(q); C-(s); D-(p)
 (c) A-(p); B-(p); C-(s); D-(r)
 (d) A-(p); B-(q); C-(p); D-(p)

NUMERICAL TYPE QUESTIONS

Answer should be rounded off upto two decimal places

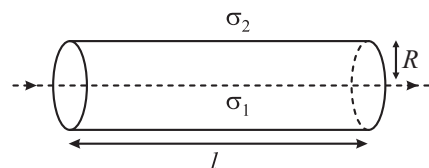
23. The circuit of a carbon arc lamp is shown in the figure (to ignite the lamp the carbon electrodes are touched together, then they are slowly drawn apart.) The voltage across the gap between the carbon rods as a function of the current can be given by the following formula: $V = A + B/I$, where A and B are constants.



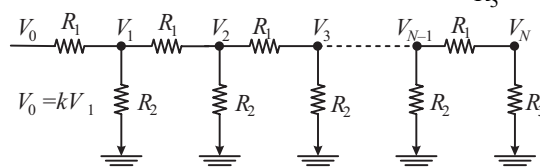
If the voltage of the battery V_0 can be considered constant what is the maximum resistance (in Ω) of the rheostat $R_{\max}$ that should be connected in series to ignite the lamp?

Data: $A = 40$ V, $B = 20$ W, $V_0 = 100$ V.

24. A long conductor of circular cross-section has radius $R = 0.5$ cm and length $l = 5$ cm as shown in the figure. The conductivity of the material near the axis is $\sigma_1 = 3 \times 10^3$ S/cm and increases linearly with the distance from axis and becomes $\sigma_2 = 5 \times 10^3$ S/cm near the surface. Find the resistance (in $\text{m}\Omega$) of the conductor if the current enters from the one end and leaves from the other end.

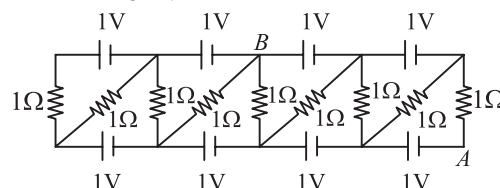


25. A network of resistances is constructed with R_1 and R_2 as shown in the figure. The potential at the points 1, 2, 3, ... N are $V_1, V_2, V_3, \dots, V_N$ respectively, each having a potential $k = 2$ times smaller than previous one. Find $\frac{R_2}{R_3}$.



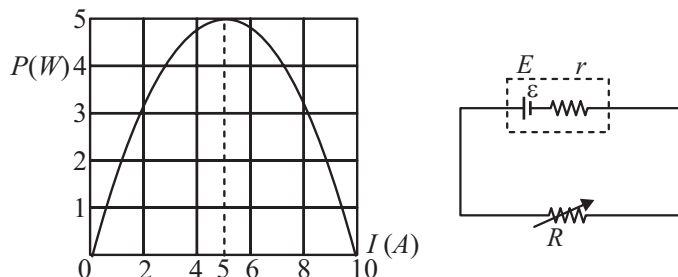
INTEGER TYPE QUESTIONS

26. The potential difference between A and B is V volts. Find the value of $18 \times V$.



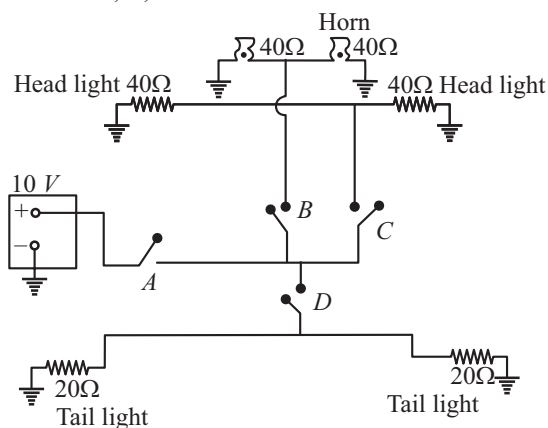
27. A total charge $Q_0 = 2C$ flows across a resistor $R = \frac{8}{\pi} \Omega$ during a time interval T in such a way that the current v/s time graph for 0 s to T s is like the loop of a sin curve in the range 0 radian to π radians. What will be the total heat generated (in mJ) in the resistor?

28. Figure shows the net power dissipated in R versus the current in a simple circuit shown.



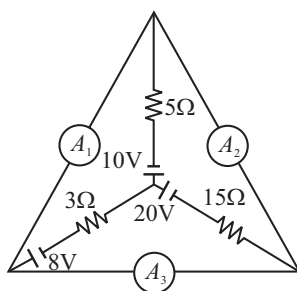
The power dissipated at $I = 2$ A is P watts. Find $10 P$.

29. Figure shows an automobile circuit. How much power (in watt) is dissipated by the automobile circuit when switches A , B , C and D are all closed.

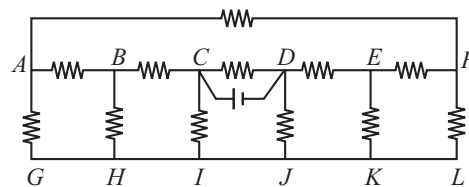


SINGLE CORRECT TYPE QUESTIONS

30. In the given circuit the ammeter A_1 and A_2 are ideal and the ammeter A_3 has a resistance of $1.9 \times 10^{-3} \Omega$. Find the sum of readings of all three meters.

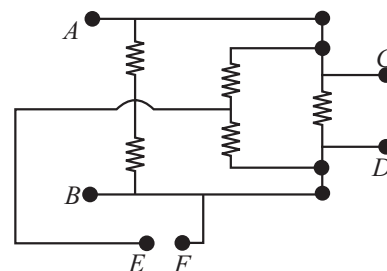


- (a) 11.6 A (b) 4.14 A (c) 1.26 A (d) 3.04 A
31. In the figure shown resistance of each resistor is same. If current in AG is 5 A, the current in CD is



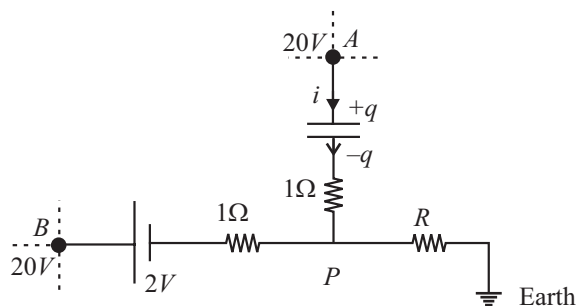
- (a) 5 A (b) 1 A
(c) 10 A (d) zero

32. Five resistors of same resistance are connected as shown in the figure. If resistances across AB , CD and EF are R_{AB} , R_{CD} and R_{EF} respectively, then



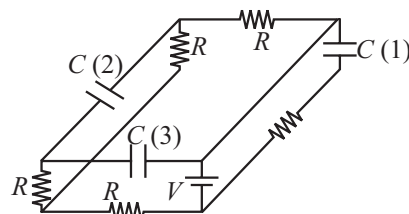
- (a) $R_{AB} = R_{CD} = R_{EF}$ (b) $R_{AB} > R_{CD} > R_{EF}$
(c) $R_{AB} = R_{CD} < R_{EF}$ (d) $R_{AB} = R_{CD} > R_{EF}$

33. The network shown in the figure is part of some bigger circuit. Charge on capacitor ($C = 1F$) at any time t (in s) is $q = 3(1 - e^{-t})$ in coulombs. If at $t = 0$ the potentials of points A and B are 20V each, then find the value of R .



- (a) 4 Ω (b) 17 Ω
(c) 4.25 Ω (d) 6.75 Ω

34. A battery of voltage $V = 10$ volt, three capacitors each of capacitance $C = 3 \mu F$ and five resistors each of resistance $R = 1 \Omega$ are arranged as shown in the figure. Arrangement is left for long time so that steady state is reached. The sum of charges on capacitors 1, 2 and 3 is

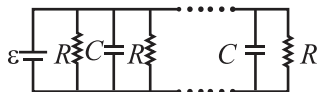


- (a) 30 μF (b) 40 μF
(c) 20 μF (d) 60 μF

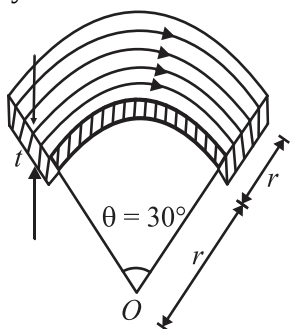
Exercise-2

1. Two non-ideal batteries are connected in parallel. Consider the following statements:
- The equivalent emf is smaller than either of the two emfs
 - The equivalent internal resistance is smaller than either of the two internal resistances
- (A) Both (a) and (b) is wrong
 (B) (a) is correct and (b) is wrong
 (C) (b) is correct but (a) is wrong
 (D) Both (a) and (b) are wrong

2. n resistance each of resistance R are joined with capacitors of capacity C (each) and a battery of emf ε as shown in the figure. In steady state condition the ratio of charge stored in the first and the last capacitor is:

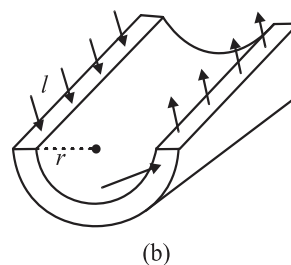
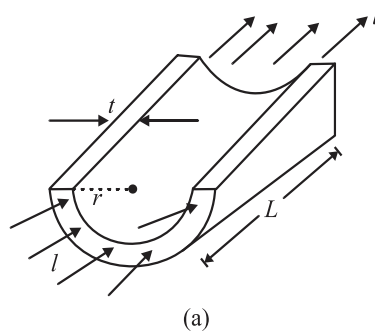


- (A) $n : 1$
 (B) $(n - 1) : R$
 (C) $\frac{n^2 + 1}{n^2 - 1}$
 (D) $1 : 1$
3. A conducting plate of thickness t is in the shape of an arc. Its inner and outer rim form a 30° arc of radius r and $2r$ respectively at point O (see figure). Electric current flows through the strip along circular arcs as indicated in the diagram. Find resistance of the plate if its material has conductivity σ .



- (A) $\frac{\pi}{6\sigma t \ln 2}$ (B) $\frac{\pi}{3\sigma t \ln(2)}$
 (C) $\frac{\pi}{\sigma t \ln(2)}$ (D) $\frac{\pi}{2\sigma \ln(2)}$
4. A conducting open pipe has shape of a half cylinder of length L . Its semicircular cross section has radius

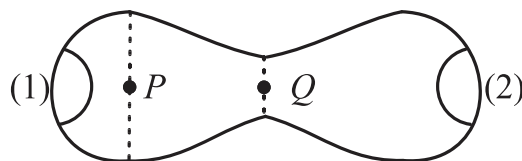
r and thickness of the conducting wall is t ($\ll r$). The resistance of the conductor when the current enters and leaves as shown in figure (a) is R_1 and its resistance is R_2 when the current is as shown in Figure (b). Find $\frac{R_1}{R_2}$.



- (A) $\frac{L^2}{\pi^2 r^2}$ (B) $\frac{\pi^2 L^2}{r^2}$
 (C) $\frac{r^2}{\pi^2 L^2}$ (D) $\frac{\pi^2 r^2}{L^2}$

Multiple Correct Type Questions

5. A metallic conductor of irregular cross-section is as shown in the figure. A constant potential difference is applied across the ends (1) and (2). Then



- (A) The current at the cross-section P equals the current at the cross-section Q
 (B) The electric field intensity at P is less than that at Q
 (C) The rate of heat generate per unit time at Q is greater than that at P
 (D) The number of electrons crossing per unit area of cross-section at P is less than that at Q .

6. The cross-section area and length of a cylindrical conductor, are A and l , respectively. The specific conductivity varies as $\sigma(x) = \sigma_0 \frac{l}{\sqrt{x}}$, where x is the distance along the axis of the cylinder from one of its ends
- (A) The resistance of the system along the cylindrical axis is $\frac{2\sqrt{l}}{3A\sigma_0}$.
- (B) The resistance of the system along the cylindrical axis is $\frac{3\sqrt{l}}{2A\sigma_0}$.
- (C) The potential drop along the cylinder is V_0 . Then electrical field at each point in the cylinder is given by $\frac{2V_0\sqrt{x}}{3l^{3/2}}$
- (D) The potential drop along the cylinder is V_0 . Then electrical field at each point in the cylinder is given by $\frac{3V_0\sqrt{x}}{2l^{3/2}}$

7. A voltmeter of resistance R_V and an ammeter of resistance R_A are connected in series across a battery of e.m.f ε and of negligible internal resistance R is connected in parallel to voltmeter, reading of ammeter increases to three times while that of voltmeter reduces to one-third. Then
- (A) $R_V = 6R$
- (B) $R_V = 8R$
- (C) $R_A = \frac{8R}{3}$
- (D) $R_A = 3R$

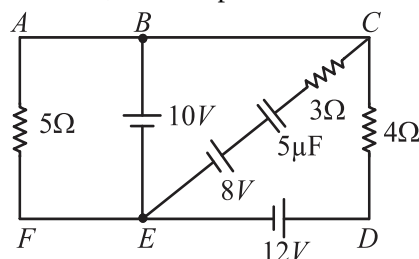
Matrix Match Type Questions

8. Three bulbs A , B and C are having rated powers P_A , P_B and P_C respectively, each bulb is designed to operate at rated voltage V . It is given that $P_A > P_B > P_C$. In list I the three bulbs are arranged in different configurations, while in list-II the information about intensities of bulbs are mentioned. Match the list-I with list-II. Neglect the variation in resistance due to change in temperature.

List-I		List-II	
I		P	A is having the minimum intensity
II		Q	C is glowing with maximum brightness
III		R	B is glowing with minimum brightness
IV		S	A is glowing with maximum brightness

- | | I | II | III | IV |
|-----|------|---------|-----|----|
| (A) | P, Q | Q, R | S | P |
| (B) | Q, P | P, Q, R | S | P |
| (C) | P, Q | P, Q, R | R | R |
| (D) | Q, P | Q, R, S | P | Q |

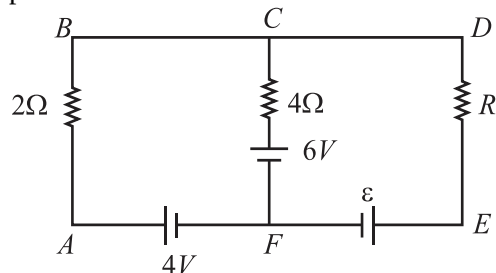
9. A network consisting of three resistors, three batteries, and a capacitor is shown in Fig.



List-I		List-II	
I	Current In branch EB is	P	$10\mu C$
II	Current In branch CB is	Q	$0.5 A$
III	Current In branch ED is	R	$1.5 A$
IV	Change on Capacitor is	S	$5 \mu Cs$

- | | I | II | III | IV |
|-----|---|----|-----|----|
| (A) | Q | R | Q | P |
| (B) | R | Q | Q | P |
| (C) | R | Q | R | S |
| (D) | Q | Q | R | S |

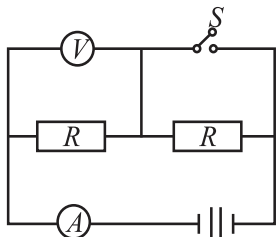
10. A circuit is shown in Fig. R is a non-zero variable with finite resistance. ε is some unknown emf with polarities as shown. Match the columns.



List-I		List-II	
I	Current passing through 4Ω resistance can be zero	P	Possible if $\varepsilon = 6V$
II	Current passing through 4Ω resistance can be from F to C direction	Q	Possible if $\varepsilon > 6V$
III	Current passing through 4Ω resistance can be from C to F direction	R	Possible if $\varepsilon < 6V$
IV	Current passing through 2Ω resistance will be from B to A direction	S	Possible for any value of ε from zero to infinite.

- | | | | | |
|-----|----------|-----------|------------|------------|
| | I | II | III | IV |
| (A) | Q | P, Q, R | Q | P, Q, R, S |
| (B) | R | Q, R, S | P | R, S |
| (C) | S | P, Q, R | Q | P, Q |
| (D) | Q | P, R, S | Q | P, R, S |

11. The diagram shows a circuit with two identical resistors. The battery has a negligible internal resistance. What will the effect on the ammeter and voltmeter be if the switch S is closed?



List-I		List-II	
I	Ammeter reading	P	Increases
II	Voltmeter reading	Q	Decreases
III	Equivalent resistance of circuit	R	Does not change
IV	Power dissipated across R in right branch	S	Becomes zero

- | | | | | |
|-----|----------|-----------|------------|-----------|
| | I | II | III | IV |
| (A) | Q | Q | P | P |
| (B) | P | Q | Q | Q |
| (C) | Q | P | P | Q, S |
| (D) | P | P | Q | Q, S |

12. List-I has four circuits each having an ammeter. List-II has four values of current in the ammeter. The ammeter has zero resistance. The voltmeter, in (B) has infinite resistance and a reading $8V$. The resistance R has not been specified. Match the circuit with its current ammeter reading.

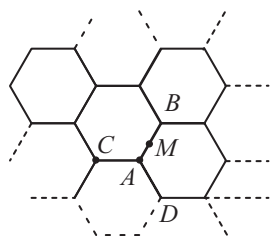
List-I		List-II	
I		P	0
II		Q	2 ampere
III		R	4 ampere
IV		S	5 ampere

- | | | | | |
|-----|----------|-----------|------------|-----------|
| | I | II | III | IV |
| (A) | Q | R | S | Q |
| (B) | S | R | S | P |
| (C) | S | R | P | Q |
| (D) | Q | P | R | S |

Exercise -3

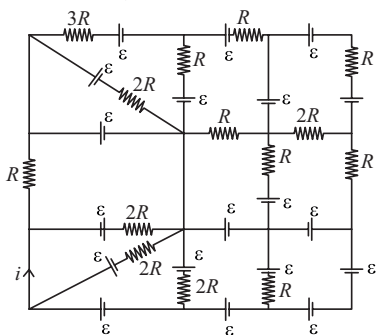
SINGLE CORRECT TYPE QUESTIONS

1. In an infinite grid of hexagonal cells, resistance of each branch is R . M is the mid-point of branch AB . If a battery of emf E is connected between A and M , find the potential difference between A and C .



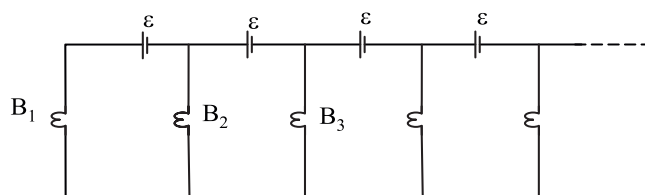
- (a) $\frac{E}{3}$ (b) $\frac{E}{4}$ (c) $\frac{E}{5}$ (d) $\frac{E}{6}$

2. In the circuit shown, current i is



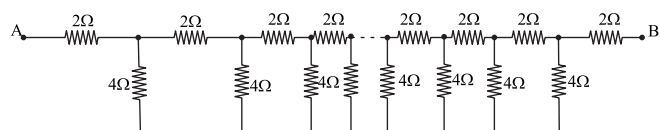
- (a) $\frac{3E}{4R}$ (b) $\frac{3E}{2R}$ (c) $\frac{E}{4R}$ (d) $\frac{5E}{2R}$

3. A infinite network is composed of identical batteries and identical bulbs of power rating 100W, 220V. Power radiated by the leftmost lamp is 100W and power radiated by any bulb is $\frac{1}{\beta}$ times the power radiated by the adjacent lamp to the left. Here $\beta = 1.44$. If the batteries are not ideal then emf of each the battery is:



- (a) 88 V (b) 66 V
(c) 44 V (d) 33 V

4. The equivalent resistance of the infinite ladder network between terminals A and B shown in the figure below is:

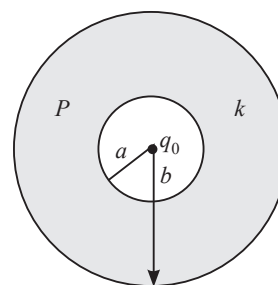


- (a) 2Ω (b) 4Ω
(c) 6Ω (d) 8Ω

5. There are six identical cylindrical resistors. The curved surfaces of resistors are thermally insulated so that heat transfer can only take place through flat surfaces. When one of them is connected across an ideal battery, its steady state temperature becomes 40°C . If other two resistors are connected in series and the pack is connected across the same battery the steady state temperature becomes 34°C . Now if the remaining three are connected in series and pack is connected across the same battery, the steady state temperature of this pack will rise to (Assume Newton's law of cooling and neglect thermal expansions and change in resistivity for the temperature range involved)

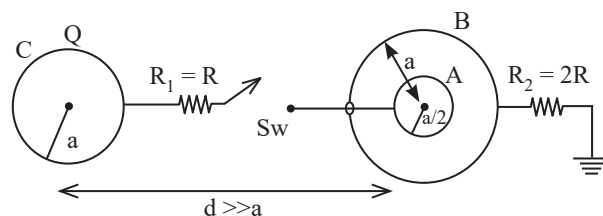
- (a) 28°C (b) 32°C
(c) 30°C (d) 31°C

6. A thick spherical shell of inner radius a and outer radius b is made of material of resistivity ρ and dielectric constant K . Initially the shell is electrically neutral and has no charge accumulation anywhere. If at $t = 0$, a point charge is placed at the center of the shell, find the charge on outer surface of the shell as function of time t .



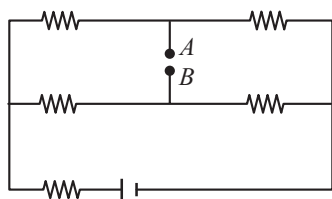
- (a) $q = q_0 \left[1 - e^{-\frac{t}{K\rho\epsilon_0}} \right]$ (b) $q = q_0 \left[1 - e^{-\frac{tb}{aK\rho\epsilon_0}} \right]$
(c) $q = q_0 \left[1 - e^{-\frac{ta}{K\rho\epsilon_0 b}} \right]$ (d) $q = q_0 \frac{a}{b} \left[1 - e^{-\frac{ta}{K\rho b\epsilon_0}} \right]$

7. In the figure shown A and B are two concentric conducting spheres of radius $\frac{a}{2}$ and a respectively. Shell A is connected through resistor R and a key to a conducting ball C of radius a . The shell B is earthed through a resistor $2R$. The distance between the ball and conducting spheres is much greater than a . If the ball is given a charge Q and key is closed then total heat dissipated in the resistor R is

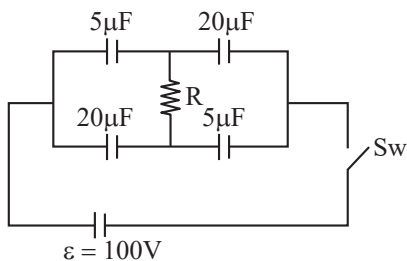


- (a) $\frac{Q^2}{16\pi\epsilon_0 a}$ (b) $\frac{Q^2}{32\pi\epsilon_0 a}$
(c) $\frac{Q^2}{48\pi\epsilon_0 a}$ (d) $\frac{Q^2}{62\pi\epsilon_0 a}$

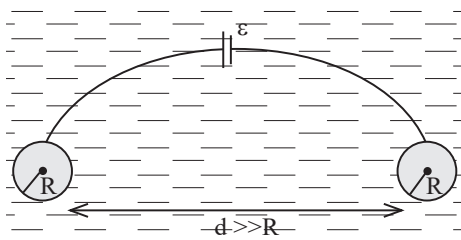
8. The network shown in the figure consists of a battery and five unknown resistors. When an ideal ammeter is connected between terminals A and B its reading is 10 A and when a resistance of 3Ω is connected in series with the ammeter its reading becomes 4A. Now if 3Ω resistance is replaced by 8Ω resistor, the reading of the ammeter will be



- (a) 3.5 A (b) 3 A
(c) 2.5 A (d) 2 A
9. In the circuit shown below the switch is open and all the capacitors are uncharged. Internal resistance of the battery is negligible. If the switch is closed find the total energy dissipated in the resistor R till the steady state is reached.



- (a) 12.5 mJ (b) 22.5 mJ
(c) 26.5 mJ (d) 32.5 mJ
10. Two identical metal spheres of radius R are immersed in a homogeneous medium. Radius of each sphere is negligible as compared to the distance between the spheres and the medium extends upto very large region around the spheres. When one sphere is connected with positive terminal of a battery and other with negative terminal of the battery using thin insulated conductor, current $i_0 = 5A$ flows through the battery. What current will flow to the battery if first sphere is replaced by a sphere of radius $2R$ and second sphere is replaced by a sphere radius $\frac{R}{2}$?



- (a) 3 A (b) 3.5 A
(c) 4 A (d) 4.5 A

Exercise -4

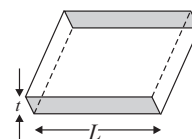
MICROSCOPIC ANALYSIS, DRIFT VELOCITY, OHM'S LAW, CONDUCTIVITY

Single Correct

1. A current of 5 A passes through a copper conductor (resistivity $= 1.7 \times 10^{-8} \Omega\text{-m}$) of radius of cross-section 5 mm. Find the mobility of the charges, if their drift velocity is $1.1 \times 10^{-3} \text{ m/s}$. (JEE Adv. 2015)

- (a) $1.5 \text{ m}^2/\text{V-s}$
(b) $1.3 \text{ m}^2/\text{V-s}$
(c) $1.0 \text{ m}^2/\text{V-s}$
(d) $1.8 \text{ m}^2/\text{V-s}$

2. Consider a thin square sheet of side L and thickness t , made of a material of resistivity ρ . The resistance between two opposite faces, shown by the shaded areas in the figure is (IIT-JEE 2010)



- (a) Directly proportional to L
(b) Directly proportional to t
(c) Independent of L
(d) Independent of t
3. A steady current flows in a metallic conductor of non-uniform cross-section. The quantity/quantities constant along the length of the conductor is/are (IIT-JEE 1997)
- (a) Current, electric field and drift speed
(b) Drift speed only
(c) Current and drift speed
(d) Current only
4. A piece of copper and another of germanium are cooled from room temperature to 80K. The resistance of (IIT-JEE 1988)
- (a) Each of them increases
(b) Each of them decreases
(c) Copper increases and germanium decreases
(d) Copper decreases and germanium increases

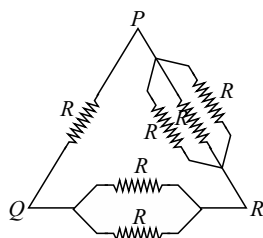
True/False

5. Electrons in a conductor have no motion in the absence of a potential difference across it. (IIT-JEE 1982)

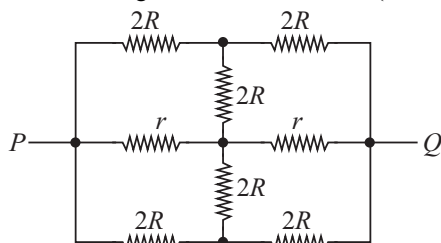
SERIES AND PARALLEL COMBINATION OF RESISTANCE

Single Correct

6. Six equal resistances are connected between points P , Q and R as shown in the figure. Then, the net resistance will be maximum between (IIT-JEE 2004)



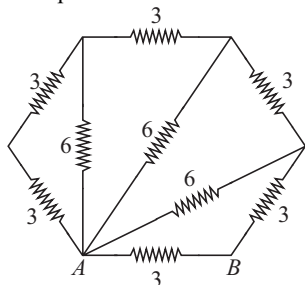
- (a) P and Q (b) Q and R
(c) P and R (d) Any two points
7. The effective resistance between points P and Q of the electrical circuit shown in the figure is (IIT-JEE 2002)



- (a) $\frac{2Rr}{R+r}$ (b) $\frac{8R(R+r)}{3R+r}$
(c) $2r+4R$ (d) $\frac{5R}{2}+2r$

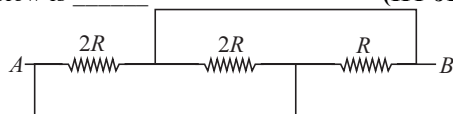
Numerical Types/Integer Types

8. When two identical batteries of internal resistance 1Ω each are connected in series across a resistor R , the rate of heat produced in R is J_1 . When the same batteries are connected in parallel across R , the rate is J_2 . If $J_1 = 2.25J_2$ then the value of R in Ω is (IIT-JEE 2010)
9. All resistances in the diagram are in ohm. Find the effective resistance between the points A and B . (IIT-JEE 1980)

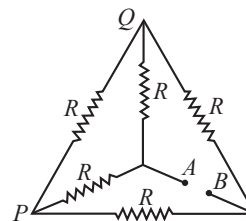


Subjective

10. The equivalent resistance between points A and B of the circuit given below is (IIT-JEE 1997)



11. If each of the resistances in the network shown in the figure is R , what is the resistance between the terminals A and B ? (IIT-JEE 1978)



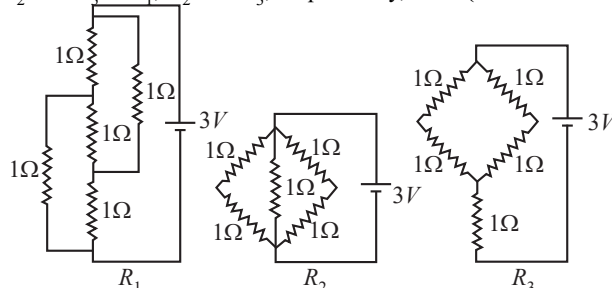
POWER AND HEAT DISSIPATED IN A CIRCUIT

Single Correct

12. Incandescent bulbs are designed by keeping in mind that the resistance of their filament increases with the increase in temperature. If at room temperature, $100W$, $60W$ and $40W$ bulbs have filament resistances R_{100} , R_{60} , and R_{40} , respectively, the relation between these resistances is (IIT-JEE 2010)

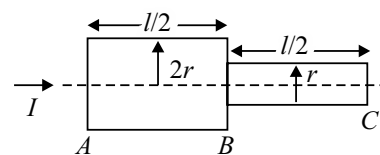
- (a) $\frac{1}{R_{100}} = \frac{1}{R_{40}} + \frac{1}{R_{60}}$ (b) $R_{100} = R_{40} + R_{60}$
(c) $R_{100} > R_{60} > R_{40}$ (d) $\frac{1}{R_{100}} > \frac{1}{R_{60}} > \frac{1}{R_{40}}$

13. Figure shows three resistor configurations R_1 , R_2 and R_3 connected to $3V$ battery. If the power dissipated by the configuration R_1 , R_2 and R_3 is P_1 , P_2 and P_3 , respectively, then (IIT-JEE 2008)



- (a) $P_1 > P_2 > P_3$ (b) $P_1 > P_3 > P_2$
(c) $P_2 > P_1 > P_3$ (d) $P_3 > P_2 > P_1$

14. Two bars of radius r and $2r$ are kept in contact as shown. An electric current I is passed through the bars. Which one of the following is correct? (IIT-JEE 2006)

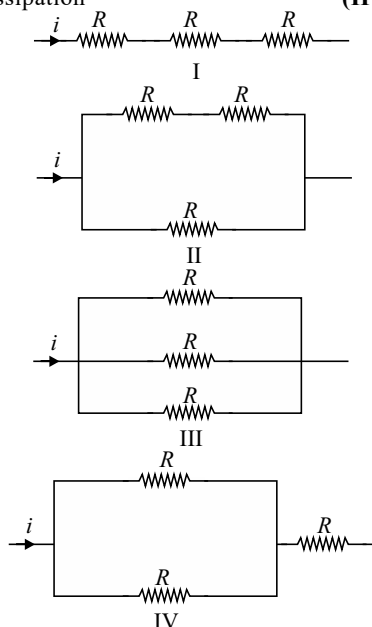


- (a) Heat produced in bar BC is 4 times the heat produced in bar AB
(b) Electric field in both halves is equal
(c) Current density across AB is double that of across BC
(d) Potential difference across AB is 4 times that of across BC

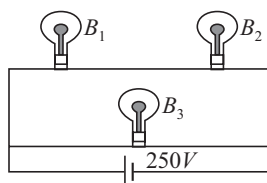
15. A rigid container with thermally insulated walls contains a coil of resistance 100Ω , carrying current $1A$. Change in internal energy after 5 min will be (JEE Adv. 2005)

- (a) zero (b) 10 kJ
(c) 20 kJ (d) 30 kJ

16. The three resistances of equal value are arranged in the different combinations shown below. Among them in increasing order of power dissipation (IIT-JEE 2003)



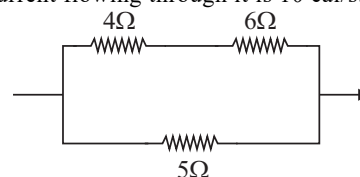
- (a) $\text{III} < \text{II} < \text{IV} < \text{I}$
 (b) $\text{II} < \text{III} < \text{IV} < \text{I}$
 (c) $\text{I} < \text{IV} < \text{III} < \text{II}$
 (d) $\text{I} < \text{III} < \text{II} < \text{IV}$
17. A 100W bulb B_1 , and two 60W bulbs B_2 and B_3 , are connected to a 250V source, as shown in the figure. Now W_1 , W_2 and W_3 are the output powers of the bulbs B_1 , B_2 and B_3 respectively. Then, (IIT-JEE 2002)



- (a) $W_1 > W_2 = W_3$
 (b) $W_1 > W_2 > W_3$
 (c) $W_1 < W_2 = W_3$
 (d) $W_1 < W_2 < W_3$
18. A wire of length L and 3 identical cells of negligible internal resistances are connected in series. Due to the current, the temperature of the wire is raised by ΔT in a time t . A number N of similar cells is now connected in series with a wire of the same material and cross-section but of length $2L$. The temperature of the wire is raised by the same amount ΔT in the same time. The value of N is (IIT-JEE 2001)
- (a) 4 (b) 6 (c) 8 (d) 10
19. An electric bulb rated for 500W at 100V is used in a circuit having a 200V supply. The resistance R that must be put in series with the bulb, so that the bulb delivers 500W is _____ Ω . (IIT-JEE 1987)

- (a) 5Ω (b) 20Ω
 (c) 10Ω (d) 30Ω

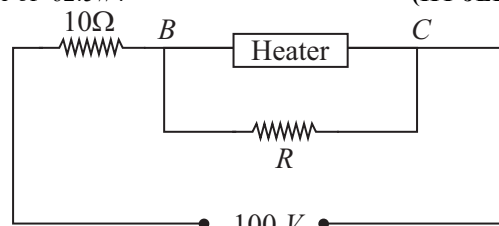
20. In the circuit shown in figure the heat produced in the 5Ω resistor due to the current flowing through it is 10 cal/s .



The heat generated in the 4Ω resistor is (IIT-JEE 1981)

- (a) 1 cal/s
 (b) 2 cal/s
 (c) 3 cal/s
 (d) 4 cal/s

21. A heater is designed to operate with a power of 1000W in a 100V line. It is connected in combination with a resistance of 10Ω and a resistance R , to a 100V mains as shown in the figure. What will be the value of R so that the heater operates with a power of 62.5W ? (IIT-JEE 1981)

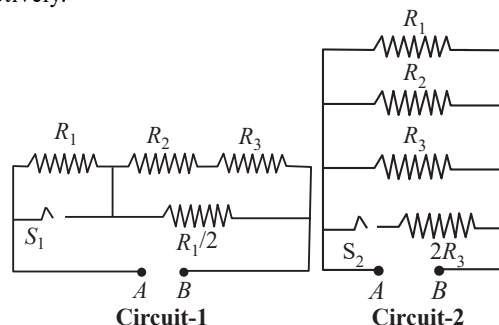


- (a) 15Ω (b) 10Ω
 (c) 5Ω (d) 25Ω

Multiple Correct

22. In Circuit-1 and Circuit-2 shown in the figures, $R_1 = 1\Omega$, $R_2 = 2\Omega$ and $R_3 = 3\Omega$. P_1 and P_2 are the power dissipations in Circuit-1 and Circuit-2 when the switches S_1 and S_2 are in open conditions, respectively.

Q_1 and Q_2 are the power dissipations in Circuit-1 and Circuit-2 when the switches S_1 and S_2 are in closed conditions, respectively.



Which of the following statements is/are correct?

C-14.39 W-23.42 UA-36.51 (JEE Adv. 2022)

- (a) When a voltage source of 6V is connected across A and B in both circuits, $P_1 < P_2$
 (b) When a constant current source of 2A is connected across A and B in both circuits, $P_1 > P_2$
 (c) When a voltage source of 6V is connected across A and B in Circuit-1, $Q_1 > P_1$
 (d) When a constant current source of 2A is connected across A and B in both circuits, $Q_2 < Q_1$

23. An incandescent bulb has a thin filament of tungsten that is heated to high temperature by passing an electric current. The hot filament emits black-body radiation. The filament is observed to break up at random locations after a sufficiently long time of operation due to non-uniform evaporation of tungsten from the filament. If the bulb is powered at constant voltage, which of the following statement(s) is/are true?

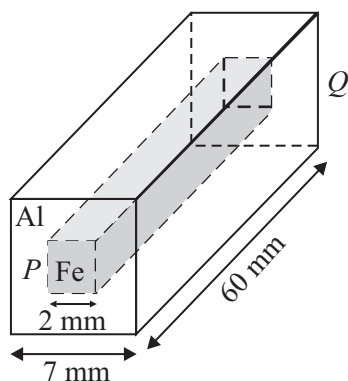
C-3.16 W-30.4 UA-55.22 PC-11.23 (JEE Adv. 2016)

- (a) The temperature distribution over the filament is uniform.
- (b) The resistance over small sections of the filament decreases with time.
- (c) The filament emits more light at higher band of frequencies before it breaks up.
- (d) The filament consumes less electrical power towards the end of the life of the bulb.

PROBLEM SOLVING USING OHM'S LAW

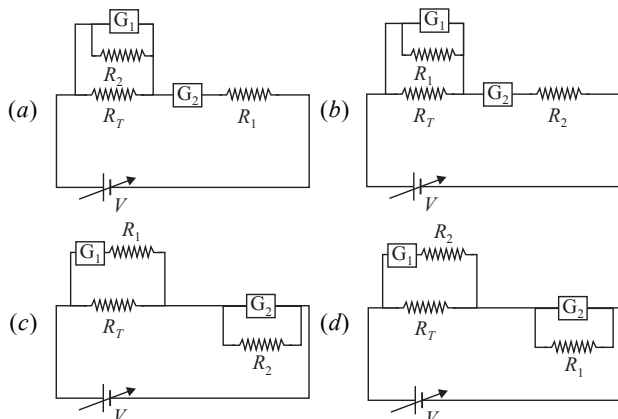
Single Correct

24. In an aluminium (Al) bar of square cross section, a square hole is drilled and is filled with iron (Fe) as shown in the figure. The electrical resistivities of Al and Fe are $27 \times 10^{-8} \Omega\text{m}$ and $10 \times 10^{-7} \Omega\text{m}$, respectively. The electrical resistance between the two faces P and Q of the composite bar is (JEE Adv. 2015)



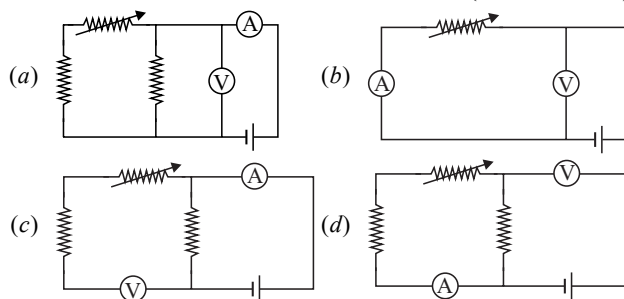
- (a) $\frac{2475}{64} \mu\Omega$ (b) $\frac{1875}{64} \mu\Omega$ (c) $\frac{1875}{49} \mu\Omega$ (d) $\frac{2475}{132} \mu\Omega$

25. To verify Ohm's law, a student is provided with a test resistor, a high resistance, a small resistance, two identical galvanometers and a variable voltage source. The correct circuit to carry out the experiment is (IIT-JEE 2010)



26. Which of the following set-up can be used to verify ohm's law?

(IIT-JEE 2003)



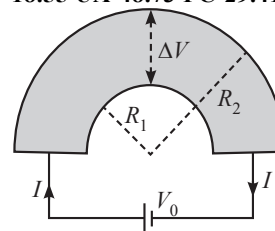
27. A copper wire is stretched to make it 0.1% longer. What is the percentage change in its resistance? (IIT-JEE 1997)

- (a) 0.2% (b) 0.5% (c) 0.1% (d) 0.05%

Multiple Correct

28. Shown in the figure is a semicircular metallic strip that has thickness t and resistivity ρ . Its inner radius is R_1 and outer radius is R_2 . If a voltage V_0 is applied between its two ends, a current I flows in it. In addition, it is observed that a transverse voltage ΔV develops between its inner and outer surfaces due to purely kinetic effects of moving electrons (ignore any role of the magnetic field due to the current). Then (figure is schematic and not drawn to scale)

C-3.31 W-18.55 UA-48.73 PC-29.41 (JEE Adv. 2020)



- (a) $I = \frac{V_0 t}{\pi \rho} \ln \left(\frac{R_2}{R_1} \right)$
- (b) the outer surface is at a higher voltage than the inner surface
- (c) the outer surface is at a lower voltage than the inner surface
- (d) $\Delta V \propto I^2$

29. Read the following statements:

Y : The resistivity of semiconductor decreases with increase of temperature.

Z : In a conducting solid, the rate of collisions between free electrons and ions increases with increase of temperature.

Select the correct statement(s) from the following: (IIT-JEE 1993)

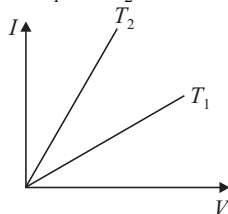
- (a) Y is true but Z is false
- (b) Y is false but Z is true
- (c) Both Y and Z are true
- (d) Y is true and Z is the correct reason for Y

30. When a potential difference is applied across, the current passing through (IIT-JEE 1999)

- (a) an insulator at 0 K is zero
- (b) a semiconductor at 0 K is zero
- (c) a metal at 0 K is finite
- (d) a p-n diode at 300 K is finite, if it is reverse biased

True/False

31. The current-voltage graphs for a given metallic wire at two different temperatures T_1 and T_2 are shown in the figure.



The temperature T_2 is greater than T_1 . (IIT-JEE 1985)

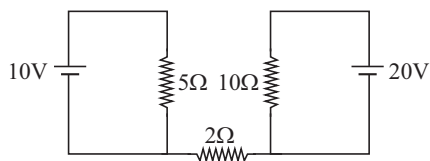
Subjective

32. Draw the circuit for experimental verification of Ohm's law using a source of variable DC voltage, a main resistance of $100\ \Omega$, two galvanometers and two resistances of values $10^6\ \Omega$ and $10^{-3}\ \Omega$ respectively. Clearly show the positions of the voltmeter and the ammeter. (IIT-JEE 2004)
33. A $25\ \text{W}$ and a $100\ \text{W}$ bulb are joined in series and connected to the mains. Which bulb will glow brighter? (IIT-JEE 1979)
34. A copper wire having cross-sectional area of $0.5\ \text{mm}^2$ and a length of $0.1\ \text{m}$ is initially at 25°C and is thermally insulated from the surrounding. If a current of $1.0\ \text{A}$ is set up in this wire,
- find the time in which the wire will start melting. The change of resistance with the temperature of the wire may be neglected.
 - What will this time be, if the length of the wire is doubled?
- Melting point of copper = 1075°C , Specific resistance of copper = $1.6 \times 10^{-8}\ \Omega\text{m}$, Density of copper = $9 \times 10^3\ \text{kg/m}^3$,
Specific heat of copper = $9 \times 10^{-2}\ \text{cal/kg}^\circ\text{C}$ (IIT-JEE 1979)

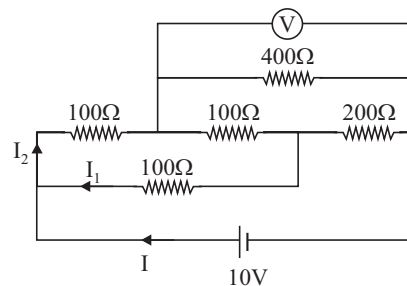
KIRCHHOFF'S LAW AND EQUIVALENT BATTERY

Single Correct

35. Find out the value of current through $2\ \Omega$ resistance for the given circuit. (IIT-JEE 2005)

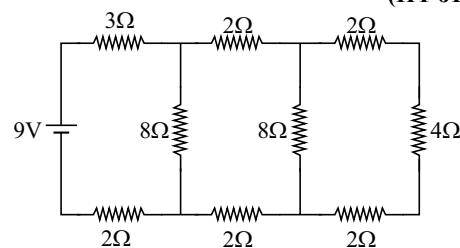


- $5\ \text{A}$
 - $2\ \text{A}$
 - Zero
 - $4\ \text{A}$
36. An electrical circuit is shown in figure. Calculate the potential difference across the resistor of $400\ \Omega$ as will be measured by the voltmeter V of resistance $400\ \Omega$ either by applying Kirchhoff's rules or otherwise. (IIT-JEE 1996)



- $20/7\ \text{V}$
- $50/3\ \text{V}$
- $10/3\ \text{V}$
- $20/3\ \text{V}$

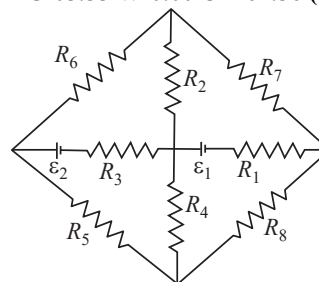
37. In the circuit shown in the figure, the current through (IIT-JEE 1995)



- The $3\ \Omega$ resistor is $0.50\ \text{A}$
- The $3\ \Omega$ resistor is $0.25\ \text{A}$
- The $4\ \Omega$ resistor is $0.50\ \text{A}$
- The $4\ \Omega$ resistor is $0.25\ \text{A}$

Multiple Correct

38. The figure shows a circuit having eight resistances of $1\ \Omega$ each, labelled R_1 to R_8 , and two ideal batteries with voltages $\epsilon_1 = 12\ \text{V}$ and $\epsilon_2 = 6\ \text{V}$. C-05.55 W-0.00 UA-64.56 (JEE Adv. 2022)



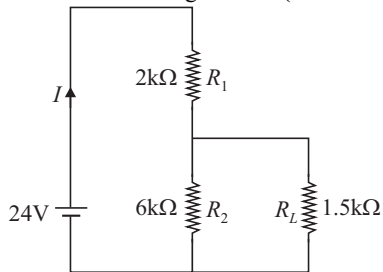
Which of the following statements is/are correct?

- The magnitude of current flowing through R_1 is $7.2\ \text{A}$
 - The magnitude of current flowing through R_2 is $1.2\ \text{A}$
 - The magnitude of current flowing through R_3 is $4.8\ \text{A}$
 - The magnitude of current flowing through R_5 is $2.4\ \text{A}$
39. Two identical moving coil galvanometer have $10\ \Omega$ resistance and full scale deflection at $2\ \mu\text{A}$ current. One of them is converted into a voltmeter of $100\ \text{mV}$ full scale reading and the other into an Ammeter of $1\ \text{mA}$ full scale current using appropriate resistors. These are then used to measure the voltage and current in the Ohm's law experiment with $R = 1000\ \Omega$ resistor by using an ideal cell. Which of the following statements is/are correct?

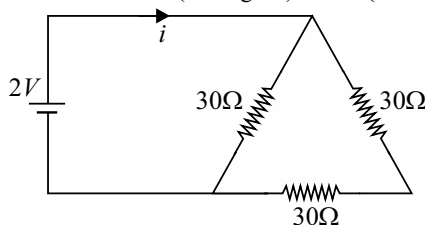
C-7 W-28 UA-44 PC-20 (JEE Adv. 2019)

- The measured value of R will be $978\ \Omega < R < 982\ \Omega$
- The resistance of the Voltmeter will be $100\ \text{k}\Omega$
- The resistance of the Ammeter will be $0.02\ \Omega$ (round off to 2nd decimal place)
- If the ideal cell is replaced by a cell having internal resistance of $5\ \Omega$ then the measured value of R will be more than $1000\ \Omega$

40. For the circuit shown in the figure (IIT-JEE 2009, 2015)



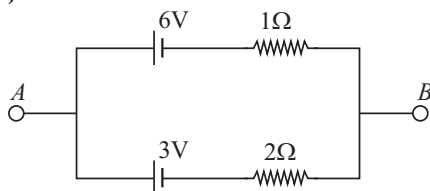
- (a) The current I through the battery is 7.5mA
 (b) The potential difference across R_L is 18V
 (c) Ratio of powers dissipated in R_1 and R_2 is 3
 (d) If R_1 and R_2 are interchanged, magnitude of the power dissipated in R_L will decrease by a factor of 9
41. The current i in the circuit (see figure) is (IIT-JEE 1983)



- (a) $1/45\text{A}$ (b) $1/15\text{A}$ (c) $1/10\text{A}$ (d) $1/5\text{A}$

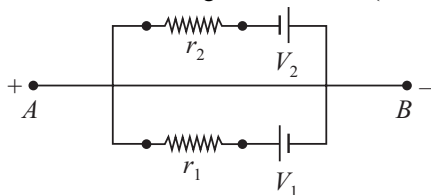
Numerical Types/Integer Types

42. Two batteries of different emfs and different internal resistances are connected as shown. The voltage across AB in volt is (IIT-JEE 2011)

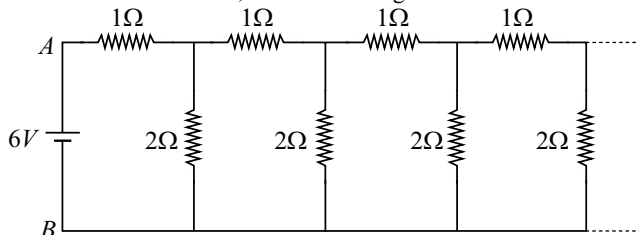


Subjective

43. Find the emf (V) and internal resistance (r) of a single battery which is equivalent to a parallel combination of two batteries of emfs V_1 and V_2 and internal resistances r_1 and r_2 respectively, with polarities as shown in figure. (IIT-JEE 1997)



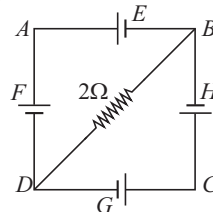
44. An infinite ladder network of resistances is constructed with 1Ω and 2Ω resistances, as shown in figure.



The 6V battery between A and B has negligible internal resistance. (IIT-JEE 1987)

- (i) Show that the effective resistance between A and B is 2Ω .
 (ii) What is the current that passes through the 2Ω resistance nearest to the battery?

45. In the circuit shown in figure E, F, G, H are cells of emf 2, 1, 3 and 1V respectively, and their internal resistances are 2, 1, 3 and 1Ω respectively.

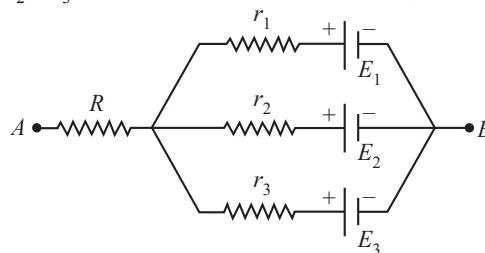


Calculate

- (i) the potential difference between B and D and
 (ii) the potential difference across the terminals of each cells G and H . (IIT-JEE 1984)

46. Two resistors, 400Ω and 800Ω are connected in series with a 6V battery. It is desired to measure the current in the circuit. An ammeter of 10Ω resistance is used for this purpose. What will be the reading in the ammeter? Similarly, if a voltmeter of 1000Ω resistance is used to measure the potential difference across the 400Ω resistor, what will be the reading in the voltmeter? (IIT-JEE 1982)

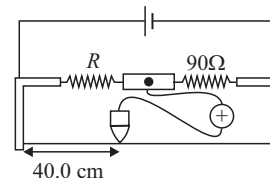
47. In the circuit shown in figure $E_1 = 3\text{V}$, $E_2 = 2\text{V}$, $E_3 = 1\text{V}$ and $R = r_1 = r_2 = r_3 = 1\Omega$. (IIT-JEE 1981)



- (a) Find the potential difference between the points A and B and the currents through each branch.
 (b) If r_2 is short-circuited and the point A is connected to point B , find the currents through E_1 , E_2 , E_3 and the resistor R .

CURRENT MEASURING INSTRUMENT, PROBLEM BASED ON WHEATSTONE BRIDGE Single Correct

48. During an experiment with a meter bridge, the galvanometer shows a null point when the jockey is pressed at 40.0cm using a standard resistance of 90Ω , as show in the scale used in the meter bridge is 1mm .

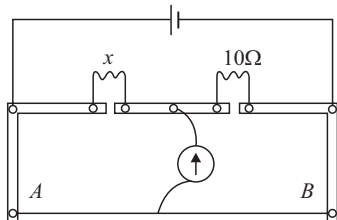


The unknown resistance is

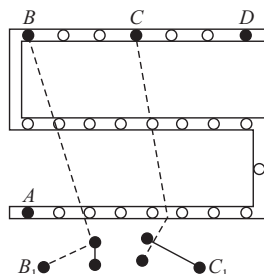
C-37.7 W-43.02 UA-19.28 (JEE Adv. 2014)

- (a) $60 \pm 0.15\Omega$ (b) $135 \pm 0.56\Omega$
 (c) $60 \pm 0.25\Omega$ (d) $135 \pm 0.23\Omega$

49. A meter bridge is set-up as shown in figure, to determine an unknown resistance X using a standard 10Ω resistor. The galvanometer shows null point when tapping-key is at 52 cm mark. The end-corrections are 1 cm and 2 cm respectively for the ends A and B . The determined value of X is (IIT-JEE 2011)

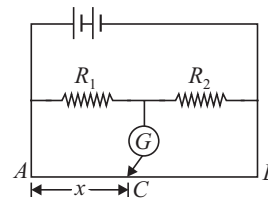


- (a) 10.2Ω (b) 10.6Ω
(c) 10.8Ω (d) 11.1Ω
50. **Statement-I:** In a meter bridge experiment, null point for an unknown resistance is measured. Now, the unknown resistance is put inside an enclosure maintained at a higher temperature. The null point can be obtained at the same point as before by decreasing the value of the standard resistance.
Statement-II: Resistance of a metal increases with increase in temperature. (IIT-JEE 2008)
- (a) If Statement-I is true, Statement-II is true; Statement-II is the correct explanation for Statement-I
(b) If Statement-I is true, Statement-II is true; Statement-II is not a correct
(c) If Statement-I is true; Statement-II is false
(d) If Statement-I is false; Statement-II is true
51. A resistance of 2Ω is connected across one gap of a meter-bridge (the length of the wire is 100 cm) and an unknown resistance, greater than 2Ω , is connected across the other gap. When these resistances are interchanged, the balance point shifts by 20 cm. Neglecting any corrections, the unknown resistance is (IIT-JEE 2007)
- (a) 3Ω (b) 4Ω
(c) 5Ω (d) 6Ω
52. A moving coil galvanometer of resistance 100Ω is used as an ammeter using a resistance 0.1Ω . The maximum deflection current in the galvanometer is $100\mu\text{A}$. Find the current in the circuit, so that the ammeter shows maximum deflection. (IIT-JEE 2005)
- (a) 100.1mA (b) 1000.1mA
(c) 10.01mA (d) 1.01mA
53. For the post office box arrangement to determine the value of unknown resistance, the unknown resistance should be connected between (IIT-JEE 2004)

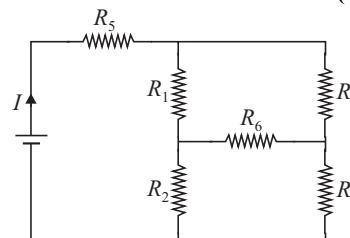


- (a) B and C (b) C and D
(c) A and D (d) B_1 and C_1

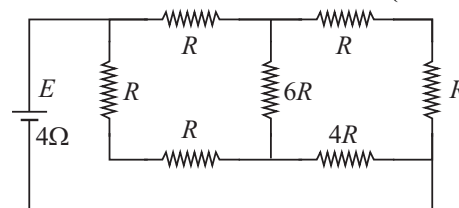
54. In the shown arrangement of the experiment of the meter bridge if AC corresponding to null deflection of galvanometer is x , what would be its value if the radius of the wire AB is doubled? (IIT-JEE 2003)



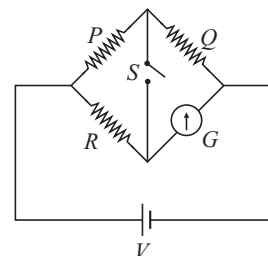
- (a) x
(b) $x/4$
(c) $4x$
(d) $2x$
55. In the given circuit, it is observed that the current I is independent of the value of the resistance R_6 . Then, the resistance values must satisfy (IIT-JEE 2001)



- (a) $R_1 R_2 R_5 = R_3 R_4 R_6$
(b) $\frac{1}{R_5} + \frac{1}{R_6} = \frac{1}{R_1 + R_2} + \frac{1}{R_3 + R_4}$
(c) $R_1 R_4 = R_2 R_3$
(d) $R_1 R_3 = R_2 R_4$
56. A battery of internal resistance 4Ω is connected to the network of resistances as shown in figure. In order that the maximum power can be delivered to the network, the value of R in Ω should be (IIT-JEE 1995)



- (a) $4/9$ (b) 2 (c) $8/3$ (d) 18
57. In the circuit shown $P \neq R$, the reading of galvanometer is same with switch S open or closed. Then (IIT-JEE 1999)



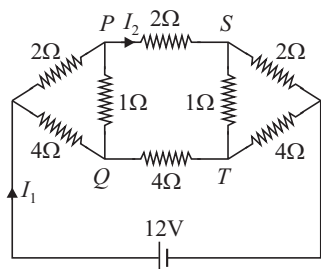
- (a) $I_R = I_G$ (b) $I_P = I_G$ (c) $I_Q = I_G$ (d) $I_Q = I_R$

Multiple Correct

58. Consider two identical galvanometers and two identical resistors with resistance R . If the internal resistance of the galvanometers $R_g < R/2$, which of the following statement(s) about any one of the galvanometers is/are true?

C-5.46 W-43.69 UA-40.95 PC-9.9 (JEE Adv. 2016)

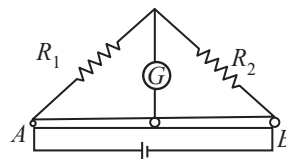
- (a) The maximum voltage range is obtained when all the components are connected in series
(b) The maximum voltage range is obtained when the two resistors and one galvanometer are connected in series, and the second galvanometer is connected in parallel to the first galvanometer
(c) The maximum current range is obtained when all the components are connected in parallel
(d) The maximum current range is obtained when the two galvanometers are connected in series, and the combination is connected in parallel with both the resistors
59. For the resistance network shown in the figure, choose the correct option(s). C-13.08 W-79.33 UA-7.6 (IIT-JEE 2012)



- (a) The current through PQ is zero
(b) $I_1 = 3A$
(c) The potential at S is less than that at Q
(d) $I_2 = 2A$
60. A micro-ammeter has a resistance of 100Ω and full scale range of $50\mu A$. It can be used as a voltmeter or as a higher range ammeter provided a resistance is added to it. Pick the correct range and resistance combination (s) (IIT-JEE 1991)
- (a) 50 V range with 10 k Ω resistance in series
(b) 10 V range with 200 k Ω resistance in series
(c) 5 mA range with 1Ω resistance in parallel
(d) 10 mA range with 1Ω resistance in parallel

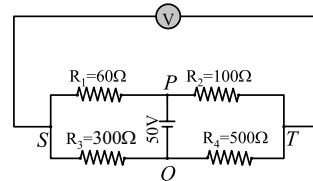
Numerical Types/Integer Types

61. Two resistances $R_1 = X\Omega$ and $R_2 = 1\Omega$ are connected to a wire AB of uniform resistivity, as shown in the figure. The radius of the wire varies linearly along its axis from 0.2 mm at A to 1 mm at B . A galvanometer (G) connected to the center of the wire, 50 cm from each end along its axis, shows zero deflection when A and B are connected to a battery. The value of x is C-9.83 W-22.8 UA-67.37 (JEE Adv. 2022)



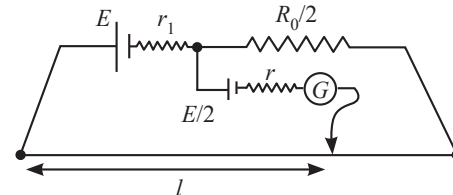
62. In the balanced condition, the values of the resistances of the four arms of a Wheatstone bridge are shown in the figure below. The resistance R_3 has temperature coefficient $0.0004\text{ }^\circ\text{C}^{-1}$. If the temperature of R_3 is increased by $100\text{ }^\circ\text{C}$, the voltage developed between S and T will be _____ volt.

C-3.01 W-72.29 UA-24.7 (JEE Adv. 2020)

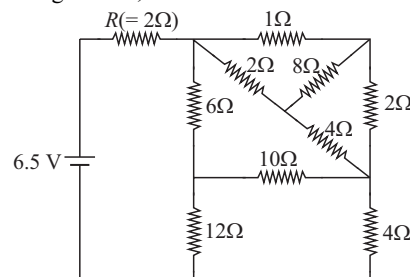


63. In order to measure the internal resistance r_1 of a cell of emf E , a meter bridge of wire resistance $R_0 = 50\Omega$, a resistance $R_0/2$, another cell of emf $E/2$ (internal resistance r) and a galvanometer G are used in a circuit, as shown in the figure. If the null point is found at $l = 72\text{ cm}$, then the value of $r_1 = \dots\Omega$.

C-9.7 W-70.34 UA-19.95 (JEE Adv. 2021)



64. In the following circuit,



the current through the resistor $R(=2\Omega)$ is I amperes. The value of I is _____.

(JEE Adv. 2015)

65. A galvanometer gives full scale deflection with 0.006 A current. By connecting it to a 4990Ω resistance, it can be converted into a voltmeter of range $0\text{--}30\text{ V}$. If connected to a $\frac{2n}{249}\Omega$ resistance, becomes an ammeter of range $0\text{--}1.5\text{ A}$. The value of n is _____.

(JEE Adv. 2014)

Subjective

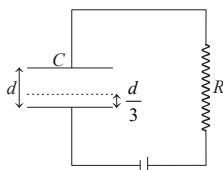
66. Show by diagram, how can we use a rheostat as the potential divider?

(IIT-JEE 2003)

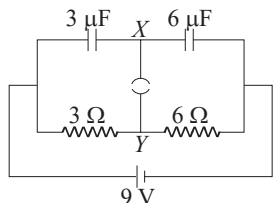
RC CIRCUIT

Single Correct

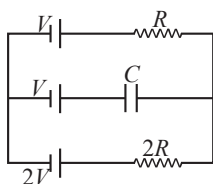
67. A parallel plate capacitor C with plates of unit area and separation d is filled with a liquid of dielectric constant $K = 2$. The level of liquid is $d/3$ initially. Suppose the liquid level decreases at a constant speed v , the time constant as a function of time t is (IIT-JEE 2008)



- (a) $\frac{6\epsilon_0 R}{5d + 3vt}$ (b) $\frac{(15d + 9vt)\epsilon_0 R}{2d^2 - 3dvt - 9v^2 t^2}$
 (c) $\frac{6\epsilon_0 R}{5d - 3vt}$ (d) $\frac{(15d - 9vt)\epsilon_0 R}{2d^2 + 3dvt - 9v^2 t^2}$
68. A circuit is connected as shown in the figure with the switch S open. When the switch is closed, the total amount of charge that flows from Y to X is (IIT-JEE 2007)



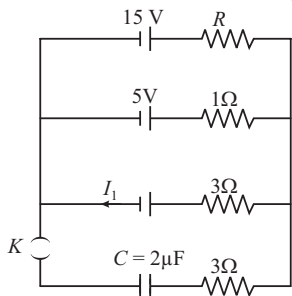
- (a) zero (b) $54 \mu\text{C}$
 (c) $27 \mu\text{C}$ (d) $81 \mu\text{C}$
69. In the given circuit, with steady current, the potential difference across the capacitor must be (IIT-JEE 2001)



Multiple Correct

70. In a circuit shown in the figure, the capacitor C is initially uncharged and the key K is open. In this condition, a current of 1A flows through the 1Ω resistor. The key is closed at time $t = t_0$. Which of the following statements is/are correct? [Given: $e^{-1} = 0.36$]

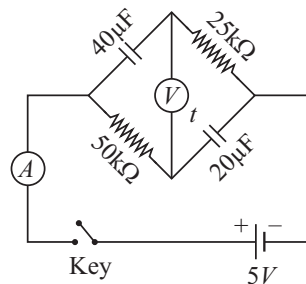
C-1.62 W-0 UA-58.73 PC-39.66 (JEE Adv. 2023)



- (a) The value of the resistance R is 3Ω .
 (b) For $t < t_0$, the value of current I_1 is 2A .
 (c) At $t = t_0 + 7.2 \mu\text{s}$, the current in the capacitor is 0.6A .
 (d) For $t \rightarrow \infty$, the charge on the capacitor is $12 \mu\text{C}$.

71. In the circuit shown below, the key is pressed at time $t = 0$. Which of the following statement(s) is/are true?

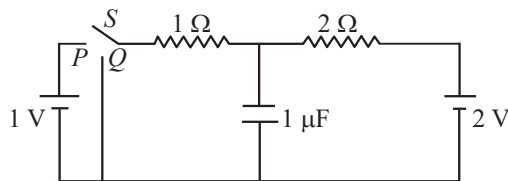
C-1.82 W-0 UA-47.97 PC-50.21 (JEE Adv. 2016)



- (a) The voltmeter displays -5V as soon as the key is pressed, and displays $+5\text{V}$ after a long time
 (b) The voltmeter will display 0V at time $t = \ln 2$ second
 (c) The current in the ammeter becomes $1/e$ of the initial value after 1 second
 (d) The current in the ammeter becomes zero after a long time

Comprehension-Based/Passage Based

Comprehension (Q.72 to 73) In the circuit shown below, the switch S is connected to position P for a long time so that the charge on the capacitor becomes $q_1 \mu\text{C}$. Then S is switched to position Q . After a long time, the charge on the capacitor is $q_2 \mu\text{C}$. (JEE Adv. 2021)



72. The magnitude of q_1 is _____. C-18.77 W-64.89 UA-16.34
 73. The magnitude of q_2 is _____. C-18.54 W-62.7 UA-18.77

Passage

Consider a simple RC circuit as shown in Figure 1.

Process 1 In the circuit, the switch S is closed at $t = 0$ and the capacitor is fully charged to voltage V_0 (i.e. charging continues for time $T \gg RC$). In the process, some dissipation (E_D) occurs across the resistance R . The amount of energy finally stored in the fully charged capacitor is E_c .

Process 2 In a different process the voltage is first set to $\frac{V_0}{3}$ and

maintained for a charging time $T \gg RC$. Then, the voltage is raised

to $\frac{2V_0}{3}$ without discharging the capacitor and again maintained for a

time $T \gg RC$. The process is repeated one more time by raising the voltage to V_0 and the capacitor is charged to the same final voltage V_0 as in process 1.

These two processes are depicted in Figure 2.

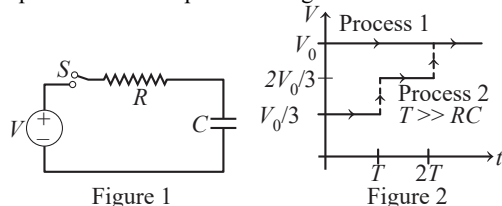


Figure 1

Figure 2

74. In process 1, the energy stored in the capacitor E_C and heat dissipated across resistance E_D are related by

(a) $E_C = E_D \ln 2$

(b) $E_C = E_D$

(c) $E_C = 2E_D$

(d) $E_C = \frac{1}{2}E_D$

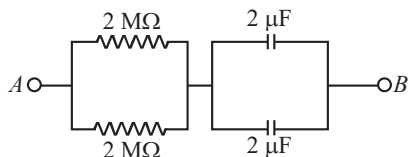
75. In process 2, total energy dissipated across the resistance E_D is
(JEE Adv. 2015)

(a) $E_D = \frac{1}{3} \left(\frac{1}{2} CV_0^2 \right)$ (b) $E_D = 3 \left(\frac{1}{2} CV_0^2 \right)$

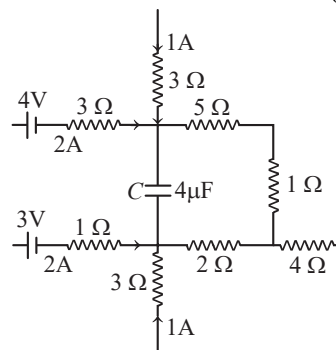
(c) $E_D = 3CV_0^2$ (d) $E_D = \frac{1}{2} CV_0^2$

Numerical Types/Integer Types

76. At time $t = 0$, a battery of 10 V is connected across points A and B in the given circuit. If the capacitors have no charge initially, at what time (in second) does the voltage across them become 4 V ?
[Take : $\ln 5 = 1.6$, $\ln 3 = 1.1$] (IIT-JEE 2010)

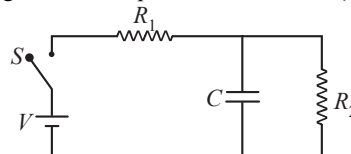


77. A part of circuit in a steady state along with the currents flowing in the branches, the values of resistances etc, is shown in the figure. Calculate the energy stored in the capacitor C ($4\mu\text{F}$). (IIT-JEE 1986)

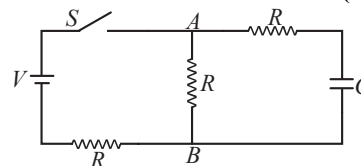


Subjective

78. At $t = 0$, switch S is closed. The charge on the capacitor is varying with time as $Q = Q_0(1 - e^{-\alpha t})$. Obtain the value of Q_0 and α in the given circuit parameters. (IIT-JEE 2005)



79. In the circuit shown in figure, the battery is an ideal one, with emf V . The capacitor is initially uncharged. The switch S is closed at time $t = 0$. (IIT-JEE 1998)



- (a) Find the charge Q on the capacitor at time t .
(b) Find the current in AB at time t . What is its limiting value as $t \rightarrow \infty$?

Answer Key (Abhedya)

Exercise -1

- | | | | | | | | | | |
|-----------------|-----------|--------------|------------|--------------|--------------|------------|-----------|----------|---------|
| 1. (a, b, c, d) | 2. (a, d) | 3. (a, c, d) | 4. (b, c) | 5. (a, c, d) | 6. (a, b, c) | 7. (a, c) | 8. (b, d) | 9. (a,c) | 10. (c) |
| 11. (a,b) | 12. (b) | 13. (b) | 14. (d) | 15. (c) | 16. (c) | 17. (a) | 18. (b) | 19. (a) | 20. (c) |
| 21. (c) | 22. (b) | 23. [45] | 24. [1.47] | 25. [2.00] | 26. [44] | 27. [4000] | 28. [32] | 29. [20] | 30. (b) |
| 31. (c) | 32. (c) | 33. (c) | 34. (d) | | | | | | |

Exercise -2

- | | | | | | | | | | |
|---------|---------|--------|--------|--------------|----------|----------|--------|--------|---------|
| 1. (c) | 2. (d) | 3. (a) | 4. (a) | 5. (a,b,c,d) | 6. (a,d) | 7. (b,c) | 8. (a) | 9. (b) | 10. (a) |
| 11. (d) | 12. (c) | | | | | | | | |

Exercise -3

- | | | | | | | | | | |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|
| 1. (c) | 2. (d) | 3. (c) | 4. (d) | 5. (b) | 6. (a) | 7. (c) | 8. (d) | 9. (b) | 10. (c) |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|

Exercise -4

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|---------------|--------------------|------------------|------------|------------|-------------|-------------|-------------|---------------|---------------|
| 1. (c) | 2. (c) | 3. (d) | 4. (d) | 5. [False] | 6. (a) | 7. (a) | 8. [4] | 9. [2] | 12. (d) |
| 13. (c) | 14. (a) | 15. (d) | 16. (a) | 17. (d) | 18. (b) | 19. (b) | 20. (b) | 21. (c) | 22. (a,b,c) |
| 23. (c, d) | 24. (b) | 25. (c) | 26. (a) | 27. (a) | 28. (a,c,d) | 29. (c) | 30. (a,b,d) | 31. [False] | 35. (c) |
| 36. (d) | 37. (d) | 38. (a, b, c, d) | 39. (a, c) | 40. (a,d) | 41. (c) | 42. [5] | 48. (c) | 49. (b) | 50. (d) |
| 51. (a) | 52. (a) | 53. (c) | 54. (a) | 55. (c) | 56. (b) | 57. (a) | 58. (a, c) | 59. (a,b,c,d) | 60. (bc) |
| 61. [5] | 62. [0.26 to 0.28] | | 63. [3] | 64. [1] | 65. [5] | 67. (a) | 68. (c) | 69. (c) | 70. (a,b,c,d) |
| 71. (a,b,c,d) | 72. [1.33] | 73. [0.67] | 74. (b) | 75. (a) | 76. [2] | 77. [0.288] | | | |